From Group Theory to Representation Theory

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Contents

Chapter 1

Rudiments of Group Theory

1 Review

Group Axioms

A group consists of a non-empty set G and a binary operation on G such that the following axioms hold:

- $\forall g, h, k \in G, (gh)k = g(hk)$ (associativity)
- $\exists e \in G$ s.t. $eg = ge = g \ \forall g \in G$ (identity)
- $\forall g \in G, \exists g^{-1} \in G$ s.t. $gg^{-1} = g^{-1}g = e$ (inverses)

Properties of Groups

- If $xy = yx$, the commutator $[x, y] = xyx^{-1}y^{-1} = e$
- If $xy = yx \ \forall x, y \in G$, then G is an **abelian** group.
- A $x \in G$ is of finite order if $\exists n \in \mathbb{N}$ s.t. $x^n = e$
- $|G|$, the number of elements in G is defined to be the order of a finite group G. Every element of a finite group is of finite order
- There are infinite groups that has every element having a finite order, but there are also infinite groups in which the identity is the only emenet of finite order (torsion-free groups such as $\mathbb{Z}$)

Properties of Subgroups

- H is a **subgroup** of G, denoted $H \leq G$, if $H \subseteq G$ and H is a group under the same operation as G. In particular, $e \in H$ and $h^{-1} \in H \ \forall h \in H$
- Every subgroup of a finite group is finite, but an infinite group always has both finite and infinite subgroups
- Every subgroup of an abelian group is abelian, but a non-abelian group can have both abelian and non-abelian subgroups

Proposition 1.1. *If $H, K \leq G$, then $H \cap K \leq G$. More generally, $H_i \leq G \Rightarrow \bigcap H_i \leq G$*

Proof. Suppose $H_i \leq G \ \forall i$
Since each $H_i \leq G$, we have $e \in H_i \ \forall i$, so $e \in \bigcap H_i$ and $\bigcap H_i \neq \emptyset$. For $x, y \in \bigcap H_i$, we have $x, y \in H_i \ \forall i$, so by closure of each H_i, $xy \in H_i \ \forall i$, giving $xy \in \bigcap H_i$. Similarly, $x^{-1} \in H_i \ \forall i$, so $x^{-1} \in \bigcap H_i$.
$\therefore \bigcap H_i \leq G$ □

Theorem (Lagrange's Theorem)**.** *Let G be a **finite** group, and let $H \leq G$. Then $|H| \mid |G|$.*

Proof. Let $x \in G$. Define $f_x \colon H \to xH$ by $f_x(h) = xh$. If $f_x(h_1) = f_x(h_2)$, then $xh_1 = xh_2$, and multiplying on the left by x^{-1} gives $h_1 = h_2$. Also, every element of xH has the form xh for some $h \in H$, so f_x is surjective. Thus f_x is a bijection, and $|xH| = |H|$.
We claim that any two sets of the form xH are either equal or disjoint. Suppose that $xH \cap yH \neq \emptyset$. Then $xh_1 = yh_2$ for some $h_1, h_2 \in H$, so $x = y(h_2h_1^{-1})$. Since $h_2h_1^{-1} \in H$, we have $x = yh$ for some $h \in H$, and so $xH = yhH = yH$. Thus the distinct sets of the form xH partition G.
Since G is finite, there are finitely many such sets, say r. Each has cardinality $|H|$, so $|G| = r|H|$.
$\therefore |H| \mid |G|$. ☐

If $X \subseteq G, \langle X \rangle \coloneqq \bigcap\{H \leq G \mid X \subseteq H\}$. By Proposition 1.1, $\langle X \rangle \leq G$, and it is the smallest subgroup of G containing X. We call $\langle X \rangle$ the **subgroup generated by X**, and we say that X **generates** $\langle X \rangle$.

Proposition 1.2. *Let $X \subseteq G$. Then $\langle X \rangle = \{e\} \cup \{x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \mid r \geq 1,\ x_i \in X,\ \epsilon_i = \pm 1,\ 1 \leq i \leq r\}$.*

Proof. Let $S = \{e\} \cup \{x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \mid r \geq 1,\ x_i \in X,\ \epsilon_i = \pm 1,\ 1 \leq i \leq r\}$.
We first show that $S \leq G$. Clearly $e \in S$. If $a = e$ or $b = e$, then $ab \in S$. Otherwise, if $a = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \in S$ and $b = y_1^{\delta_1} \cdots y_s^{\delta_s} \in S$, then $ab = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} y_1^{\delta_1} \cdots y_s^{\delta_s} \in S$. Also $e^{-1} = e$, and if $a = x_1^{\epsilon_1} \cdots x_r^{\epsilon_r}$, then $a^{-1} = x_r^{-\epsilon_r} \cdots x_1^{-\epsilon_1} \in S$. Hence $S \leq G$.
Let $\mathcal{H}_X = \{H \leq G \mid X \subseteq H\}$. By Proposition 1.1, $\langle X \rangle = \bigcap_{H \in \mathcal{H}_X} H \leq G$. Since $X \subseteq S$, we have $S \in \mathcal{H}_X$, and so $\langle X \rangle = \bigcap_{H \in \mathcal{H}_X} H \subseteq S$.
Conversely, let $H \in \mathcal{H}_X$. Since $x_i \in X \subseteq H$, we have $x_i^{\epsilon_i} \in H$, so $x_1^{\epsilon_1} \cdots x_r^{\epsilon_r} \in H$. Also $e \in H$, hence $S \subseteq H$. Thus $S \subseteq H$ for all $H \in \mathcal{H}_X$, so $S \subseteq \bigcap_{H \in \mathcal{H}_X} H = \langle X \rangle$.
$\therefore S = \langle X \rangle$. ☐

Cyclic Groups

- A group G is **cyclic** if $G = \langle g \rangle$ for some $g \in G$. By Proposition 1.2, $\langle g \rangle = \{g^n \mid n \in \mathbb{Z}\}$ s.t. $|\langle g \rangle| = n$ and $o(g) = n$
- All cyclic groups are abelian
- If g is not of finite order, then $\langle g \rangle$ is a torsion-free infinite abelian group, and $\langle g \rangle \cong \mathbb{Z}$
- If g is of finite order, then $\langle g \rangle \cong \mathbb{Z}/n\mathbb{Z}$
- If $g \in G$ and $o(g) = n \in \mathbb{Z}$, $\langle g \rangle \leq G$, so by Lagrange's Theorem, $n \mid |G|$. Thus, the order of an element of a finite group must divide the order of that group

If $X, Y \subseteq G, XY \coloneqq \{xy \mid x \in X, y \in Y\} \subseteq G$. Also, $X^{-1} \coloneqq \{x^{-1} \mid x \in X\} \subseteq G$. If $H \neq \emptyset$, then $H \leq G \iff HH = H$ and $H^{-1} = H$

Proposition 1.3. *Let $H, K \leq G$. Then $HK \leq G \iff HK = KH$*

Proof. ($\Rightarrow$) Suppose $HK \leq G$. Since HK is a subgroup, $(HK)^{-1} = HK$. Also $H^{-1} = H$ and $K^{-1} = K$, since $H, K \leq G$. Hence $(HK)^{-1} = K^{-1}H^{-1} = KH$. Therefore $HK = (HK)^{-1} = KH$.
($\Leftarrow$) Suppose $HK = KH$. Since $e \in H \cap K$, we have $e = ee \in HK$. Let $h_1k_1, h_2k_2 \in HK$. Since $k_1h_2 \in KH = HK$, write $k_1h_2 = h_3k_3$ for some $h_3 \in H,\ k_3 \in K$. Then $(h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2 = h_1h_3k_3k_2 \in HK$. Also $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$ for all $h \in H,\ k \in K$.
$\therefore HK \leq G$. ☐

Theorem 1.4. *Let $G = \langle g \rangle$ be a cyclic group of order n*

- *For each $d \mid n$, there is exactly one subgroup of order d, $\langle g^{\frac{n}{d}} \rangle$*
- *If $d, e \mid n$, then $\langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\gcd(d,e)}} \rangle$*
- *If $d, e \mid n$, then $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\mathrm{lcm}(d,e)}} \rangle$*

Proof. First note that if $m \mid n$, then $o(g^m) = \frac{n}{m}$. Indeed, $(g^m)^{\frac{n}{m}} = g^n = e$, and if $(g^m)^t = e$, then $n \mid mt$, so $\frac{n}{m} \mid t$.
Let $d \mid n$. Taking $m = \frac{n}{d}$, we get $o(g^{\frac{n}{d}}) = d$, so $\langle g^{\frac{n}{d}} \rangle$ is a subgroup of order d.
For uniqueness, let $H \leq G$ with $|H| = d$, and let $a = \min\{m \in \mathbb{N} \mid g^m \in H\}$. This set is non-empty since $g^n = e \in H$. If $g^b \in H$, write $b = qa + r$, $0 \leq r < a$. Then $g^r = g^b(g^a)^{-q} \in H$, so by minimality of

a, $r = 0$. Hence $a \mid b$, so $H \subseteq \langle g^a \rangle$. Since $g^a \in H$, $\langle g^a \rangle \subseteq H$, and hence $H = \langle g^a \rangle$. Applying the same division argument to $n = qa + r$ gives $a \mid n$, so $|H| = |\langle g^a \rangle| = \frac{n}{a} = d$. Thus $a = \frac{n}{d}$, and $H = \langle g^{\frac{n}{d}} \rangle$.
Now let $d, e \mid n$. An element $g^b \in \langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle$ iff $\frac{n}{d} \mid b$ and $\frac{n}{e} \mid b$, iff $\mathrm{lcm}(\frac{n}{d}, \frac{n}{e}) = \frac{n}{\gcd(d,e)}$ divides b. Hence $\langle g^{\frac{n}{d}} \rangle \cap \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\gcd(d,e)}} \rangle$.
Finally, $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \{ g^{\frac{n}{d} r + \frac{n}{e} s} \mid r, s \in \mathbb{Z} \}$. Since $\frac{n}{d}\mathbb{Z} + \frac{n}{e}\mathbb{Z} = \gcd(\frac{n}{d}, \frac{n}{e})\mathbb{Z} = \frac{n}{\mathrm{lcm}(d,e)}\mathbb{Z}$, we get $\langle g^{\frac{n}{d}} \rangle \langle g^{\frac{n}{e}} \rangle = \langle g^{\frac{n}{\mathrm{lcm}(d,e)}} \rangle$.
$\therefore$ all three assertions hold. $\square$

Cosets

- If $H \leq G$ and $x \in G, xH$ is called a left coset of H in G and Hx is called a right coset of H in g
- There is a bijective correspondence between left and right cosets of H in G, sending xH to its inverse $(xH)^{-1} = Hx^{-1}$
- Any two cosets of are equal or disjoint, with $xH = yH \iff y^{-1}x \in H$
- A $x \in G$ lies in exacty one coset of H, xH
- $\forall x \in G, \exists$ a bijective correspondence between H and xH
- We define the index of H in G, $|G : H|$ to be the number of cosets of H in G
- The cosets of H in G partition G into $|G : H|$ disjoint sets of cardinality $|H|$, and we have $|G| = |G : H||H|$ (Lagrange's Theorem)
- All subgroups of a finite group are of finite index, while subgroups of an infinite group may be of finite or infinite index

Theorem 1.5. *Let $G = \langle g \rangle$ be an infinite cyclic group*

- *For each $d \in \mathbb{N}, \exists$ exactly one subgroup of G of index d, $\langle g^d \rangle$. Furthermore, every non-trivial subgroup of G is of finite index*
- *The intersection of the subgroups of indices d and e is the subgroup of index* $\mathrm{lcm}(d, e)$
- *The product of subgroups of indices d and e is the subgroup of index* $\gcd(d, e)$

Proof. As in the proof of Theorem 1.4, let $1 \neq H \leq G$, and choose $a = \min\{m \in \mathbb{N} \mid g^m \in H\}$. This set is non-empty since H contains some $g^b \neq e$, and then either $b > 0$ or $g^{-b} \in H$ with $-b > 0$. If $g^c \in H$, write $c = qa + r$, $0 \leq r < a$. Then $g^r = g^c(g^a)^{-q} \in H$, so minimality of a gives $r = 0$. Hence $a \mid c$, so $H = \langle g^a \rangle$.
For $d \in \mathbb{N}$, the cosets of $\langle g^d \rangle$ are $g^i \langle g^d \rangle$, $0 \leq i < d$. Hence $|G : \langle g^d \rangle| = d$. By the previous paragraph, every non-trivial subgroup is $\langle g^a \rangle$ for some $a \in \mathbb{N}$, so it has finite index a, and $\langle g^d \rangle$ is the unique subgroup of index d.
If $d, e \in \mathbb{N}$, then $g^b \in \langle g^d \rangle \cap \langle g^e \rangle \iff d \mid b$ and $e \mid b \iff \mathrm{lcm}(d, e) \mid b$. Thus $\langle g^d \rangle \cap \langle g^e \rangle = \langle g^{\mathrm{lcm}(d,e)} \rangle$, which has index $\mathrm{lcm}(d, e)$.
Since G is abelian, $\langle g^d \rangle \langle g^e \rangle \leq G$ by Proposition 1.3. Moreover $\langle g^d \rangle \langle g^e \rangle = \{ g^{dr+es} \mid r, s \in \mathbb{Z} \} = \langle g^{\gcd(d,e)} \rangle$, since $d\mathbb{Z} + e\mathbb{Z} = \gcd(d, e)\mathbb{Z}$. Hence the product has index $\gcd(d, e)$.
$\therefore$ all three assertions hold. $\square$

Theorem 1.6. *If $K \leq H \leq G$, then $|G : K| = |G : H||H : K|$*

Proof. By the coset-counting identity in Lagrange's Theorem, the coefficient of $|H|$ in $|G|$ is $|G : H|$, and the coefficient of $|K|$ in $|H|$ is $|H : K|$. Hence $|G| = |G : H||H|$ and $|H| = |H : K||K|$.
Substituting, $|G| = |G : H||H : K||K|$. But applying the same identity directly to $K \leq G$ gives $|G| = |G : K||K|$.
$\therefore |G : K| = |G : H||H : K|$. $\square$

Let $H \leq G$, and let $\mathcal{I}$ be an indexing set that is in bijective correspondence with G/H. $T = \{t_i \mid i \in \mathcal{I}\} \subseteq G$ is said to be a (*left*) *transversal* for H (or a set of (*left*) *coset representatives* of H in G) if the sets t_iH are precisely the distinct cosets of H in G

Normal Subgroups

- N is a *normal subgroup* of G if $xN = Nx\ \forall x \in G$, or equivalently if $xNx^{-1} \subseteq N$ for all $x \in G$
- If G is abelian, then every subgroup of G is normal.
- If {e} and G are the only normal subgroups of G, then we say that G is *simple*.
- If $H \trianglelefteq G$ and $K \trianglelefteq H$, then it is not necessarily true that $K \trianglelefteq G$. However, it is clearly true that if $K \trianglelefteq G$ and $K \leq H \leq G$, then $K \trianglelefteq H$.

Proposition 1.7. *Let $H, K \leq G$. If $K \trianglelefteq G$, then $HK \leq G$ and $H \cap K \trianglelefteq H$; if also $H \trianglelefteq G$, then $HK \trianglelefteq G$ and $H \cap K \trianglelefteq G$.*

Proof. Let $H, K \leq G$. Suppose that $K \trianglelefteq G$.
Then we have $gK = Kg\ \forall g \in G$. Since $H \leq G$, we have $hK = Kh\ \forall h \in H$.
Take $x \in HK$, so $x = hk$ for some $h \in H, k \in K$. Since $hK = Kh, \exists$ some $h' \in H$ such that $x = kh'$, so we have $HK \subseteq KH$. By a similar argument, we have $KH \subseteq HK$, so we conclude $HK = KH$. By Proposition 1.3, we have $HK \leq G$.
Take $y \in H \cap K$ and $h \in H$. Since $y \in H$, we have $hyh^{-1} \in H$. Since $y \in K$ and $K \trianglelefteq G$, we have $hKh^{-1} \subseteq K \Rightarrow hyh^{-1} \in K$. It follows that $hyh^{-1} \in H \cap K$, so $H \cap K \trianglelefteq H$.
Suppose also that $H \trianglelefteq G$. Take $hk \in HK$ and $g \in G$. Then $g(hk)g^{-1} = (ghg^{-1})(gkg^{-1})$, where $ghg^{-1} \in H$ since $H \trianglelefteq G$ and $gkg^{-1} \in K$ since $K \trianglelefteq G$. Hence $g(hk)g^{-1} \in HK$, so $gHKg^{-1} \subseteq HK \Rightarrow HK \trianglelefteq G$.
Take $y \in H \cap K$ and $g \in G$. Since $H \trianglelefteq G$, $gyg^{-1} \in H$, and since $K \trianglelefteq G$, $gyg^{-1} \in K$. So $gyg^{-1} \in H \cap K \Rightarrow H \cap K \trianglelefteq G$. □

Proposition 1.8. *Any subgroup of index 2 is normal*

Proof. Let $H \leq G$ and suppose that $|G : H| = 2$
Then there are 2 left cosets, H and G - H
It follows that $x \in H \iff xH = H = Hx$, and $x \notin H \iff xH = G - H = Hx$
$\therefore H \trianglelefteq G$ □

Theorem 1.9. *If $N \trianglelefteq G$, then G/N forms the quotient group under the operation $(xN)(yN) = (xy)N$*

Proof. First, we show that the operation is well-defined. Suppose $xN = x'N$ and $yN = y'N$. Then $x' = xn_1$ and $y' = yn_2$ for some $n_1, n_2 \in N$. Since $N \trianglelefteq G$, $y^{-1}n_1y \in N$, so $n_1y = y(y^{-1}n_1y)$. Hence $x'y' = xn_1yn_2 = xy(y^{-1}n_1y)n_2 \in xyN$, and so $x'y'N = xyN$. Thus $(xN)(yN) = (xy)N$ is well-defined.
Associativity follows from associativity in G, since $((xN)(yN))(zN) = ((xy)z)N$ and $(xN)((yN)(zN)) = (x(yz))N$.
The identity is $N = eN$, since $N(xN) = xN = (xN)N$. Also, $(xN)(x^{-1}N) = N = (x^{-1}N)(xN)$, so $x^{-1}N$ is the inverse of xN.
$\therefore G/N$ is a group under the operation $(xN)(yN) = (xy)N$. □

The identity element of G/N is N, and the inverse is $x^{-1}N$. If G is abelian, then G/N is also abelian

Conjugation

- The conjugate of an element x by g is defined as gxg^{-1}
- x and y are conjugate if $\exists g \in G$ s.t. $y = gxg^{-1}$
- No 2 distinct elements of an abelian group can be conjugate
- $N \trianglelefteq G \iff \forall n \in N, gng^{-1} \in N$

Permutations

- A permutation of a set X is a bijective set map from X to X. The set of permutations of X forms a group under composition
- If $X = \{1, \ldots, n\}$, then this groups is called the symmetric group, S_n
- S_n is a finite group of order $n!$

- $\rho \in S_n$ is called a r-cycle if $\exists$ distinct integers $1 \leq a_1, \dots, a_r \leq n$ s.t. $\rho(a_i) = a_{i+1} \forall\ 1 \leq i \leq r, \rho(a_r) = a_1, \rho(b) = b\ \forall 1 \leq b \leq n$ which is not equal to some a_i
- Two cycles are disjoint if there is no number that is moved by both cycles
- Every element of S_n can be written as a product of disjoint cycles
- The cycle structure refers to the partion of n consisting of the lengths of the cycles that appear in a disjoint cycle decomposition of ρ

Proposition 1.10. *Let $n \in \mathbb{N}$. Then two elements of S_n are conjugate iff they have the same cycle structure*

Proof. Let $\alpha, \beta \in S_n$.
($\Rightarrow$) Suppose that α and β are conjugate, so $\beta = \sigma\alpha\sigma^{-1}$ for some $\sigma \in S_n$. We show that $\sigma\alpha\sigma^{-1}$ is the permutation with the same cycle structure as α obtained by applying σ to the symbols in α. Let π be this permutation.
Suppose that α fixes some i, so $\sigma(i)$ resides in a 1-cycle of π, and π fixes $\sigma(i)$. Then $\sigma\alpha\sigma^{-1}(\sigma(i)) = \sigma\alpha(i) = \sigma(i)$, so $\sigma\alpha\sigma^{-1}$ fixes $\sigma(i)$ as well.
Suppose that α moves i, and let $\alpha(i) = j$. Take the complete factorization of α to be

$$\alpha = \gamma_1\gamma_2 \cdots (\cdots i\ j \cdots) \cdots \gamma_t$$

Take $\sigma(i) = k$ and $\sigma(j) = l$, so $\pi : k \mapsto l$. Since $\sigma\alpha\sigma^{-1} : k \mapsto i \mapsto j \mapsto l$, we have $\sigma\alpha\sigma^{-1}(k) = \pi(k)$. So π and $\sigma\alpha\sigma^{-1}$ agree on all symbols of the form $k = \sigma(i)$. Since σ is a surjection, it follows that $\pi = \sigma\alpha\sigma^{-1} = \beta$, so α and β have the same cycle structure.
($\Leftarrow$) Suppose that α and β have the same cycle structure. Place the complete factorization of α over that of β so that cycles of the same length correspond, and let $\gamma \in S_n$ be the function sending the top to the bottom. For example, if

$$\alpha = \gamma_1\gamma_2 \cdots (\cdots i\ j \cdots) \cdots \gamma_t$$
$$\beta = \delta_1\delta_2 \cdots (\cdots k\ l \cdots) \cdots \delta_t$$

then $\gamma(i) = k, \gamma(j) = l$, etc. γ is a permutation, since every i between 1 and n occurs exactly once in a complete factorization. By the forward direction, $\gamma\alpha\gamma^{-1} = \beta$, so α and β are conjugate. ☐

- A **transposition** in S_n is a 2-cycle
- Every element of S_n can be written as a (not necessarily disjoint) product of transpositions in many ways. However, any two such expressions use the same number, modulo 2, of transpositions
- A permutation is **even** (resp. **odd**) if it is a product of an even (resp. odd) number of transpositions; every permutation is exactly one of these. Since an r-cycle is a product of $r - 1$ transpositions, an r-cycle is even iff r is odd
- The subset of S_n consisting of all even permutations is a subgroup of index 2, hence is normal in S_n by Proposition 1.8; this is the **alternating group** of degree n, denoted A_n

Homomorphisms

- Let G and H be groups. A **homomorphism** is a map $\varphi\colon G \to H$ satisfying $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$
- If φ is a homomorphism, then $\varphi(1) = 1$ and $\varphi(x^{-1}) = \varphi(x)^{-1}$ for any $x \in G$
- The **trivial homomorphism** from G to H sends every element of G to the identity of H
- φ is a **monomorphism** if injective, an **epimorphism** if surjective, and an **isomorphism** if bijective. If φ is an isomorphism, then so is $\varphi^{-1} : H \to G$
- A homomorphism $\varphi : G \to G$ is an **endomorphism** of G; a bijective endomorphism is an **automorphism**
- If there is an isomorphism $\varphi : G \to H$, we say G and H are **isomorphic** and write $G \cong H$. Isomorphism is an equivalence relation on groups, so we can speak of the *isomorphism class* of a group.

Standard examples of homomorphisms and isomorphisms:

- Let $G = \langle g \rangle$ and $H = \langle h \rangle$ be cyclic groups of order n. The map $\varphi : G \to H$ by $\varphi(g^a) = h^a$ is an isomorphism. Hence any two finite cyclic groups of the same order are isomorphic.
- Let G be a group, $H \leq G$, $g \in G$. The **conjugate** of H by g is $gHg^{-1} = \{ghg^{-1} \mid h \in H\}$, and $gHg^{-1} \leq G$. The map $\varphi : H \to gHg^{-1}$ by $\varphi(h) = ghg^{-1}$ is an isomorphism, so conjugate subgroups are isomorphic. (However, isomorphic subgroups need not be conjugate)
- Let $X = \{x_1, \dots, x_n\}$ and let S_X be the group of permutations of X. The map $\varphi : S_n \to S_X$ by $\varphi(\rho)(x_i) = x_{\rho(i)}$ is an isomorphism
- Let G be a group, $N \trianglelefteq G$. The **natural map** $\eta : G \to G/N$ defined by $\eta(x) = xN$ is an epimorphism

Kernel and image:

- If $\varphi : G \to H$ is a homomorphism, the **kernel** of φ is $\ker\varphi = \{g \in G \mid \varphi(g) = 1\}$, and the **image** is $\operatorname{im}\varphi = \{\varphi(g) \mid g \in G\}$
- We use $\varphi(K) = \{\varphi(k) \mid k \in K\}$ for the image of $K \leq G$
- E.g., if $N \trianglelefteq G$ and $\eta : G \to G/N$ is the natural map, then $\ker\eta = N$ and $\eta(K) = KN/N$ for any $K \leq G$. ($\eta(K) = K/N$ if K contains N.)

Proposition 1.11. *Let G and H be groups, and let $\varphi : G \to H$ be a homomorphism. Then* $\ker\varphi \trianglelefteq G$, *and $\varphi(K) \leq H$ for any $K \leq G$.*

Proof. First, $\varphi(1) = 1$, so $1 \in \ker\varphi$. If $x, y \in \ker\varphi$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = 1$, so $xy^{-1} \in \ker\varphi$. Hence $\ker\varphi \leq G$.
Let $x \in \ker\varphi$ and $g \in G$. Then $\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g^{-1}) = \varphi(g)\varphi(g)^{-1} = 1$. Thus $gxg^{-1} \in \ker\varphi$, so $\ker\varphi \trianglelefteq G$.
Now let $K \leq G$. Since $1 \in K$, we have $1 = \varphi(1) \in \varphi(K)$. If $a, b \in \varphi(K)$, then $a = \varphi(k_1)$ and $b = \varphi(k_2)$ for some $k_1, k_2 \in K$. Hence $ab^{-1} = \varphi(k_1)\varphi(k_2)^{-1} = \varphi(k_1k_2^{-1}) \in \varphi(K)$.
$\therefore \varphi(K) \leq H$. □

Theorem (Fundamental Theorem on Homomorphisms)**.** *If G and H are groups and $\varphi : G \to H$ is a homomorphism, then there is an isomorphism $\psi : G/K \to \varphi(G)$ such that $\varphi = \psi \circ \eta$, where $K = \ker\varphi$ and $\eta : G \to G/K$ is the natural map; moreover, the map ψ is uniquely determined.*

Proof. Define $\psi \colon G/K \to \varphi(G)$ by $\psi(xK) = \varphi(x)$. This is well-defined: if $xK = yK$, then $y^{-1}x \in K$, so $1 = \varphi(y^{-1}x) = \varphi(y)^{-1}\varphi(x)$, and hence $\varphi(x) = \varphi(y)$.
The map ψ is a homomorphism, since $\psi((xK)(yK)) = \psi(xyK) = \varphi(xy) = \varphi(x)\varphi(y) = \psi(xK)\psi(yK)$.
It is surjective by definition of $\varphi(G)$: if $h \in \varphi(G)$, then $h = \varphi(x) = \psi(xK)$ for some $x \in G$. It is injective since $\psi(xK) = 1 \Rightarrow \varphi(x) = 1 \Rightarrow x \in K \Rightarrow xK = K$. Hence ψ is an isomorphism.
For all $x \in G$, $(\psi \circ \eta)(x) = \psi(xK) = \varphi(x)$, so $\varphi = \psi \circ \eta$.
Finally, suppose $\theta \colon G/K \to \varphi(G)$ also satisfies $\varphi = \theta \circ \eta$. For any $xK \in G/K$, $\theta(xK) = \theta(\eta(x)) = \varphi(x) = \psi(xK)$. Thus $\theta = \psi$, so ψ is uniquely determined. □

Theorem (First Isomorphism Theorem)**.** *Let G be a group. If $N \trianglelefteq G$ and $H \leq G$, then $HN/N \cong H/(H \cap N)$.*

Proof. By Proposition 1.7, $HN \leq G$ and $H \cap N \trianglelefteq H$. Since $N \trianglelefteq G$ and $HN \leq G$, we also have $N \trianglelefteq HN$, so HN/N is a group.
Let $\eta \colon G \to G/N$ be the natural map $\eta(g) = gN$, and let $\varphi = \eta|_H \colon H \to G/N$. Then φ is a homomorphism, and $\ker\varphi = \{h \in H \mid hN = N\} = H \cap N$.
For the natural map, $\eta(K) = KN/N$ for any $K \leq G$. Taking $K = H$, we get $\varphi(H) = \eta(H) = HN/N$, so we may regard φ as a homomorphism $H \to HN/N$.
By the Fundamental Theorem on Homomorphisms, $H/\ker\varphi \cong \varphi(H)$. Thus $H/(H \cap N) \cong HN/N$.
$\therefore HN/N \cong H/(H \cap N)$. □

Theorem (Correspondence Theorem). *Let G and H be groups, and let $\varphi : G \to H$ be an epimorphism having kernel N. Then there is a bijective correspondence given by φ between the set of subgroups of G that contain N and the set of subgroups of H. If K is a subgroup of G containing N, then this correspondence sends K to $\varphi(K)$; if L is a subgroup of H, then the subgroup of G sent to L under this correspondence is $\varphi^{-1}(L) = \{x \in G \mid \varphi(x) \in L\}$. Moreover, if K_1 and K_2 are subgroups of G containing N, then:*

- *$K_2 \leq K_1$ iff $\varphi(K_2) \leq \varphi(K_1)$, and in this case we have $|K_1 : K_2| = |\varphi(K_1) : \varphi(K_2)|$.*
- *$K_2 \trianglelefteq K_1$ iff $\varphi(K_2) \trianglelefteq \varphi(K_1)$, and in this case the map from K_1/K_2 to $\varphi(K_1)/\varphi(K_2)$ sending xK_2 to $\varphi(x)\varphi(K_2)$ is an isomorphism.*

Proof. Let $\mathcal{S} = \{K \leq G \mid N \leq K\}$ and $\mathcal{T} = \{L \leq H\}$. For $K \in \mathcal{S}$, Proposition 1.11 gives $\varphi(K) \leq H$. If $L \in \mathcal{T}$, then $\varphi^{-1}(L) \leq G$ by the subgroup test, and $N \leq \varphi^{-1}(L)$, so $\varphi^{-1}(L) \in \mathcal{S}$.
Define $\Phi\colon \mathcal{S} \to \mathcal{T}$ by $\Phi(K) = \varphi(K)$, and define $\Psi\colon \mathcal{T} \to \mathcal{S}$ by $\Psi(L) = \varphi^{-1}(L)$. Since φ is surjective, $\Phi(\Psi(L)) = \varphi(\varphi^{-1}(L)) = L$. If $K \in \mathcal{S}$, then $K \subseteq \Psi(\Phi(K))$. Conversely, if $x \in \Psi(\Phi(K))$, then $\varphi(x) = \varphi(k)$ for some $k \in K$, so $k^{-1}x \in \ker\varphi = N \leq K$, and hence $x \in K$. Thus $\Psi(\Phi(K)) = K$. Hence Φ and Ψ are inverse bijections.
Let $K_1, K_2 \in \mathcal{S}$. If $K_2 \leq K_1$, then $\varphi(K_2) \leq \varphi(K_1)$. Conversely, if $\varphi(K_2) \leq \varphi(K_1)$, then $K_2 = \Psi(\Phi(K_2)) \leq \Psi(\Phi(K_1)) = K_1$.
Now suppose $K_2 \leq K_1$. Define $\alpha\colon K_1/K_2 \to \varphi(K_1)/\varphi(K_2)$ by $\alpha(xK_2) = \varphi(x)\varphi(K_2)$. If $xK_2 = yK_2$, then $y^{-1}x \in K_2$, so $\varphi(y)^{-1}\varphi(x) \in \varphi(K_2)$, and $\alpha(xK_2) = \alpha(yK_2)$. Thus α is well-defined. It is surjective by definition. If $\alpha(xK_2) = \alpha(yK_2)$, then $\varphi(y)^{-1}\varphi(x) = \varphi(k)$ for some $k \in K_2$. Hence $k^{-1}y^{-1}x \in N \leq K_2$, so $y^{-1}x \in K_2$, and $xK_2 = yK_2$. Thus α is bijective, so $|K_1 : K_2| = |\varphi(K_1) : \varphi(K_2)|$.
If $K_2 \trianglelefteq K_1$, then for $x \in K_1$, $k \in K_2$, we have $\varphi(x)\varphi(k)\varphi(x)^{-1} = \varphi(xkx^{-1}) \in \varphi(K_2)$, so $\varphi(K_2) \trianglelefteq \varphi(K_1)$.
Conversely, suppose $\varphi(K_2) \trianglelefteq \varphi(K_1)$. If $x \in K_1$, $k \in K_2$, then $\varphi(xkx^{-1}) \in \varphi(K_2)$, so $xkx^{-1} \in \Psi(\Phi(K_2)) = K_2$. Hence $K_2 \trianglelefteq K_1$.
In this case, α is a homomorphism, since $\alpha((xK_2)(yK_2)) = \alpha(xyK_2) = \varphi(xy)\varphi(K_2) = \varphi(x)\varphi(y)\varphi(K_2) = \alpha(xK_2)\alpha(yK_2)$. Since α is already bijective, it is an isomorphism.
$\therefore$ the correspondence has all the stated properties. □

As a special case of the Correspondence Theorem, if G is a group and $N \trianglelefteq G$, then every subgroup of G/N has the form K/N for some subgroup $K \leq G$ with $N \leq K$. Here we take φ to be the natural map $G \to G/N$.

Theorem (Second Isomorphism Theorem). *Let H and K be normal subgroups of a group G. If H contains K, then $G/H \cong (G/K)/(H/K)$.*

Proof. Let $\eta\colon G \to G/K$ be the natural map. Then $\ker\eta = K$. Since $K \leq H \leq G$, the Correspondence Theorem applies to $H \leq G$.
Since $H \trianglelefteq G$, applying the normality and quotient-isomorphism parts with $K_1 = G$ and $K_2 = H$ gives $\eta(H) \trianglelefteq \eta(G)$ and $G/H \cong \eta(G)/\eta(H)$.
Now $\eta(G) = G/K$ and $\eta(H) = H/K$. Hence $G/H \cong (G/K)/(H/K)$.
$\therefore G/H \cong (G/K)/(H/K)$. □

Exercises

Exercise 1.1. Prove, or complete the sketched proof of, each result in this section.

Solution. The sketched proofs are completed above in:

- Proposition 1.1, Lagrange's Theorem, Proposition 1.2, and Proposition 1.3
- Theorem 1.4, Theorem 1.5, and Theorem 1.6
- Proposition 1.7, Proposition 1.8, Theorem 1.9, Proposition 1.10, and Proposition 1.11
- Fundamental Theorem on Homomorphisms, First Isomorphism Theorem, Correspondence Theorem, and Second Isomorphism Theorem

□

Exercise 1.2. We say that a group G has *exponent* e if e is the smallest positive integer such that $x^e = 1$ for every $x \in G$. Show that if G has exponent 2, then G is abelian. For what integers e is a group having exponent e necessarily abelian?

Solution. First suppose that G has exponent 2. Let $x, y \in G$. Then $x^2 = y^2 = (xy)^2 = 1$, so $xyxy = 1$. Multiplying on the left by x and on the right by y, and using $x^2 = y^2 = 1$, gives $yx = xy$. Since x, y were arbitrary, G is abelian.
If $e = 1$, then $x = 1$ for all $x \in G$, so $G = \{1\}$, which is abelian. We claim that these are the only possibilities.
Let $e \geq 3$. If e is even, take the dihedral group

$$D_e = \langle r, s \mid r^e = s^2 = 1,\ srs^{-1} = r^{-1} \rangle.$$

This group is non-abelian, since $srs^{-1} = r^{-1} \neq r$. Its rotations have order dividing e, and its reflections have order 2. Since e is even, the exponent is $\operatorname{lcm}(e, 2) = e$.
Now suppose that $e \geq 3$ is odd. Let

$$U_e = \left\{ M(a,b,c) = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \;\middle|\; a, b, c \in \mathbb{Z}/e\mathbb{Z} \right\},$$

under matrix multiplication, with entries computed in $\mathbb{Z}/e\mathbb{Z}$. Direct multiplication gives

$$M(a,b,c)M(a',b',c') = M(a+a', b+b', c+c'+ab').$$

Hence U_e is closed under multiplication. The identity is $M(0,0,0)$, and

$$M(a,b,c)^{-1} = M(-a, -b, ab - c).$$

Associativity follows from associativity of matrix multiplication, so U_e is a group. It is non-abelian, since

$$M(1,0,0)M(0,1,0) = M(1,1,1), \qquad M(0,1,0)M(1,0,0) = M(1,1,0).$$

For $n \in \mathbb{N}$, induction using the multiplication formula gives

$$M(a,b,c)^n = M\left(na, nb, nc + \binom{n}{2}ab\right).$$

Taking $n = e$, we get

$$M(a,b,c)^e = M\left(0, 0, \binom{e}{2}ab\right).$$

Since e is odd, $\binom{e}{2} = e(e-1)/2$ is divisible by e, so $M(a,b,c)^e = M(0,0,0)$ for every $M(a,b,c) \in U_e$. Thus every element has order dividing e. However $M(1,0,0)^n = M(n,0,0)$, so $M(1,0,0)$ has order e. Hence U_e is a non-abelian group of exponent e.
$\therefore$ a group of exponent e is necessarily abelian iff $e = 1$ or $e = 2$. □

Exercise 1.3. Let G be a finite group, and suppose that the map $\varphi : G \to G$ defined by $\varphi(x) = x^3$ for $x \in G$ is a homomorphism. Show that if 3 does not divide $|G|$, then G must be abelian. (See [2] for a generalization.)

Solution. Let $x, y \in G$. Since φ is a homomorphism, we have

$$x^3y^3 = \varphi(x)\varphi(y) = \varphi(xy) = (xy)^3 = xyxyxy.$$

Cancelling x on the left and y on the right gives

$$(yx)^2 = x^2y^2.$$

Swapping x and y, we also have

$$(xy)^2 = y^2x^2.$$

Using this in the original equation,

$$x^3y^3 = (xy)^3 = (xy)(xy)^2 = xy(y^2x^2) = xy^3x^2.$$

Cancelling x on the left gives
$$y^3x^2 = x^2y^3.$$
Hence every cube commutes with every square.
Now let $z \in \ker\varphi$. Then $z^3 = 1$, so $o(z) \mid 3$. Then we also have $o(z) \mid |G|$. Since $3 \nmid |G|$, we must have $o(z) = 1$, and hence $z = 1$. Thus $\ker\varphi = \{1\}$, so φ is injective. Since G is finite, φ is also surjective. Therefore every element of G is a cube.
Let $g \in G$. Since every element of G is a cube, write $g = y^3$ for some $y \in G$. Then for every $x \in G$, the relation above gives
$$gx^2 = y^3x^2 = x^2y^3 = x^2g.$$
Thus every square is central. In particular, x^2 and y^2 commute for all $x, y \in G$. Since $(xy)^2 = y^2x^2$, it follows that
$$(xy)^2 = x^2y^2.$$
Hence $xyxy = xxyy$. Cancelling x on the left and y on the right gives $yx = xy$. Since x, y were arbitrary, G is abelian. □

Exercise 1.4. Let g be an element of a group G, and suppose that $|G| = mn$ where m and n are coprime. Show that there are unique elements x and y of G such that $xy = g = yx$ and $x^m = 1 = y^n$. (In the case where m is a power of some prime p, we call x the p-part of g and y the p'-part of g; more generally, if π is a set of primes which includes all prime divisors of m and no prime divisors of n, then x and y are called the π-part and π'-part, respectively, of g.)

Solution. Since $\gcd(m, n) = 1$, choose $a, b \in \mathbb{Z}$ such that $am + bn = 1$. Since $\langle g\rangle \leq G$, Lagrange's Theorem gives $o(g) = |\langle g\rangle| \mid |G| = mn$, so $g^{mn} = 1$. Define
$$x = g^{bn}, \qquad y = g^{am}.$$
Then x and y commute, since both are powers of g. Moreover,
$$xy = g^{bn}g^{am} = g^{am+bn} = g,$$
and similarly $yx = g$. Also,
$$x^m = g^{bmn} = (g^{mn})^b = 1, \qquad y^n = g^{amn} = (g^{mn})^a = 1.$$
Thus such elements $x, y \in G$ exist.
We now show that they are unique. Suppose that $x', y' \in G$ also satisfy
$$x'y' = g = y'x', \qquad (x')^m = 1, \qquad (y')^n = 1.$$
Since x' and y' commute, we may take powers of $x'y'$ termwise. Thus
$$g^{bn} = (x'y')^{bn} = (x')^{bn}(y')^{bn}.$$
But $(y')^{bn} = ((y')^n)^b = 1$, so
$$g^{bn} = (x')^{bn}.$$
Since $bn = 1 - am$, we get
$$(x')^{bn} = (x')^{1-am} = x'((x')^m)^{-a} = x'.$$
Therefore $x' = g^{bn} = x$. Similarly,
$$g^{am} = (x'y')^{am} = (x')^{am}(y')^{am} = (y')^{am},$$
and since $am = 1 - bn$,
$$(y')^{am} = (y')^{1-bn} = y'((y')^n)^{-b} = y'.$$
Therefore $y' = g^{am} = y$. Hence the required elements x and y exist and are unique.

□

Exercise 1.5. Let r, s, and t be positive integers greater than 1. Show that there is a finite group G having elements x and y such that x has order r, y has order s, and xy has order t.

Solution. First approach: a linear construction. Let

$$N = 2rst.$$

Choose a prime $p \nmid N$. By taking a suitable power $q = p^d$, we may arrange that

$$q \equiv 1 \pmod{N},$$

since p is invertible modulo N. Thus $2r, 2s, 2t$ all divide $q-1$. Since $\mathbb{F}_q^\times$ is cyclic, we may choose elements $u, v, w \in \mathbb{F}_q^\times$ of orders $2r, 2s, 2t$, respectively. Define

$$A = \begin{pmatrix} u & 1 \\ 0 & u^{-1} \end{pmatrix}, \qquad B = \begin{pmatrix} v & 0 \\ \lambda & v^{-1} \end{pmatrix},$$

where

$$\lambda = w + w^{-1} - uv - u^{-1}v^{-1}.$$

Then $A, B \in \mathrm{SL}(2, \mathbb{F}_q)$. The characteristic polynomial of A is

$$X^2 - (u + u^{-1})X + 1,$$

with distinct roots u, u^{-1}, since u has order $2r > 2$. Hence A is conjugate over $\mathbb{F}_q$ to $\mathrm{diag}(u, u^{-1})$, and so A has order $2r$. Similarly, B has characteristic polynomial

$$X^2 - (v + v^{-1})X + 1$$

with distinct roots v, v^{-1}, and therefore B has order $2s$. Finally,

$$AB = \begin{pmatrix} uv + \lambda & v^{-1} \\ u^{-1}\lambda & u^{-1}v^{-1} \end{pmatrix},$$

so

$$\mathrm{tr}(AB) = uv + \lambda + u^{-1}v^{-1} = w + w^{-1}.$$

Since $\det(AB) = 1$, the characteristic polynomial of AB is

$$X^2 - (w + w^{-1})X + 1,$$

with distinct roots w, w^{-1}. Thus AB has order $2t$.
Now pass to the finite quotient

$$G = \mathrm{SL}(2, \mathbb{F}_q)/\{\pm I\}.$$

The element $-I$ is the unique element of order 2 in $\mathrm{SL}(2, \mathbb{F}_q)$: indeed, if $C^2 = I$, then C is diagonalizable with eigenvalues in $\{\pm 1\}$, and the condition $\det C = 1$ forces either $C = I$ or $C = -I$. Hence, if an element $g \in \mathrm{SL}(2, \mathbb{F}_q)$ has order $2n$, its image modulo $\{\pm I\}$ has order n. For g^n is the unique element of order 2, hence $g^n = -I$, so the image has order dividing n; and if a smaller positive power became trivial in the quotient, then $g^m = \pm I$ for some $m < n$. The case $g^m = I$ contradicts $o(g) = 2n$, while $g^m = -I = g^n$ gives $g^{n-m} = I$, again impossible. Therefore the images x, y of A, B in G have orders r, s, respectively, and xy, the image of AB, has order t.

Second approach: finite quotients of triangle groups. We use one standard fact about a particular group given by generators and relations. For $a, b, c > 1$, the triangle group $\Delta(a, b, c)$ is the abstract group

$$\Delta(a, b, c) = \langle U, V \mid U^a = 1,\ V^b = 1,\ (UV)^c = 1 \rangle$$

generated by two formal elements U and V, subject to the relations $U^a = 1$, $V^b = 1$, and $(UV)^c = 1$. The word “triangle” is only the standard name for this presentation; no geometry is needed here. Equivalently, $\Delta(a, b, c)$ is the universal group with these three relations: whenever a group H has elements $\alpha, \beta \in H$ satisfying

$$\alpha^a = 1, \qquad \beta^b = 1, \qquad (\alpha\beta)^c = 1,$$

there is a homomorphism $\Delta(a, b, c) \to H$ sending U to α and V to β.
The standard fact we use is that U, V, UV have exact orders a, b, c, respectively, in $\Delta(a, b, c)$, and that $\Delta(a, b, c)$ is residually finite. Here residually finite means that if $g \neq 1$, then there is a homomorphism

from $\Delta(a, b, c)$ to some finite group in which the image of g is not 1.
Now apply this fact with $a = r$, $b = s$, and $c = t$. Put

$$\Gamma = \langle X, Y \mid X^r = 1,\ Y^s = 1,\ (XY)^t = 1\rangle.$$

In Γ, the elements X, Y, XY have exact orders r, s, t. Hence every element of the finite set

$$\mathcal{S} = \{X^i \mid 1 \le i < r\} \cup \{Y^j \mid 1 \le j < s\} \cup \{(XY)^k \mid 1 \le k < t\}$$

is non-identity. By residual finiteness, for each $g \in \mathcal{S}$, there is a finite group Q_g and a homomorphism

$$\varphi_g \colon \Gamma \to Q_g$$

such that $\varphi_g(g) \neq 1$. Now define

$$Q = \prod_{g \in \mathcal{S}} Q_g, \qquad \Phi\colon \Gamma \to Q, \qquad \Phi(h) = \big(\varphi_g(h)\big)_{g \in \mathcal{S}}.$$

Let $G = \Phi(\Gamma) \le Q$, and set

$$x = \Phi(X), \qquad y = \Phi(Y).$$

Since Q is finite, G is finite. Also, since $X^r = 1$, $Y^s = 1$, and $(XY)^t = 1$ in Γ, we have

$$x^r = 1, \qquad y^s = 1, \qquad (xy)^t = 1.$$

Thus $o(x) \mid r$, $o(y) \mid s$, and $o(xy) \mid t$.
It remains to show that no smaller positive powers become trivial in this finite quotient. Suppose $1 \le i < r$. Since $X^i \in \mathcal{S}$, the X^i-coordinate of $\Phi(X^i)$ is $\varphi_{X^i}(X^i)$, which is not 1. Hence $\Phi(X^i) \neq 1$, so $x^i \neq 1$. Therefore $o(x) = r$.
Similarly, if $1 \le j < s$, then the Y^j-coordinate of $\Phi(Y^j)$ is $\varphi_{Y^j}(Y^j) \neq 1$, so $y^j \neq 1$. Hence $o(y) = s$. Finally, if $1 \le k < t$, then the $(XY)^k$-coordinate of $\Phi((XY)^k)$ is $\varphi_{(XY)^k}((XY)^k) \neq 1$, so $(xy)^k \neq 1$. Hence $o(xy) = t$.

Third approach: Miller's permutation construction. We now give Miller's construction in modern notation. Miller's "substitutions" are permutations, and his "circular substitutions" are cycles. The proof is based on the following explicit overlap calculation. Let two cycles share a consecutive block

$$a_1, a_2, \ldots, a_c$$

and have no other letters in common. If $c = 2h + 1$ is odd, arrange the two cycles as

$$S_1 = (a_1\ \cdots\ a_{2j}\ a_{2j+1}\ \cdots\ a_{2h+1}\ b_1\ \cdots\ b_p),$$

$$S_2 = (a_{2j+1}\ \cdots\ a_1\ a_{2j+2}\ \cdots\ a_{2h+1}\ d_1\ \cdots\ d_q).$$

Tracing the product gives

$$S_1S_2 = (a_{2j+1}\ a_{2j+3}\ \cdots\ a_{2h+1}\ b_1\ \cdots\ b_p\ a_{2j+2}\ a_{2j+4}\ \cdots\ a_{2h}\ d_1\ \cdots\ d_q),$$

so S_1S_2 is one cycle and the omitted common letters are exactly

$$a_1, \ldots, a_{2j}.$$

Thus an odd overlap lets us omit any prescribed even number of common letters. Similarly, if $c = 2h$ is even and the same calculation is made with the first omitted block $a_1, \ldots, a_{2j-1}$, then the product is one cycle omitting exactly $2j - 1$ common letters. Thus an even overlap lets us omit any prescribed odd number of common letters.
If the required parity is opposite to the parity supplied by the two-cycle calculation, Miller uses one cycle together with a permutation of the form

(one transposition)(one disjoint cycle).

The same trace shows that this changes the parity of the number of omitted common letters. This is harmless whenever the corresponding order is even, since the least common multiple of 2 and an even cycle length is still the required even order. When both relevant orders are odd, no transposition is inserted; instead, one uses two adjacent odd cycles and performs the same overlap twice. The first overlap changes the parity, and the second restores the required odd order. This is the only extra bookkeeping in Miller's all-odd case.

We now use this calculation to build permutations L, M, N of prescribed orders $\ell, m, n > 1$ with $LM = N$. Since

$$LM = N \quad \Longleftrightarrow \quad MN^{-1} = L^{-1} \quad \Longleftrightarrow \quad N^{-1}L = M^{-1},$$

we may construct the triple after arranging the desired orders as $\ell \le m \le n$, and then translate back to the original order. Write

$$n = am + b, \qquad 0 \le b < m.$$

Take M to be a product of a disjoint m-cycles,

$$M = (a_1 \, \cdots \, a_m)(a_{m+1} \, \cdots \, a_{2m}) \cdots (a_{(a-1)m+1} \, \cdots \, a_{am}).$$

The permutation L is chosen so that its cycles join these m-cycles end-to-end. If $a = 1$, this is exactly the overlap calculation above, since $m \le n < m + \ell$. For instance, when ℓ is even and $a > 1$, the joining part may be written schematically as

$$\begin{aligned} L = {} & (a_1 \, a_{am+1})(a_2 \, a_{am+2}) \cdots (a_b \, a_{am+b}) \\ & \cdot (a_m \, a_{m+1})(a_{2m} \, a_{2m+1}) \cdots \\ & \cdot (a_{(a-1)m} \, a_{(a-1)m+1} \, \cdots \, a_{(a-1)m+\ell-1}). \end{aligned}$$

where the first row is omitted if $b = 0$. Here the displayed transpositions perform the ordinary joins, and the final ℓ-cycle makes $o(L) = \ell$. A direct trace explains the construction: starting at a_1, the product LM first follows the first m-cycle of M; when it reaches the end of that cycle, the next factor of L sends it to the beginning of the next m-cycle; the same process repeats until the trace reaches the final block. At the final block the overlap calculation above omits exactly $m - b$ common letters, so the resulting product has precisely

$$am + b = n$$

letters. Hence LM is an n-cycle.

If ℓ is odd, the same threading is done with ℓ-cycles rather than transpositions. The only issue is parity of the number of omitted common letters in the final overlap. When m is even, the transposition-plus-cycle variant above supplies the missing parity while keeping $o(M) = m$. When both ℓ and m are odd, two neighboring odd overlaps are used: the first overlap changes the parity of the omitted block and the second overlap joins the next m-cycle back into the same long trace. Since every component of L is then an ℓ-cycle and every component of M is an m-cycle, the orders remain

$$o(L) = \ell, \qquad o(M) = m.$$

The same trace as before shows that the product is still a single n-cycle. Thus in all cases we have constructed permutations L, M, N such that

$$o(L) = \ell, \qquad o(M) = m, \qquad o(N) = n, \qquad LM = N.$$

We now return to r, s, t. If t is the largest of the three orders, take $\ell = r$, $m = s$, and $n = t$, and set $x = L$, $y = M$. If r is the largest, take $\ell = s$, $m = t$, and $n = r$. Then $LM = N$, so $N^{-1}L = M^{-1}$; setting $x = N^{-1}$ and $y = L$ gives $o(x) = r$, $o(y) = s$, and $o(xy) = t$. If s is the largest, take $\ell = t$, $m = r$, and $n = s$. Then $LM = N$, so $MN^{-1} = L^{-1}$; setting $x = M$ and $y = N^{-1}$ gives $o(x) = r$, $o(y) = s$, and $o(xy) = t$. In all cases the permutations act on a finite set Ω. Let

$$G = \langle x, y \rangle \le \mathrm{Sym}(\Omega).$$

Then G is finite, and by construction $o(x) = r$, $o(y) = s$, and $o(xy) = t$. $\therefore$ the required finite group exists. $\square$

Exercise 1.6. Let X and Y be subsets of a group G. Are $\langle X \rangle \cap \langle Y \rangle$ and $\langle X \cap Y \rangle$ necessarily equal? Are $\langle \langle X \rangle \cup \langle Y \rangle \rangle$ and $\langle X \cup Y \rangle$ necessarily equal?

Solution. The first pair need not be equal. Let $G = C_3 = \langle g \rangle$, $X = \{1, g\}$, and $Y = \{1, g^2\}$. Then $\langle X \rangle = G$, since $g \in X$, and $\langle Y \rangle = G$, since g^2 also generates C_3. Hence

$$\langle X \rangle \cap \langle Y \rangle = G.$$

However, $X \cap Y = \{1\}$, so

$$\langle X \cap Y \rangle = \{1\}.$$

Thus $\langle X \rangle \cap \langle Y \rangle$ and $\langle X \cap Y \rangle$ are not necessarily equal.
The second pair is always equal. Since $X \subseteq \langle X \rangle$ and $Y \subseteq \langle Y \rangle$, we have

$$X \cup Y \subseteq \langle X \rangle \cup \langle Y \rangle.$$

Therefore

$$\langle X \cup Y \rangle \leq \langle \langle X \rangle \cup \langle Y \rangle \rangle.$$

Conversely, since $X \subseteq X \cup Y$ and $Y \subseteq X \cup Y$, we have

$$\langle X \rangle \leq \langle X \cup Y \rangle, \qquad \langle Y \rangle \leq \langle X \cup Y \rangle.$$

Hence

$$\langle X \rangle \cup \langle Y \rangle \subseteq \langle X \cup Y \rangle.$$

Since $\langle X \cup Y \rangle$ is a subgroup containing $\langle X \rangle \cup \langle Y \rangle$, it contains the subgroup generated by that union, so

$$\langle \langle X \rangle \cup \langle Y \rangle \rangle \leq \langle X \cup Y \rangle.$$

$\therefore \langle \langle X \rangle \cup \langle Y \rangle \rangle = \langle X \cup Y \rangle$. □

Exercise 1.7. Let G be a finite group and let $H \leq G$. Show that there is a subset T of G which is simultaneously a left transversal for H and a right transversal for H.

Solution. We use the following elementary matching fact. Suppose $A_1, \ldots, A_n$ are finite sets of possible choices. If, for every $I \subseteq \{1, \ldots, n\}$,

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|,$$

then we can choose distinct representatives $a_i \in A_i$.
Let

$$L_1, \ldots, L_n$$

be the distinct left cosets of H, and let

$$R_1, \ldots, R_n$$

be the distinct right cosets of H. There are the same number n of each, since both left and right cosets partition G into subsets of size $|H|$. For each i, define

$$A_i = \{ j \in \{1, \ldots, n\} \mid L_i \cap R_j \neq \emptyset \}.$$

Thus A_i is the set of right cosets which the left coset L_i meets. We check the matching condition. Choose any m left cosets, say those indexed by I, where $|I| = m$. Suppose these m left cosets meet q right cosets. Since distinct left cosets are disjoint,

$$\left| \bigcup_{i \in I} L_i \right| = m|H|.$$

This union is contained in the union of the q right cosets it meets, whose total size is $q|H|$. Hence

$$m|H| \leq q|H|,$$

so $m \leq q$. Therefore any m left cosets meet at least m right cosets. In terms of the sets A_i, this says exactly that

$$\left| \bigcup_{i \in I} A_i \right| \geq |I|.$$

By the matching fact, choose distinct $j_1, \ldots, j_n$ such that $j_i \in A_i$ for every i. Since $j_i \in A_i$,

$$L_i \cap R_{j_i} \neq \emptyset,$$

so choose

$$t_i \in L_i \cap R_{j_i}.$$

Set

$$T = \{t_1, \ldots, t_n\}.$$

Because $t_i \in L_i$, the set T contains one element from each left coset. Since the L_i are disjoint, it contains exactly one element from each left coset, so T is a left transversal for H.
Because the indices $j_1, \ldots, j_n$ are distinct, the right cosets $R_{j_1}, \ldots, R_{j_n}$ are all distinct. There are n of them, and there are only n right cosets in total, so they are all the right cosets. Since $t_i \in R_{j_i}$, the set T also contains exactly one element from each right coset. Hence T is a right transversal for H.
$\therefore$ there exists a subset $T \subseteq G$ which is simultaneously a left transversal and a right transversal for H. $\square$

Exercise 1.8. Suppose that $\mathcal{C}$ is a family of subsets of a group G which forms a partition of G, and suppose further that $gC \in \mathcal{C}$ for any $g \in G$ and $C \in \mathcal{C}$. (Recall that a *partition* of a set S is a collection $\mathcal{S}$ of subsets of S with the property that every element of S lies in exactly one member of $\mathcal{S}$.) Show that $\mathcal{C}$ is the set of cosets of some subgroup of G.

Solution. Since $\mathcal{C}$ is a partition of G, there is a unique member $H \in \mathcal{C}$ such that $1 \in H$. We first show that $H \leq G$.
Let $h \in H$. By the hypothesis, $hH \in \mathcal{C}$. Also, since $1 \in H$, we have $h = h1 \in hH$. Thus both H and hH are members of $\mathcal{C}$ containing h. Since $\mathcal{C}$ is a partition, we must have

$$hH = H.$$

Hence for any $h, k \in H$, we have $hk \in hH = H$, so H is closed under multiplication. Also, since $1 \in H = hH$, there exists $u \in H$ such that $hu = 1$. Thus $u = h^{-1}$, so $h^{-1} \in H$. Therefore $H \leq G$.
Now let $C \in \mathcal{C}$. Choose $g \in C$. By the hypothesis, $gH \in \mathcal{C}$, and since $1 \in H$, we have $g \in gH$. Thus both C and gH are members of $\mathcal{C}$ containing g. Since $\mathcal{C}$ is a partition, $C = gH$. Therefore every member of $\mathcal{C}$ is a left coset of H. Conversely, for every $g \in G$, the set gH belongs to $\mathcal{C}$ by the hypothesis.
$\therefore \mathcal{C} = \{gH \mid g \in G\}$, the set of left cosets of the subgroup H. $\square$

Exercise 1.9. Suppose that $\mathcal{C}$ is a family of subsets of a group G which forms a partition of G, and suppose further that $XY \in \mathcal{C}$ for any $X, Y \in \mathcal{C}$. Show that exactly one of the sets belonging to $\mathcal{C}$ is a subgroup of G, that this subgroup is normal in G, and that $\mathcal{C}$ consists of its cosets.

Solution. Since $\mathcal{C}$ is a partition of G, there is a unique member $H \in \mathcal{C}$ such that $1 \in H$. We first show that $H \leq G$.
By the hypothesis, $HH \in \mathcal{C}$. Since $1 \in H$, we have $1 = 1 \cdot 1 \in HH$. Thus H and HH are both members of $\mathcal{C}$ containing 1, so $HH = H$. Hence H is closed under multiplication.
Now let $h \in H$. Let $K \in \mathcal{C}$ be the unique member containing h^{-1}. By the hypothesis, $HK \in \mathcal{C}$. Since $h \in H$ and $h^{-1} \in K$, we have $1 = hh^{-1} \in HK$. Thus HK and H are both members of $\mathcal{C}$ containing 1, so $HK = H$. Since $1 \in H$ and $h^{-1} \in K$, it follows that $h^{-1} = 1h^{-1} \in HK = H$. Therefore $H \leq G$.
We now show that every member of $\mathcal{C}$ is a left coset of H. Let $X \in \mathcal{C}$, and choose $x \in X$. By the hypothesis, $XH \in \mathcal{C}$. Since $1 \in H$, we have $x = x1 \in XH$. Thus XH and X are both members of $\mathcal{C}$ containing x, so $XH = X$. Hence $xH \subseteq XH = X$.
For the reverse inclusion, let $X' \in \mathcal{C}$ be the unique member containing x^{-1}. By the hypothesis, $X'X \in \mathcal{C}$. Since $x^{-1}x = 1 \in X'X$, we have $X'X = H$. If $y \in X$, then $x^{-1}y \in X'X = H$, and hence

$$y = x(x^{-1}y) \in xH.$$

Thus $X \subseteq xH$. Therefore $X = xH$, and so every member of $\mathcal{C}$ is a left coset of H. Conversely, if $g \in G$, and $X \in \mathcal{C}$ contains g, then $X = gH$. Hence

$$\mathcal{C} = \{gH \mid g \in G\}.$$

The subgroup H is the only member of $\mathcal{C}$ which is a subgroup, since it is the only member containing 1. It remains to show that H is normal. Let $g \in G$. Since $gH, g^{-1}H \in \mathcal{C}$, the product $(gH)(g^{-1}H)$ is a member of $\mathcal{C}$. It contains 1, since $g \in gH$, $g^{-1} \in g^{-1}H$, and $gg^{-1} = 1$. Thus

$$(gH)(g^{-1}H) = H.$$

Now let $h \in H$. Since $gh \in gH$ and $g^{-1} \in g^{-1}H$, we have

$$ghg^{-1} \in (gH)(g^{-1}H) = H.$$

Thus $gHg^{-1} \subseteq H$. Replacing g by g^{-1}, we also have $g^{-1}Hg \subseteq H$, and conjugating this inclusion by g gives $H \subseteq gHg^{-1}$. Hence $gHg^{-1} = H$ for all $g \in G$. Therefore $H \trianglelefteq G$.
$\therefore$ exactly one member of $\mathcal{C}$ is a subgroup, this subgroup is normal in G, and $\mathcal{C}$ consists of its cosets. $\square$

Exercise 1.10. Prove the following generalization of Proposition 1.8: If G is a finite group and $H \leq G$ is such that $|G : H|$ is equal to the smallest prime divisor of $|G|$, then $H \trianglelefteq G$.

Solution. Let $p = |G : H|$. By hypothesis, p is the smallest prime divisor of $|G|$. Let

$$\Omega = \{xH \mid x \in G\}$$

be the set of left cosets of H. Then $|\Omega| = p$. Define

$$\varphi \colon G \to S_\Omega$$

by

$$\varphi(g)(xH) = gxH$$

for all $g, x \in G$. This is well-defined: if $xH = yH$, then $gxH = gyH$. For each $g \in G$, the map $\varphi(g)$ is a permutation of Ω, with inverse $\varphi(g^{-1})$. Also,

$$\varphi(g_1g_2)(xH) = g_1g_2xH = \varphi(g_1)(g_2xH) = \big(\varphi(g_1)\varphi(g_2)\big)(xH),$$

so φ is a homomorphism.
Let $K = \ker\varphi$. Then $K \trianglelefteq G$. We claim that $K \leq H$. If $k \in K$, then $\varphi(k)$ fixes every left coset, so in particular

$$H = \varphi(k)(H) = kH.$$

Thus $k \in H$, and hence $K \leq H$.
By the Fundamental Theorem on Homomorphisms, $G/K \cong \operatorname{im}\varphi \leq S_\Omega$. Hence

$$|G : K| = |\operatorname{im}\varphi| \mid |S_\Omega| = p!.$$

Also, by Lagrange's Theorem, $|G : K| \mid |G|$. Therefore every prime divisor of $|G : K|$ is a prime divisor of $|G|$, and hence is at least p. On the other hand, since $|G : K| \mid p!$, every prime divisor of $|G : K|$ is at most p. Thus the only possible prime divisor of $|G : K|$ is p.
Since $K \leq H \leq G$, we have

$$|G : K| = |G : H|\,|H : K| = p|H : K|.$$

Thus $p \mid |G : K|$. Since the only possible prime divisor of $|G : K|$ is p, we may write $|G : K| = p^a$ for some $a \geq 1$. But $|G : K| \mid p!$, and $p^2 \nmid p!$, so $a = 1$. Hence

$$|G : K| = p.$$

It follows that $p|H : K| = p$, so $|H : K| = 1$. Therefore $H = K$. Since $K \trianglelefteq G$, we conclude that $H \trianglelefteq G$. $\square$

Further Exercises

If $K \trianglelefteq H \leq G$, then H/K is called a *section* of G. We say that two sections H_1/K_1 and H_2/K_2 of G are *incident* if every coset of K_1 in H_1 has non-empty intersection with exactly one coset of K_2 in H_2, and vice versa. (In other words, two sections are incident if the relation of non-empty intersection gives a bijective correspondence between their elements.)

Exercise 1.11. Show that incident sections are isomorphic.

Solution. Let H_1/K_1 and H_2/K_2 be incident sections of G. Define

$$\varphi\colon H_1/K_1 \to H_2/K_2$$

by letting $\varphi(xK_1)$ be the unique coset yK_2 such that

$$xK_1 \cap yK_2 \neq \varnothing.$$

This is well-defined by the definition of incidence. We first show that φ is bijective. If $\varphi(xK_1) = \varphi(x'K_1) = yK_2$, then yK_2 has non-empty intersection with both xK_1 and $x'K_1$. Since incidence also says that each coset of K_2 in H_2 has non-empty intersection with exactly one coset of K_1 in H_1, we have $xK_1 = x'K_1$. Hence φ is injective. If $yK_2 \in H_2/K_2$, then, again by incidence, yK_2 has non-empty intersection with some coset $xK_1 \in H_1/K_1$, and then $\varphi(xK_1) = yK_2$. Hence φ is surjective.

It remains to show that φ is a homomorphism. Suppose

$$\varphi(xK_1) = yK_2 \quad \text{and} \quad \varphi(x'K_1) = y'K_2.$$

Choose

$$a \in xK_1 \cap yK_2 \quad \text{and} \quad b \in x'K_1 \cap y'K_2.$$

Then $a = xk$ and $b = x'k'$ for some $k, k' \in K_1$. Hence

$$ab = xkx'k' = xx'(x'^{-1}kx')k' \in xx'K_1,$$

because $K_1 \trianglelefteq H_1$ and $x' \in H_1$. Similarly, writing $a = y\ell$ and $b = y'\ell'$ with $\ell, \ell' \in K_2$, we get

$$ab = y\ell y'\ell' = yy'(y'^{-1}\ell y')\ell' \in yy'K_2,$$

because $K_2 \trianglelefteq H_2$ and $y' \in H_2$. Therefore

$$xx'K_1 \cap yy'K_2 \neq \varnothing.$$

By the uniqueness in the definition of incidence, this implies

$$\varphi(xx'K_1) = yy'K_2.$$

On the other hand,

$$\varphi(xK_1)\varphi(x'K_1) = (yK_2)(y'K_2) = yy'K_2.$$

Thus

$$\varphi((xK_1)(x'K_1)) = \varphi(xK_1)\varphi(x'K_1).$$

Therefore φ is a bijective homomorphism, so $H_1/K_1 \cong H_2/K_2$. □

Exercise 1.12. (cont.) Suppose that $N \trianglelefteq G$ and $H \leq G$. Show that HN/N and $H/(H \cap N)$ are incident. (Exercises 1.11 and 1.12 provide an alternate proof of the First Isomorphism Theorem.)

Solution. Let $K = H \cap N$. By Proposition 1.7, $HN \leq G$ and $K \trianglelefteq H$. Since $N \trianglelefteq G$ and $HN \leq G$, we also have $N \trianglelefteq HN$, so HN/N and H/K are sections of G. We show that they are incident.
First let $xN \in HN/N$. Since $x \in HN$, we may write $x = hn$ for some $h \in H$ and $n \in N$. Hence

$$xN = hnN = hN.$$

Since $1 \in K$, we have $h \in hN \cap hK$, so $hN \cap hK \neq \varnothing$. Thus every coset of N in HN has non-empty intersection with at least one coset of K in H. We now prove uniqueness. Suppose that

$$hN \cap h'K \neq \varnothing$$

for some $h' \in H$. Choose $z \in hN \cap h'K$. Then $z = hn_1 = h'k$ for some $n_1 \in N$ and $k \in K$, and hence

$$h^{-1}h' = n_1k^{-1}.$$

Since $n_1 \in N$ and $k \in K \leq N$, we have $n_1k^{-1} \in N$. Since also $h, h' \in H$, we have $h^{-1}h' \in H$. Hence $h^{-1}h' \in H \cap N = K$, so $hK = h'K$. Therefore each coset of N in HN intersects exactly one coset of K in H.

Conversely, let $hK \in H/K$. Since $h \in hK \cap hN$, the coset hK has non-empty intersection with at least one coset of N in HN. Suppose that $hK \cap xN \neq \varnothing$ for some $xN \in HN/N$. As above, we may write $xN = h'N$ for some $h' \in H$. Choose $z \in hK \cap h'N$. Then $z = hk = h'n_2$ for some $k \in K$ and $n_2 \in N$, and hence

$$h^{-1}h' = kn_2^{-1}.$$

Since $k \in K \leq N$ and $n_2 \in N$, we get $h^{-1}h' \in N$. Therefore $h'N = hN$. Thus each coset of K in H intersects exactly one coset of N in HN.
$\therefore HN/N$ and $H/(H \cap N)$ are incident. □

If L/M is a section of G and $H \leq G$, then the *projection* of H on L/M is the subset of L/M consisting of those cosets of M in L which contain elements of H.

Exercise 1.13. (cont.) Show that the projection of H on L/M is the subgroup $(L \cap H)M/M$ of L/M.

Solution. Let

$$P = \{lM \in L/M \mid lM \cap H \neq \varnothing\}$$

be the projection of H on L/M. Since L/M is a section of G, we have $M \trianglelefteq L$. Let $\eta\colon L \to L/M$ be the natural map. By Proposition 1.11,

$$\eta(L \cap H) = (L \cap H)M/M$$

is a subgroup of L/M. It remains to show that $P = \eta(L \cap H)$.
First let $lM \in P$. Then $lM \cap H \neq \varnothing$, so we may choose $h \in lM \cap H$. Since $h \in lM \subseteq L$, and also $h \in H$, we have $h \in L \cap H$. Moreover $h \in lM$, so $hM = lM$. Hence

$$lM = hM \in \eta(L \cap H).$$

Thus $P \subseteq \eta(L \cap H)$.
Conversely, let $xM \in \eta(L \cap H)$. Then $xM = hM$ for some $h \in L \cap H$. Since $h \in H$ and $h \in hM = xM$, we have $xM \cap H \neq \varnothing$. Hence $xM \in P$. Thus $\eta(L \cap H) \subseteq P$.
$\therefore P = \eta(L \cap H) = (L \cap H)M/M$, so the projection of H on L/M is the subgroup $(L \cap H)M/M$ of L/M. □

Let H_1/K_1 and H_2/K_2 be sections of a group G.

Exercise 1.14. (cont.) Show that the projection of K_2 on H_1/K_1 is a normal subgroup of the projection of H_2 on H_1/K_1. The quotient group obtained thereby is called the projection of H_2/K_2 on H_1/K_1.

Solution. By Exercise 1.13, the projection of K_2 on H_1/K_1 is

$$(H_1 \cap K_2)K_1/K_1,$$

and the projection of H_2 on H_1/K_1 is

$$(H_1 \cap H_2)K_1/K_1.$$

We first show that $H_1 \cap K_2 \trianglelefteq H_1 \cap H_2$. Let $x \in H_1 \cap K_2$ and $y \in H_1 \cap H_2$. Since $x, y \in H_1$, we have $yxy^{-1} \in H_1$. Since $x \in K_2$, $y \in H_2$, and $K_2 \trianglelefteq H_2$, we also have $yxy^{-1} \in K_2$. Hence

$$yxy^{-1} \in H_1 \cap K_2.$$

Therefore $H_1 \cap K_2 \trianglelefteq H_1 \cap H_2$.
Now take

$$xK_1 \in (H_1 \cap K_2)K_1/K_1 \quad \text{and} \quad yK_1 \in (H_1 \cap H_2)K_1/K_1,$$

where $x \in H_1 \cap K_2$ and $y \in H_1 \cap H_2$. Then

$$(yK_1)(xK_1)(yK_1)^{-1} = (yxy^{-1})K_1.$$

By the first paragraph, $yxy^{-1} \in H_1 \cap K_2$, so

$$(yxy^{-1})K_1 \in (H_1 \cap K_2)K_1/K_1.$$

Thus

$$(H_1 \cap K_2)K_1/K_1 \trianglelefteq (H_1 \cap H_2)K_1/K_1.$$

$\therefore$ the projection of K_2 on H_1/K_1 is a normal subgroup of the projection of H_2 on H_1/K_1. □

Exercise 1.15. (cont.) Show that the projection of H_1/K_1 on H_2/K_2 and the projection of H_2/K_2 on H_1/K_1 are incident. Deduce that, if $H_1, H_2 \le G$, $K_1 \trianglelefteq H_1$, and $K_2 \trianglelefteq H_2$, then

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

Solution. Let

$$A = H_1 \cap H_2, \quad B_1 = (H_1 \cap K_2)K_1, \quad B_2 = (K_1 \cap H_2)K_2.$$

By Exercise 1.13 and Exercise 1.14, the projection of H_2/K_2 on H_1/K_1 is

$$AK_1/B_1,$$

and the projection of H_1/K_1 on H_2/K_2 is

$$AK_2/B_2.$$

Every coset of B_1 in AK_1 has the form aB_1 for some $a \in A$, since $K_1 \le B_1$. Similarly, every coset of B_2 in AK_2 has the form aB_2 for some $a \in A$, since $K_2 \le B_2$. For each $a \in A$, the two cosets aB_1 and aB_2 have non-empty intersection, since both contain a.
We now prove uniqueness. Suppose that

$$aB_1 \cap bB_2 \neq \varnothing$$

for some $a, b \in A$. Choose $z \in aB_1 \cap bB_2$. Then $z = as = bt$ for some $s \in B_1$ and $t \in B_2$. Write

$$s = ck, \quad c \in H_1 \cap K_2, \quad k \in K_1,$$

and

$$t = d\ell, \quad d \in K_1 \cap H_2, \quad \ell \in K_2.$$

From $ack = bd\ell$, we get

$$k = c^{-1}a^{-1}bd\ell.$$

Since $a, b \in A \le H_2$, $c \in K_2 \le H_2$, $d \in H_2$, and $\ell \in K_2 \le H_2$, it follows that $k \in H_2$. Thus $k \in K_1 \cap H_2$. Since $c \in K_2$ and $K_2 \trianglelefteq H_2$, we have

$$s = ck = k(k^{-1}ck) \in (K_1 \cap H_2)K_2 = B_2.$$

Since $t \in B_2$, the equality $as = bt$ gives

$$b^{-1}a = ts^{-1} \in B_2.$$

Hence $aB_2 = bB_2$. Therefore each coset of B_1 in AK_1 intersects at most one coset of B_2 in AK_2.
Conversely, suppose that

$$aB_2 \cap bB_1 \neq \varnothing$$

for some $a, b \in A$. Choose $z \in aB_2 \cap bB_1$. Then $z = at = bs$ with $t \in B_2$ and $s \in B_1$. Write

$$t = d\ell, \quad d \in K_1 \cap H_2, \quad \ell \in K_2,$$

and

$$s = ck, \quad c \in H_1 \cap K_2, \quad k \in K_1.$$

From $ad\ell = bck$, we get

$$\ell = d^{-1}a^{-1}bck.$$

Since $a, b \in A \le H_1$, $d \in K_1 \le H_1$, $c \in H_1$, and $k \in K_1 \le H_1$, it follows that $\ell \in H_1$. Thus $\ell \in H_1 \cap K_2$. Since $d \in K_1$ and $K_1 \trianglelefteq H_1$, we have

$$t = d\ell = \ell(\ell^{-1}d\ell) \in (H_1 \cap K_2)K_1 = B_1.$$

Since $s \in B_1$, the equality $at = bs$ gives

$$b^{-1}a = st^{-1} \in B_1.$$

Hence $aB_1 = bB_1$. Therefore each coset of B_2 in AK_2 intersects at most one coset of B_1 in AK_1.
Thus the projections AK_1/B_1 and AK_2/B_2 are incident. $\square$

Theorem (Third Isomorphism Theorem)**.** *Let $H_1, H_2 \leq G$, let $K_1 \trianglelefteq H_1$, and let $K_2 \trianglelefteq H_2$. Then*

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

Proof. By Exercise 1.15, the sections

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \quad \text{and} \quad (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2$$

are incident. By Exercise 1.11, incident sections are isomorphic. Hence

$$(H_1 \cap H_2)K_1/(H_1 \cap K_2)K_1 \cong (H_1 \cap H_2)K_2/(K_1 \cap H_2)K_2.$$

□

(This result is also called the fourth isomorphism theorem, or Zassenhaus' lemma — after its discoverer, who proved it as a student at the age of 21 — or even the butterfly lemma. The last name refers to the shape of the diagram showing the inclusion relations between the many subgroups involved.)

2 Automorphisms

The Automorphism Group

- The set of automorphisms of a group G is denoted $\operatorname{Aut}(G)$.
- If $\varphi, \rho \in \operatorname{Aut}(G)$, then $\varphi \circ \rho \in \operatorname{Aut}(G)$, so composition is a binary operation on $\operatorname{Aut}(G)$.
- Under composition, $\operatorname{Aut}(G)$ is a group: identity is the trivial automorphism, and the inverse of φ is its set-map inverse φ^{-1}.
- Call $\operatorname{Aut}(G)$ with this operation the *automorphism group* of G; we may write $\varphi\rho$ for $\varphi \circ \rho$.

Inner Automorphisms, $\operatorname{Inn}(G)$, and the Center

- Each $g \in G$ defines a conjugation homomorphism $\varphi_g : G \to G$ by $\varphi_g(x) = gxg^{-1}$.
- Each φ_g is an automorphism: for $x \in G$, $x = \varphi_g(g^{-1}xg)$, and if $\varphi_g(x) = \varphi_g(y)$ then $x = y$ by cancellation. These are the *inner automorphisms* of G.
- $\varphi_g\varphi_h = \varphi_{gh}$ for $g, h \in G$, since $g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1}$. Hence there is a homomorphism $G \to \operatorname{Aut}(G)$ sending g to φ_g.
- Its image is the *inner automorphism group* of G, denoted $\operatorname{Inn}(G)$; its kernel is the *center* of G, denoted $Z(G)$. Observe
$$Z(G) = \{g \in G \mid \varphi_g(x) = x \text{ for all } x \in G\} = \{g \in G \mid gx = xg \text{ for all } x \in G\},$$
so $Z(G)$ consists of those elements of G which commute with every element of G.
- Clearly G is abelian iff $Z(G) = G$.

Outer Automorphisms, $\operatorname{Out}(G)$

- If $\sigma \in \operatorname{Aut}(G)$ and $\varphi_g \in \operatorname{Inn}(G)$, then $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. Hence $\operatorname{Inn}(G) \trianglelefteq \operatorname{Aut}(G)$.
- The quotient $\operatorname{Aut}(G)/\operatorname{Inn}(G)$ is the *outer automorphism group* of G, denoted $\operatorname{Out}(G)$.
- "Outer automorphism" refers to automorphisms of G which are not inner and which thus have non-trivial image in $\operatorname{Out}(G)$.
- If G is abelian, then all non-trivial automorphisms of G are outer in this sense, since $\operatorname{Inn}(G) = 1$.

Automorphism Groups of Cyclic Groups

- Let $G = \langle x \rangle \cong \mathbb{Z}$, and let φ be an automorphism of G. Then $\varphi(x)$ must generate G; the only generators are x and x^{-1}. So φ either fixes each element or sends each to its inverse, and $\operatorname{Aut}(G) \cong \mathbb{Z}_2$.
- Let $n \in \mathbb{N}$ and $G = \langle x \rangle \cong \mathbb{Z}_n$. Suppose φ is an endomorphism of G. Then $\varphi(x) = x^m$ for some $0 \le m < n$, so φ sends every element to its mth power. Hence G has exactly n endomorphisms, namely the mth power maps σ_m for $0 \le m < n$.

Proposition 2.1. *Let $G = \langle x \rangle \cong \mathbb{Z}_n$ for $n \in \mathbb{N}$, and for each $0 \le m < n$ let σ_m be the endomorphism of G sending x to x^m. Then $\operatorname{Aut}(G)$ consists precisely of those σ_m for which $m \neq 0$ and $\gcd(m, n) = 1$. Furthermore, $\operatorname{Aut}(G)$ is abelian and is isomorphic with the group $(\mathbb{Z}/n\mathbb{Z})^\times$ of units of the ring $\mathbb{Z}/n\mathbb{Z}$.*

Proof. The map σ_0 has trivial image, so it is not an automorphism. Let $1 \le m < n$ and consider σ_m. Suppose first that $\gcd(m, n) = 1$. Then there exist $a, b \in \mathbb{Z}$ such that $am + bn = 1$, and hence
$$\sigma_m(x^a) = x^{am} = x^{1-bn} = x(x^n)^{-b} = x.$$
Thus $x \in \operatorname{im}\sigma_m$, so σ_m is surjective. Since G is finite, σ_m is also injective, and therefore $\sigma_m \in \operatorname{Aut}(G)$. Conversely, suppose that $\sigma_m \in \operatorname{Aut}(G)$. Since σ_m is surjective, there is some $a \in \mathbb{Z}$ such that $x = \sigma_m(x^a) = x^{am}$. Hence $x^{am-1} = 1$, so $am - 1 = bn$ for some $b \in \mathbb{Z}$. It follows that any common divisor

of m and n divides 1, and therefore $\gcd(m, n) = 1$. This proves the first assertion.
Now let $1 \le m_1, m_2 < n$ with $\gcd(m_1, n) = \gcd(m_2, n) = 1$. If t is chosen with $1 \le t < n$ and $m_1 m_2 \equiv t \pmod{n}$, then

$$(\sigma_{m_1}\sigma_{m_2})(x) = \sigma_{m_1}(x^{m_2}) = x^{m_1 m_2} = x^t.$$

Hence $\sigma_{m_1}\sigma_{m_2} = \sigma_t = \sigma_{m_2}\sigma_{m_1}$, so $\mathrm{Aut}(G)$ is abelian. Finally, the map

$$\mathrm{Aut}(G) \to (\mathbb{Z}/n\mathbb{Z})^\times, \qquad \sigma_m \mapsto m + n\mathbb{Z}$$

is a bijection by the first assertion, and the computation above shows that it preserves multiplication. Hence $\mathrm{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$. □

The Totient and Primitive Roots

- The *totient* of $n \in \mathbb{N}$ is the number of positive integers less than n and coprime to n (the value at n of the Euler phi-function).
- If $n = p_1^{a_1} \cdots p_r^{a_r}$ with the p_i distinct primes, then the totient of n is $(p_1^{a_1} - p_1^{a_1-1}) \cdots (p_r^{a_r} - p_r^{a_r-1})$. To see this, note that m is coprime to n iff no p_i divides m. Among $1, \dots, n$, exactly n/p_i integers are divisible by p_i, exactly $n/(p_i p_j)$ are divisible by both p_i and p_j, and so on. Hence inclusion-exclusion gives
$$\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^{r} \left(p_i^{a_i} - p_i^{a_i-1}\right).$$
- By Proposition 2.1, the order of $\mathrm{Aut}(\mathbb{Z}_n)$ is the totient of n. In particular, $|\mathrm{Aut}(\mathbb{Z}_p)| = p - 1$ when p is prime.

Proposition 2.2. *Let p be a prime. Then* $\mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$.

Proof. Let $F = \mathbb{F}_p$. By Proposition 2.1, $\mathrm{Aut}(\mathbb{Z}_p)$ is isomorphic with $F^\times = (\mathbb{Z}/p\mathbb{Z})^\times$. Thus it suffices to show that $F^\times$ is cyclic of order $p - 1$.
For each divisor d of $p - 1$, define

$$f_d = |\{x \in F^\times \mid o(x) = d\}|, \qquad z_d = |\{x \in \mathbb{Z}_{p-1} \mid o(x) = d\}|.$$

We first show that $f_d \le z_d$ for every $d \mid (p - 1)$. If $x \in F^\times$ has order dividing d, then $x^d = 1$, so x is a root of $T^d - 1 \in F[T]$. Since this polynomial has degree d, it has at most d roots in F. Now suppose $F^\times$ has an element x of order d. Then the d elements of $\langle x\rangle$ are all roots of $T^d - 1$, so they are all the roots. Hence every element of $F^\times$ of order d lies in $\langle x\rangle \cong \mathbb{Z}_d$. Therefore either $f_d = 0$, or f_d is the number of elements of order d in a cyclic group of order d. By Theorem 1.4, the cyclic group $\mathbb{Z}_{p-1}$ has a unique subgroup of order d; every element of order d generates a subgroup of order d, so all elements of order d in $\mathbb{Z}_{p-1}$ lie in this unique subgroup. This subgroup is cyclic of order d, and hence $f_d \le z_d$.
Since every element of $F^\times$ has order dividing $|F^\times| = p-1$, and every element of $\mathbb{Z}_{p-1}$ has order dividing $p - 1$, we have

$$\sum_{d|(p-1)} f_d = |F^\times| = p - 1 = |\mathbb{Z}_{p-1}| = \sum_{d|(p-1)} z_d.$$

Since $f_d \le z_d$ for every $d \mid (p - 1)$ and the sums are equal, we must have $f_d = z_d$ for every $d \mid (p - 1)$. In particular, $f_{p-1} = z_{p-1}$. But $\mathbb{Z}_{p-1}$ has an element of order $p - 1$, so $z_{p-1} > 0$, and hence $f_{p-1} > 0$. Therefore $F^\times$ has an element of order $p - 1$, so $F^\times \cong \mathbb{Z}_{p-1}$.
$\therefore \mathrm{Aut}(\mathbb{Z}_p) \cong \mathbb{Z}_{p-1}$. □

Primitive Roots Modulo n

- Let $G = \langle x\rangle \cong \mathbb{Z}_n$, $1 \le m \le n$, $\gcd(m, n) = 1$. Induction gives $(\sigma_m)^k(x) = x^{m^k}$.
- The order of σ_m is the least positive k such that $x^{m^k} = x$, i.e. the smallest $k \in \mathbb{N}$ with $m^k \equiv 1 \pmod{n}$.
- If the order of σ_m equals the totient of n, we call m a *primitive root modulo n*.
- $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff there is a primitive root modulo n.

- For composite n, the structure of $\mathrm{Aut}(\mathbb{Z}_n)$ belongs to number theory more than group theory.

Theorem 2.3. *For $n > 1$, $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff $n = 2$ or 4, or $n = p^k$ or $2p^k$ for some odd prime p and some $k \in \mathbb{N}$.*

Proof. By Proposition 2.1,
$$\mathrm{Aut}(\mathbb{Z}_n) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$
Under this isomorphism, σ_m corresponds to the unit $m + n\mathbb{Z}$, and the order of σ_m is the least $k \in \mathbb{N}$ such that $m^k \equiv 1 \pmod n$. Hence $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic iff there is a primitive root modulo n. It remains to prove the number-theoretic classification of those $n > 1$ which have primitive roots.
We use Euler's theorem in the following form: if $\gcd(a, q) = 1$, then $a^{\varphi(q)} \equiv 1 \pmod q$. First suppose $q = 2^s$ with $s \geq 3$. If a is odd, then $a^2 \equiv 1 \pmod 8$. We claim that
$$a^{2^{s-2}} \equiv 1 \pmod{2^s}$$
for all $s \geq 3$. The case $s = 3$ is the congruence just observed. If $a^{2^{s-2}} = 1 + b2^s$, then
$$a^{2^{s-1}} = (1 + b2^s)^2 \equiv 1 \pmod{2^{s+1}},$$
so the claim follows by induction. But $\varphi(2^s) = 2^{s-1}$, so every odd residue modulo 2^s has order at most $2^{s-2} < \varphi(2^s)$. Thus 2^s has no primitive root for $s \geq 3$.
Next suppose $\gcd(u, v) = 1$ with $u, v > 2$. Let a be coprime to uv, and put
$$h = \mathrm{lcm}(\varphi(u), \varphi(v)).$$
Since $u, v > 2$, both $\varphi(u)$ and $\varphi(v)$ are even (the reduced residue classes pair as b and $-b$), so
$$h \leq \frac{\varphi(u)\varphi(v)}{2} = \frac{\varphi(uv)}{2}.$$
Euler's theorem gives $a^h \equiv 1 \pmod u$ and $a^h \equiv 1 \pmod v$; since $\gcd(u, v) = 1$, this implies $a^h \equiv 1 \pmod{uv}$. Hence no unit modulo uv can have order $\varphi(uv)$, and uv has no primitive root.
These two obstructions leave only 2, 4, odd prime powers, and twice odd prime powers. Indeed, if
$$n = 2^a p_1^{e_1} \cdots p_t^{e_t}$$
is the prime factorization of $n > 1$, then two distinct odd prime factors are ruled out by the previous paragraph, as is $2^a p^e$ with $a \geq 2$ and p odd. The pure powers 2^a with $a \geq 3$ were ruled out in the paragraph before that. Therefore any $n > 1$ with a primitive root must be one of 2, 4, p^e, or $2p^e$ with p odd.
It remains to see that all numbers in this list actually have primitive roots. Clearly 1 is a primitive root modulo 2, and 3 is a primitive root modulo 4. Now let p be an odd prime. By Proposition 2.2, there is a primitive root r modulo p. We may choose such an r with
$$r^{p-1} \not\equiv 1 \pmod{p^2}.$$
For if a chosen primitive root r satisfies $r^{p-1} \equiv 1 \pmod{p^2}$, then $r + p$ is still primitive modulo p, and
$$(r + p)^{p-1} \equiv r^{p-1} + (p-1)pr^{p-2} \equiv 1 - pr^{p-2} \not\equiv 1 \pmod{p^2}.$$
We claim that this r is primitive modulo p^e for every $e \in \mathbb{N}$. For $e = 1$ this is how r was chosen. For $e \geq 2$, we first show by induction that
$$r^{p^{e-2}(p-1)} \not\equiv 1 \pmod{p^e}.$$
The case $e = 2$ is our choice of r. If the assertion holds for e, then Euler's theorem modulo p^{e-1} gives
$$r^{p^{e-2}(p-1)} = 1 + cp^{e-1}$$
for some integer c not divisible by p. Raising both sides to the pth power gives
$$r^{p^{e-1}(p-1)} = (1 + cp^{e-1})^p \equiv 1 + cp^e \not\equiv 1 \pmod{p^{e+1}},$$

proving the induction step. Now let d be the order of r modulo p^e. Then $d \mid \varphi(p^e) = p^{e-1}(p-1)$, and reducing modulo p shows $p-1 \mid d$. Hence $d = p^j(p-1)$ for some $0 \le j \le e-1$. If $j < e-1$, then $d \mid p^{e-2}(p-1)$, contradicting the congruence just proved. Thus $d = p^{e-1}(p-1) = \varphi(p^e)$, so r is primitive modulo p^e.
Finally, let r be primitive modulo p^e. Replacing r by $r + p^e$ if necessary, we may assume that r is odd; this does not change its residue modulo p^e. Let d be the order of r modulo $2p^e$. Since $r^d \equiv 1 \pmod{2p^e}$ implies $r^d \equiv 1 \pmod{p^e}$, we have $\varphi(p^e) \mid d$. Also $d \mid \varphi(2p^e) = \varphi(p^e)$. Therefore $d = \varphi(2p^e)$, and r is primitive modulo $2p^e$.
We have proved that primitive roots exist exactly for $n = 2$, 4, p^e, and $2p^e$ with p odd. Therefore $\mathrm{Aut}(\mathbb{Z}_n)$ is cyclic exactly for the same values of n. □

The number-theoretic primitive-root classification used in this proof follows [9, Section 8.3].

Fixed and Characteristic Subgroups

- Let $\varphi \in \mathrm{Aut}(G)$ and $H \le G$. We say H is *fixed by* φ if $\varphi(H) = H$; in this case the restriction is an automorphism of H.
- If $L \le \mathrm{Aut}(G)$, H is *fixed by* L if H is fixed by every $\varphi \in L$.
- H is normal in G iff H is fixed by $\mathrm{Inn}(G)$.
- H is a *characteristic subgroup* of G (or H *characteristic* in G) if H is fixed by $\mathrm{Aut}(G)$. (Some authors write $H \operatorname{char} G$.)
- Example: $Z(G)$ is always characteristic. For if $x \in Z(G)$ and $\varphi \in \mathrm{Aut}(G)$, then $\varphi(x)y = \varphi(x\varphi^{-1}(y)) = \varphi(\varphi^{-1}(y)x) = y\varphi(x)$ for any $y \in G$, showing $\varphi(x) \in Z(G)$.
- Characteristic subgroups are normal, but the converse fails. In fact, an infinite abelian group need not have any non-trivial proper characteristic subgroups (Exercise 2.5).

Lemma 2.4. *If K is a characteristic subgroup of H and H is a characteristic subgroup of G, then K is a characteristic subgroup of G.*

Proof. Let $\varphi \in \mathrm{Aut}(G)$. We need to show that $\varphi(K) = K$. Since H is characteristic in G, we have $\varphi(H) = H$. Hence the restriction $\varphi|_H \colon H \to H$ is an automorphism of H: it is still a homomorphism, it is injective because φ is injective, and it is surjective onto H because $\varphi(H) = H$.
Now K is characteristic in H, so every automorphism of H fixes K. Applying this to $\varphi|_H$, we obtain $(\varphi|_H)(K) = K$. But $K \le H$, so $(\varphi|_H)(K) = \varphi(K)$. Therefore $\varphi(K) = K$. Since $\varphi \in \mathrm{Aut}(G)$ was arbitrary, K is characteristic in G. □

The Derived Group

- Normality is not transitive: if $N \trianglelefteq G$, the restriction to N of an element of $\mathrm{Inn}(G)$ lies in $\mathrm{Aut}(N)$ but need not lie in $\mathrm{Inn}(N)$.
- For $x, y \in G$ the *commutator* is $[x, y] = xyx^{-1}y^{-1}$.
- The *derived group* G' of G is the subgroup generated by all commutators: $G' = \langle \{[x, y] \mid x, y \in G\} \rangle$.
- G is abelian iff $G' = 1$.
- If $H \le G$, then $H' \le G'$.
- In general, G' contains more than just the commutators of elements of G.
- Since $[x, y]^{-1} = (xyx^{-1}y^{-1})^{-1} = yxy^{-1}x^{-1} = [y, x]$, by Proposition 1.2 an arbitrary element of G' is a finite product of commutators of elements of G.

Lemma 2.5. *Let G be a group. Then G' is characteristic in G.*

Proof. Let $\varphi \in \mathrm{Aut}(G)$. For any $x, y \in G$, we have

$$\varphi([x,y]) = \varphi(xyx^{-1}y^{-1}) = \varphi(x)\varphi(y)\varphi(x)^{-1}\varphi(y)^{-1} = [\varphi(x), \varphi(y)].$$

Thus φ sends each commutator of elements of G to another commutator of elements of G. Now let $g \in G'$. By Proposition 1.2 and the preceding observation about inverses of commutators, we may write

$$g = [x_1, y_1] \cdots [x_r, y_r]$$

for some $x_i, y_i \in G$. Applying φ, we obtain

$$\varphi(g) = [\varphi(x_1), \varphi(y_1)] \cdots [\varphi(x_r), \varphi(y_r)] \in G'.$$

Hence $\varphi(G') \leq G'$. Applying the same argument to φ^{-1} gives $\varphi^{-1}(G') \leq G'$, and therefore $G' \leq \varphi(G')$. Thus $\varphi(G') = G'$. Since φ was arbitrary, G' is characteristic in G. □

Proposition 2.6. *Let G be a group, and let $N \trianglelefteq G$. Then G/N is abelian iff $G' \leq N$.*

Proof. By Lemma 2.5, G' is characteristic in G, and hence $G' \trianglelefteq G$. We first observe that G/G' is abelian. Indeed, for any $x, y \in G$,

$$[xG', yG'] = [x,y]G' = G',$$

since $[x,y] \in G'$. Thus every commutator in G/G' is trivial, so G/G' is abelian.
Suppose first that $G' \leq N$. Since $G' \leq N$, the set N/G' is a subgroup of G/G'. Moreover, it is normal in G/G': if $g \in G$ and $n \in N$, then

$$(gG')(nG')(gG')^{-1} = gng^{-1}G' \in N/G',$$

because $N \trianglelefteq G$. Thus $N/G' \trianglelefteq G/G'$, and by Second Isomorphism Theorem,

$$G/N \cong (G/G')/(N/G').$$

Since G/G' is abelian, every quotient of G/G' is abelian. Hence G/N is abelian.
Conversely, suppose that G/N is abelian. For any $x, y \in G$, the commutator of xN and yN is trivial, so

$$N = [xN, yN] = [x,y]N.$$

Therefore $[x,y] \in N$ for all $x, y \in G$. Since N contains every commutator of elements of G, it contains the subgroup generated by all such commutators. Hence $G' \leq N$. □

External Direct Products

- Let $n \in \mathbb{N}$ and let $G_1, \ldots, G_n$ be groups. The Cartesian product $G_1 \times \cdots \times G_n$ becomes a group under componentwise multiplication $(g_1, \ldots, g_n)(g_1', \ldots, g_n') = (g_1 g_1', \ldots, g_n g_n')$. The identity is $(1, \ldots, 1)$ and inverses are $(g_1^{-1}, \ldots, g_n^{-1})$.
- This is the (*external*) *direct product* of $G_1, \ldots, G_n$.
- Order of factors is irrelevant: $G_1 \times \cdots \times G_n \cong G_{\rho(1)} \times \cdots \times G_{\rho(n)}$ for any $\rho \in S_n$.
- For $G = G_1 \times \cdots \times G_n$:
  - Each $H_i = \{(1, \ldots, g_i, \ldots, 1) \mid g_i \in G_i\}$ is a normal subgroup of G isomorphic with G_i. Moreover G/H_i is isomorphic with the direct product of the remaining G_j.
  - Every $g \in G$ has a unique expression $g = h_1 \cdots h_n$ with $h_i \in H_i$; if $g = (g_1, \ldots, g_n)$ then $h_i = (1, \ldots, g_i, \ldots, 1)$.
  - If the G_i are finite, then $|G| = |G_1| \cdots |G_n|$.

Internal Direct Products

- Suppose G has subgroups $H_1, \dots, H_n$ such that:
  1. $H_i \trianglelefteq G$ for each $1 \le i \le n$.
  2. Every $g \in G$ has a unique expression $g = h_1 \cdots h_n$ where $h_i \in H_i$.

  Conditions (1) and (2) imply:

  3. $G = H_1 \cdots H_n$.
  4. $H_i \cap H_1 \cdots H_{i-1} H_{i+1} \cdots H_n = 1$ for each i.
  5. If $i \neq j$, then elements of H_i commute with elements of H_j.
  6. If $g = h_1 \cdots h_n$ and $g' = h_1' \cdots h_n'$ with $h_i, h_i' \in H_i$, then $gg' = (h_1 h_1') \cdots (h_n h_n')$.
- Under these conditions, there is a unique isomorphism from G to the external direct product $H_1 \times \cdots \times H_n$ sending H_i to $1 \times \cdots \times H_i \times \cdots \times 1$. Call G the (*internal*) *direct product* of $H_1, \dots, H_n$ and write $G = H_1 \times \cdots \times H_n$ (slight abuse of notation).
- If (1) holds, then (2) holds iff both (3) and (4) hold; so to check a direct product, it suffices to verify either (1) and (2) or (1), (3), and (4). (For $n = 2$, (4) reduces to $H_1 \cap H_2 = 1$.)

Lemma 2.7. *Let G be a group having normal subgroups H and K such that $G = HK$. Then $G/H \cap K \cong H/H \cap K \times K/H \cap K$.*

Proof. Put $M = H \cap K$. By Proposition 1.7, we have $M \trianglelefteq G$, and hence G/M, H/M, and K/M are quotient groups. By Correspondence Theorem, the subgroups H/M and K/M are normal in G/M.
We first compute their intersection. Suppose that $xM \in (H/M) \cap (K/M)$. Since $xM \in H/M$, there is some $h \in H$ such that $xM = hM$, and because $M \le H$ this implies $x \in H$. Similarly, $x \in K$. Thus $x \in H \cap K = M$, and so $xM = M$. Therefore

$$(H/M) \cap (K/M) = 1.$$

Next let $gM \in G/M$. Since $G = HK$, we may write $g = hk$ with $h \in H$ and $k \in K$. Then

$$gM = hkM = (hM)(kM),$$

so $(G/M) = (H/M)(K/M)$. Hence G/M is the internal direct product of H/M and K/M, and therefore is isomorphic with the corresponding external direct product. Thus

$$G/(H \cap K) \cong H/(H \cap K) \times K/(H \cap K).$$

□

Lemma 2.8. *Let $n \in \mathbb{N}$, and write $n = p_1^{a_1} \cdots p_r^{a_r}$ where the p_i are distinct primes and the a_i are positive integers. Then we have $\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}$.*

Proof. If $n = 1$, there is nothing to prove, so assume $n > 1$. For each i, let

$$P_i = \langle x_i \rangle \cong \mathbb{Z}_{p_i^{a_i}},$$

and put $P = P_1 \times \cdots \times P_r$. Then

$$|P| = |P_1| \cdots |P_r| = p_1^{a_1} \cdots p_r^{a_r} = n.$$

Let $x = (x_1, \dots, x_r) \in P$. For $k \in \mathbb{N}$, we have

$$x^k = (1, \dots, 1)$$

iff $x_i^k = 1$ for every i, iff $p_i^{a_i} \mid k$ for every i. Since the prime powers $p_i^{a_i}$ are pairwise coprime, this is equivalent to $n \mid k$. Hence $o(x) = n$.
Therefore $\langle x \rangle$ has n elements and lies in P, which also has n elements. Thus $\langle x \rangle = P$, so P is cyclic of order n. It follows that $P \cong \mathbb{Z}_n$, and substituting the groups $P_i \cong \mathbb{Z}_{p_i^{a_i}}$ gives

$$\mathbb{Z}_n \cong \mathbb{Z}_{p_1^{a_1}} \times \cdots \times \mathbb{Z}_{p_r^{a_r}}.$$

□

Corollary 2.9. *If* $\gcd(a,b) = 1$*, then* $\mathbb{Z}_{ab} \cong \mathbb{Z}_a \times \mathbb{Z}_b$*.*

Proof. If $a = 1$ or $b = 1$, the assertion is immediate because $\mathbb{Z}_1$ is the trivial group. Thus assume $a, b > 1$. Write

$$a = p_1^{\alpha_1} \cdots p_s^{\alpha_s}, \qquad b = q_1^{\beta_1} \cdots q_t^{\beta_t}$$

as prime-power factorizations. Since $\gcd(a,b) = 1$, the primes $p_1, \ldots, p_s$ are distinct from the primes $q_1, \ldots, q_t$. Hence

$$ab = p_1^{\alpha_1} \cdots p_s^{\alpha_s} q_1^{\beta_1} \cdots q_t^{\beta_t}$$

is the prime-power factorization of ab. Applying Lemma 2.8 to ab gives

$$\mathbb{Z}_{ab} \cong \mathbb{Z}_{p_1^{\alpha_1}} \times \cdots \times \mathbb{Z}_{p_s^{\alpha_s}} \times \mathbb{Z}_{q_1^{\beta_1}} \times \cdots \times \mathbb{Z}_{q_t^{\beta_t}}.$$

Applying Lemma 2.8 separately to a and to b, and regrouping the factors, gives

$$\mathbb{Z}_{ab} \cong \mathbb{Z}_a \times \mathbb{Z}_b.$$

□

Proposition 2.10. *Suppose that a finite group* G *is the direct product of its subgroups* $H_1, \ldots, H_n$*, where the orders* $|H_i|$ *are pairwise coprime. Then any subgroup* L *of* G *is the direct product of* $L \cap H_1, \ldots, L \cap H_n$*.*

Proof. We first prove the case $n = 2$. Write $G = H \times K$, where $\gcd(|H|, |K|) = 1$, and let $L \leq G$. Put

$$A = L \cap H, \qquad B = L \cap K.$$

Since $H, K \trianglelefteq G$, we have $A, B \trianglelefteq L$. Also $A \cap B \leq H \cap K = 1$, so $A \cap B = 1$. It remains to show that $L = AB$.
Let $g \in L$. Since $G = H \times K$, we may write $g = hk$ with $h \in H$ and $k \in K$, and h and k commute. By Lagrange's Theorem, $o(h) \mid |H|$ and $o(k) \mid |K|$, so $o(h)$ and $o(k)$ are coprime. Put $a = o(h)$ and $b = o(k)$. If $(hk)^m = 1$, then, since h and k commute,

$$h^m = k^{-m} \in H \cap K = 1.$$

Hence $a \mid m$ and $b \mid m$, and because $\gcd(a,b) = 1$, we have $ab \mid m$. Conversely, $(hk)^{ab} = 1$, so $o(hk) = ab$. Now $\langle hk \rangle \leq \langle h \rangle \langle k \rangle$. The subgroup $\langle h \rangle \langle k \rangle$ is the internal direct product of $\langle h \rangle$ and $\langle k \rangle$, so it has order ab. Since $\langle hk \rangle$ also has order ab, we have

$$\langle hk \rangle = \langle h \rangle \langle k \rangle.$$

Thus $h, k \in \langle hk \rangle = \langle g \rangle \leq L$. Therefore $h \in A$ and $k \in B$, and so $g = hk \in AB$. This proves $L \leq AB$, while $AB \leq L$ is immediate. Hence $L = AB$. Since A and B are normal in L, have trivial intersection, and commute with one another, $L = A \times B$.
We now argue by induction on n. The case $n = 1$ is immediate, and the case $n = 2$ has just been proved. Suppose $n > 2$ and that the result is known for direct products with $n - 1$ factors. Let

$$M = H_1 \times \cdots \times H_{n-1}.$$

Then $G = M \times H_n$, and pairwise coprimeness of the orders $|H_i|$ gives $\gcd(|M|, |H_n|) = 1$. Applying the $n = 2$ case to $G = M \times H_n$ gives

$$L = (L \cap M) \times (L \cap H_n).$$

Inside M, the subgroup $L \cap M$ is a subgroup of the direct product $H_1 \times \cdots \times H_{n-1}$, whose factor orders are still pairwise coprime. By the induction hypothesis,

$$L \cap M = (L \cap H_1) \times \cdots \times (L \cap H_{n-1}).$$

Substituting this into the preceding display gives

$$L = (L \cap H_1) \times \cdots \times (L \cap H_n),$$

as required. □

Internal Semidirect Products

- Suppose G has subgroup H and normal subgroup N with $G = NH$ and $N \cap H = 1$. Then G is the (*internal*) *semidirect product* of N by H, written $G = N \rtimes H$. (Notation common but not standard; alternatives include $N \rtimes H$, $H \ltimes N$, no notation.)
- If in addition $H \trianglelefteq G$, then G is the direct product of N and H.
- Example: $G = S_3$, $N = A_3$, $H = \langle (1\,2) \rangle$. Then $G = N \rtimes H$, but H is not normal, so G is not the direct product.
- Properties of $G = N \rtimes H$:
  - $H \cong H/(N \cap H) \cong NH/N = G/N$ by First Isomorphism Theorem; if G is finite, $|G| = |N||G : N| = |N||H|$.
  - Each $x \in G$ has a unique expression $x = nh$ with $n \in N$, $h \in H$.
  - If $x = n_1 h_1$, $y = n_2 h_2$, then $xy = n_1 h_1 n_2 h_1^{-1} \cdot h_1 h_2 = n'h'$ with $n' = n_1 h_1 n_2 h_1^{-1} \in N$ and $h' = h_1 h_2 \in H$.
  - For $h \in H$, conjugation by h maps N to N, so $\varphi_h : N \to N$, $\varphi_h(n) = hnh^{-1}$, is an automorphism of N, and $\varphi_h \circ \varphi_{h'} = \varphi_{hh'}$. Hence the *conjugation homomorphism* $\varphi : H \to \operatorname{Aut}(N)$, $\varphi(h) = \varphi_h$.
  - $(n_1 h_1)(n_2 h_2) = n_1 \varphi(h_1)(n_2) \cdot h_1 h_2$, so the multiplication of G is determined by the operations of N and H together with φ.
  - If φ is trivial, then $hnh^{-1} = n$ for all $n \in N$, $h \in H$, so $H \trianglelefteq G$ and $G = N \times H$. Conversely, if $G = N \times H$, then H commutes with N, so φ is trivial.
  - If φ is non-trivial, G is non-abelian: there exist $h \in H$, $n \in N$ with $hnh^{-1} = \varphi(h)(n) \neq n$.

External Semidirect Products

- Given groups N, H and a homomorphism $\varphi : H \to \operatorname{Aut}(N)$, define a binary operation on $N \times H$ by
$$(n_1, h_1)(n_2, h_2) = (n_1 \varphi(h_1)(n_2), h_1 h_2).$$
This makes $N \times H$ a group with identity $(1, 1)$ and inverse $(n, h)^{-1} = (\varphi(h^{-1})(n^{-1}), h^{-1})$.
- Call this the (*external*) *semidirect product* of N by H corresponding to φ, denoted $G = N \rtimes_\varphi H$.
- This generally differs from the direct product unless φ is trivial.
- $\mathcal{N} = N \times 1$ and $\mathcal{H} = 1 \times H$ in G are isomorphic copies of N, H respectively. Computation shows $\mathcal{N} \trianglelefteq G$, $G = \mathcal{N}\mathcal{H}$, $\mathcal{N} \cap \mathcal{H} = 1$, so G is the internal semidirect product of $\mathcal{N}$ by $\mathcal{H}$.
- Identifying $\mathcal{N}$ with N and $\mathcal{H}$ with H, the conjugation homomorphism from $\mathcal{H}$ to $\operatorname{Aut}(\mathcal{N})$ corresponds to the original φ.
- Write $N \rtimes_\varphi H = \{nh \mid n \in N, h \in H\}$ with multiplication $(n_1 h_1)(n_2 h_2) = n_1 \varphi(h_1)(n_2) \cdot h_1 h_2$. Note $hnh^{-1} = \varphi(h)(n)$.
- Distinct $\varphi, \psi : H \to \operatorname{Aut}(N)$ need not yield isomorphic semidirect products. The next two propositions are partial converses.

Proposition 2.11. *Let H be a cyclic group and let N be an arbitrary group. If φ and ψ are monomorphisms from H to $\operatorname{Aut}(N)$ such that $\varphi(H) = \psi(H)$, then we have $N \rtimes_\varphi H \cong N \rtimes_\psi H$.*

Proof. Write $H = \langle x \rangle$. Since $\varphi(H) = \psi(H)$, the automorphisms $\varphi(x)$ and $\psi(x)$ generate the same cyclic subgroup of $\operatorname{Aut}(N)$. Hence there exist $a, b \in \mathbb{Z}$ such that
$$\varphi(x)^a = \psi(x), \qquad \psi(x)^b = \varphi(x).$$
It follows that, for every $h \in H$,
$$\varphi(h^a) = \psi(h), \qquad \psi(h^b) = \varphi(h).$$

Indeed, if $h = x^m$, then $\varphi(h^a) = \varphi(x^{am}) = \varphi(x)^{am} = \psi(x)^m = \psi(h)$, and the second equality is similar. Define

$$\tau\colon N \rtimes_\psi H \to N \rtimes_\varphi H, \qquad \tau(nh) = nh^a.$$

We show that τ is a homomorphism. Let $n_1, n_2 \in N$ and $h_1, h_2 \in H$. Multiplication in $N \rtimes_\psi H$ gives

$$(n_1h_1)(n_2h_2) = n_1\psi(h_1)(n_2)h_1h_2.$$

Since H is cyclic, it is abelian, so $(h_1h_2)^a = h_1^a h_2^a$. Therefore

$$\tau((n_1h_1)(n_2h_2)) = n_1\psi(h_1)(n_2)(h_1h_2)^a = n_1\varphi(h_1^a)(n_2)h_1^a h_2^a.$$

But this is exactly the product of $n_1h_1^a$ and $n_2h_2^a$ in $N \rtimes_\varphi H$, so

$$\tau((n_1h_1)(n_2h_2)) = \tau(n_1h_1)\tau(n_2h_2).$$

Thus τ is a homomorphism. Similarly, the map

$$\lambda\colon N \rtimes_\varphi H \to N \rtimes_\psi H, \qquad \lambda(nh) = nh^b$$

is a homomorphism.
It remains to check that these two homomorphisms are inverse to one another. Since

$$\varphi(x) = \psi(x)^b = (\varphi(x)^a)^b = \varphi(x^{ab}),$$

and φ is injective, we have $x = x^{ab}$. Hence $h = h^{ab}$ for every $h \in H$. Similarly, using the injectivity of ψ, we get $h = h^{ba}$ for every $h \in H$. Therefore

$$(\tau \circ \lambda)(nh) = \tau(nh^b) = nh^{ba} = nh, \qquad (\lambda \circ \tau)(nh) = \lambda(nh^a) = nh^{ab} = nh.$$

Thus τ and λ are mutually inverse isomorphisms, and so $N \rtimes_\varphi H \cong N \rtimes_\psi H$. □

Proposition 2.12. *Let N and H be groups, let $\psi : H \to \mathrm{Aut}(N)$ be a homomorphism, and let $f \in \mathrm{Aut}(N)$. If $\hat{f}$ is the inner automorphism of $\mathrm{Aut}(N)$ induced by f, then $N \rtimes_{\hat{f}\circ\psi} H \cong N \rtimes_\psi H$.*

Proof. By definition, the inner automorphism $\hat{f}$ of $\mathrm{Aut}(N)$ sends $\alpha \in \mathrm{Aut}(N)$ to $f \circ \alpha \circ f^{-1}$. Define

$$\theta\colon N \rtimes_\psi H \to N \rtimes_{\hat{f}\circ\psi} H, \qquad \theta(nh) = f(n)h.$$

We show that θ is a homomorphism. Let $n_1, n_2 \in N$ and $h_1, h_2 \in H$. Multiplication in $N \rtimes_\psi H$ gives

$$(n_1h_1)(n_2h_2) = n_1\psi(h_1)(n_2)h_1h_2.$$

Hence

$$\theta((n_1h_1)(n_2h_2)) = f(n_1\psi(h_1)(n_2))h_1h_2 = f(n_1)f(\psi(h_1)(n_2))h_1h_2.$$

On the other hand, multiplication in $N \rtimes_{\hat{f}\circ\psi} H$ gives

$$\theta(n_1h_1)\theta(n_2h_2) = (f(n_1)h_1)(f(n_2)h_2) = f(n_1)(\hat{f} \circ \psi)(h_1)(f(n_2))h_1h_2.$$

Since

$$(\hat{f} \circ \psi)(h_1)(f(n_2)) = (f \circ \psi(h_1) \circ f^{-1})(f(n_2)) = f(\psi(h_1)(n_2)),$$

the two expressions are equal. Thus θ is a homomorphism.
Finally, the map

$$N \rtimes_{\hat{f}\circ\psi} H \to N \rtimes_\psi H, \qquad nh \mapsto f^{-1}(n)h$$

is inverse to θ. Therefore θ is an isomorphism, and so $N \rtimes_{\hat{f}\circ\psi} H \cong N \rtimes_\psi H$. □

Dihedral Groups

- Let $N \cong \mathbb{Z}_n$ ($n \in \mathbb{N}$), $H \cong \mathbb{Z}_2$, $\varphi : H \to \operatorname{Aut}(N)$ send the generator of H to the automorphism sending each element of N to its inverse (so $\varphi(h) = \sigma_{n-1}$).
- The group $N \rtimes_\varphi H$ is the *dihedral group* of order $2n$, denoted D_{2n}. (Some authors write D_n.)
- D_{2n} is non-abelian iff $n > 2$; when $n = 2$, φ is trivial and $D_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.
- Geometrically D_{2n} is the symmetry group of a regular n-gon: the generator of N is rotation by $2\pi/n$, the generator of H is reflection through a fixed axis.
- Also: the *infinite dihedral group* $D_\infty = N \rtimes_\varphi H$ where $N \cong \mathbb{Z}$ and H, φ as above.

Proposition 2.13. *Let s and t be elements of order 2 in a group G. (Such elements are called* involutions.*) Then $\langle s, t\rangle$ is a dihedral group; in particular, $\langle s, t\rangle = \langle st\rangle \rtimes \langle s\rangle$.*

Proof. Put
$$L = \langle s, t\rangle, \qquad r = st, \qquad N = \langle r\rangle, \qquad H = \langle s\rangle.$$
Since s and t have order 2, we have $s^{-1} = s$ and $t^{-1} = t$. Therefore
$$srs^{-1} = s(st)s = ts = (st)^{-1} = r^{-1}.$$
Similarly,
$$trt^{-1} = t(st)t = ts = r^{-1}.$$
Thus both generators s and t of L conjugate r to r^{-1}. Hence both generators normalize $N = \langle r\rangle$, and so
$$N \trianglelefteq L.$$
Also $H = \langle s\rangle \le L$. We next show that $L = NH$. Since $r = st \in N$ and $s \in H$, we have
$$t = sr \in HN.$$
But $N \trianglelefteq L$, so $HN = NH$. Hence both s and t lie in NH, and therefore
$$L = \langle s, t\rangle \le NH.$$
The reverse inclusion is immediate from $N, H \le L$, so $L = NH$.
It remains to check that $N \cap H = 1$. Since $H = \langle s\rangle$ and s has order 2, the only possible non-identity element of H is s. Suppose, for contradiction, that $s \in N$. Then s is a power of r. Since $t = sr$, it follows that $t \in N$ as well. Hence $L = \langle s, t\rangle \le N$, so $L = N$ is cyclic. But a cyclic group has at most one element of order 2, so $s = t$. Then $r = st = 1$, and hence $N = \langle r\rangle = 1$, contradicting $s \in N$. Therefore $s \notin N$, and so $N \cap H = 1$.
We have shown that
$$L = NH, \qquad N \trianglelefteq L, \qquad N \cap H = 1.$$
Hence L is the internal semidirect product $N \rtimes H$. Finally, the conjugation action of H on N sends the generator r of N to r^{-1}, so it sends every element of N to its inverse. This is precisely the action used in the definition of a dihedral group. Thus $\langle s, t\rangle$ is dihedral; more explicitly, if $r = st$ has finite order n, then $L \cong D_{2n}$, while if r has infinite order, then $L \cong D_\infty$. □

Reference note. Reference [7] is Aschbacher's *Finite Group Theory*; the local reference copy is the second edition, and Section 45 is titled "Involutions in finite groups." The point of the reference is that the small fact proved above becomes a major tool once one studies finite simple groups. In a finite group, two involutions x and y generate a dihedral group whose rotation subgroup is $\langle xy\rangle$. This means that the product xy controls the subgroup $\langle x, y\rangle$, and the elementary geometry of dihedral groups gives information about conjugacy among involutions.

Aschbacher begins Section 45 by using this exact observation. If $D = \langle x, y\rangle$ and $n = o(xy)$, then D has order $2n$. When n is odd, all involutions of D are conjugate in D, so x and y are already conjugate inside the small subgroup they generate. When n is even, there is a unique involution in the rotation subgroup $\langle xy\rangle$, and this central involution gives an additional conjugacy possibility. Thus the parity of $o(xy)$ tells us how involutions can fuse, meaning how they may become conjugate in the larger group.

This is important because involutions are unavoidable in non-abelian finite simple groups: by the Feit–Thompson Odd Order Theorem, every non-abelian finite simple group has even order, and therefore has involutions. Section 45 uses dihedral subgroups generated by pairs of involutions to count and control such elements. Two central results in that section are the Brauer–Fowler Theorem and the Thompson Order Formula. The Brauer–Fowler Theorem says, roughly, that fixing the centralizer of an involution leaves only finitely many possible finite simple groups. This suggests that finite simple groups can be attacked by studying the centralizers of their involutions.

The Thompson Order Formula is more explicit. When a finite group has at least two conjugacy classes of involutions, it expresses the order of the group in terms of centralizers of involutions and the fusion of involutions inside those centralizers. In one form, if $x_1, \dots, x_k$ represent the conjugacy classes of involutions and n_i counts certain ordered pairs of involutions whose product-generated subgroup contains x_i, then

$$|G| = |C_G(x_1)|\,|C_G(x_2)| \sum_{i=1}^{k} \frac{n_i}{|C_G(x_i)|}.$$

The important point is not the formula itself at this stage, but the method behind it: because pairs of involutions generate dihedral groups, their products and conjugacy relations can be counted very precisely. Thus Proposition 2.13 is not just a cute exercise. It is one of the basic mechanisms behind the local analysis of finite simple groups, where one tries to understand a group by studying centralizers and normalizers of special subgroups.

Exercises

Exercise 2.1. Let H be a subgroup of a cyclic group G. Show that every automorphism of H is the restriction to H of an automorphism of G.

Solution. □

Exercise 2.2. Show that $\mathrm{Aut}(\mathbb{Z}_8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Solution. □

Exercise 2.3. Show that $\mathrm{Aut}(\mathbb{Z}_{p^2}) \cong \mathbb{Z}_{p^2-p}$ for p a prime. (HINT: Let m be a primitive root modulo p, and show that either m or $m+p$ be a primitive root modulo p^2.)

Solution. □

Exercise 2.4. Show that if $H \trianglelefteq G$, then any characteristic subgroup of H is normal in G.

Solution. □

Exercise 2.5. Let F be a field, and consider F as a group under its additive operation. Show that F has no non-trivial proper characteristic subgroups.

Solution. □

Exercise 2.6. Verify the claim made on page 19 that if (1) holds, then (2) holds iff (3) and (4) hold.

Solution. □

Exercise 2.7. Verify the claim made on page 19 that (1) and (2) together imply (3) through (6).

Solution. □

Exercise 2.8. Let G_1 and G_2 be groups, and let $H \leq G_1 \times G_2$. Define

$$P_1 = \{x \in G_1 \mid (x, y) \in H \text{ for some } y \in G_2\},$$

$$I_1 = \{x \in G_1 \mid (x, 1) \in H\},$$

and analogously define subsets P_2 and I_2 of G_2.

(a) Show for $i = 1, 2$ that $P_i \leq G_i$ and $I_i \trianglelefteq P_i$.

(b) Show that there is a unique isomorphism from P_1/I_1 to P_2/I_2 sending xI_1 to yI_2, where y is any element of G_2 such that $(x, y) \in H$.

(c) Prove *Goursat's theorem*: There is a bijective correspondence between subgroups of $G_1 \times G_2$ and triples (S_1, S_2, φ), where S_i is a section of G_i $(i = 1, 2)$ and $\varphi : S_1 \to S_2$ is an isomorphism. (Recall that a section of a group G is a group L/M, where $M \trianglelefteq L \leq G$.)

Solution. □

Exercise 2.9. (cont.) Use Exercise 2.8 to give a different proof of Proposition 2.10.

Solution. □

Exercise 2.10. Let $G = N \rtimes H$, and suppose that $N \leq K \leq G$. Show that $K = N \rtimes (K \cap H)$.

Solution. □

Further Exercises

Let N and H be groups. An *extension* of N by H is a group E along with a monomorphism $i : N \to E$ and an epimorphism $\pi : E \to H$ such that $i(N) = \ker \pi$ (so that N imbeds in E as a normal subgroup, with the quotient group being isomorphic with H). We shall usually refer to an extension (E, i, π) simply by the group E; however, the nature of the maps i and π are important in distinguishing between extensions. We identify N with its image under i, and H with the quotient of E by N. As an example, let $\varphi : H \to \mathrm{Aut}(N)$ be a homomorphism; then the semidirect product $N \rtimes_\varphi H$ is an extension of N by H in an obvious way, taking i to be the inclusion map sending $n \in N$ to $(n, 1)$ and π to be the projection map sending (n, h) to h.

Exercise 2.11. We say that an extension E of N by H is a *split extension* if there is a homomorphism $t : H \to E$ (called a *splitting map* for the extension) such that $\pi \circ t$ is the identity map on H, in which case $t(H)$ will be a transversal for N in E. Show that E is a split extension iff it is a semidirect product of N by H.

Solution. □

Exercise 2.12. (cont.) Let Q be the quaternion group of order 8. (We can consider Q as the set $\{\pm 1, \pm i, \pm j, \pm k\}$ with multiplication given by the rules $i^2 = j^2 = k^2 = -1$ and $ij = k = -ji$.) Show that Q can be realized as a non-trivial extension in four ways—thrice as an extension of $\mathbb{Z}_4$ by $\mathbb{Z}_2$, and once as an extension of $\mathbb{Z}_2$ by $\mathbb{Z}_2 \times \mathbb{Z}_2$—but that none of these extensions is split. (In other words, Q cannot be written non-trivially as a semidirect product.)

Solution. □

If E is an extension of N by H, then we cannot expect to find a homomorphism $t : H \to E$ such that $t(H)$ will be a transversal for N in E, for if such a t existed then E would be split. However, since $H \cong E/N$, we can always find a set map $t : H \to E$ whose image is a transversal for N; such a map is called a *section* of the extension. Moreover, we can always choose t so that $t(1) = 1$, in which case we say that t is *normalized*. (We use normalized sections instead of arbitrary sections in order to keep the notational complexity to a minimum.)

Exercise 2.13. (cont.) Let t be a normalized section of an extension E. Let $\Psi : E \to \mathrm{Aut}(E)$ be the homomorphism sending an element of E to the corresponding inner automorphism of E. We shall, for $x \in E$, regard $\Psi(x)$ as being an automorphism of N, which is possible since $N \trianglelefteq E$. Define set maps $f : H \times H \to N$ and $\varphi : H \to \mathrm{Aut}(N)$ by

$$f(\alpha, \beta) = t(\alpha)t(\beta)t(\alpha\beta)^{-1},$$

$$\varphi(\alpha) = \Psi(t(\alpha)).$$

We call (f, φ) the *factor pair* arising from t. Show that (f, φ) has the following properties:

1 $f(\alpha, 1) = f(1, \alpha) = 1$ for every $\alpha \in H$, and $\varphi(1)$ is the identity in $\mathrm{Aut}(N)$.

2 $\varphi(\alpha)\varphi(\beta) = \Psi(f(\alpha, \beta))\varphi(\alpha\beta)$ for $\alpha, \beta \in H$.

3 $f(\alpha,\beta)f(\alpha\beta,\gamma) = \varphi(\alpha)(f(\beta,\gamma))f(\alpha,\beta\gamma)$ for $\alpha,\beta,\gamma \in H$.

Solution. □

Exercise 2.14. (cont.) Just as we were able to externalize the notion of semidirect product, so should we be able to externalize the notion of extension; that is, given groups N and H and appropriate additional data, we should be able to construct an extension of N by H. Using Exercise 2.13 as a guide, formulate such an external construction and prove that it works.

Solution. □

We shall return to these ideas in the further exercises to Section 9.

3 Group Actions

Defining group actions

- A (*left*) *action* of G on a set X is a map $G \times X \to X$, the image of (g, x) denoted gx, satisfying $1x = x$ for every $x \in X$ and $(g_1 g_2)x = g_1(g_2 x)$ for all $g_1, g_2 \in G$, $x \in X$.
- Right actions are defined analogously, but in this book virtually all actions considered are left actions.
- If G acts on X, we say X is a G-set; for any $H \leq G$, X is also an H-set by restriction.
- For $H \leq G$, the coset space G/H is a G-set under left multiplication: $(g, xH) \mapsto gxH$.

Proposition 3.1. *There is a natural bijective correspondence between the set of actions of G on a set X and the set of homomorphisms from G to S_X.*

Proof. □

Faithful actions and Cayley's Theorem

- If G has a proper subgroup H with $|G : H| = n$, then the action of G on G/H gives a non-trivial homomorphism $G \to S_n$ via Proposition 3.1; if G is simple, this homomorphism must be injective.
- The action of G on X is *faithful* (or G acts *faithfully*) if the corresponding homomorphism $G \to S_X$ is injective; equivalently, the only $g \in G$ satisfying $gx = x$ for every $x \in X$ is the identity.
- If G acts faithfully on X, we call G a *permutation group* on X, since G is then imbedded isomorphically in S_X.

Theorem (Cayley's Theorem)**.** *G is isomorphic with a subgroup of S_G; in particular, if G is finite with $|G| = n$, then G is isomorphic with a subgroup of S_n.*

Proof. □

G-set homomorphisms and orbits

- If X, Y are G-sets, $\varphi : X \to Y$ is a *G-set homomorphism* if it commutes with the actions: $\varphi(gx) = g\varphi(x)$ for all $g \in G, x \in X$. A bijective such φ is a *G-set isomorphism*; we then write $X \cong Y$.
- For each $x \in X$, the *orbit* of x is $Gx = \{gx \mid g \in G\} \subseteq X$, and the *stabilizer* of x is $G_x = \{g \in G \mid gx = x\} \leq G$.
- Gx is itself a G-set under the induced action; a subset of X is a G-set under the induced action iff it is a union of orbits.

Lemma 3.2. *If X is a G-set, then $G_{gx} = gG_x g^{-1}$ for any $g \in G$ and $x \in X$.*

Proof. □

Transitive actions and the orbit decomposition

- The action of G on X is *transitive* (or G acts *transitively* on X) if there is some $x \in X$ such that $Gx = X$, equivalently, if for any $x_1, x_2 \in X$ there exists some $g \in G$ such that $gx_1 = x_2$.
- A subset of X is a transitive G-set under the induced action iff it consists of a single orbit.
- For $H \leq G$, the action of G on G/H is transitive: $(yx^{-1})xH = yH$.

Proposition 3.3. *Any G-set has a unique partition consisting of transitive G-sets, namely its partition into orbits.*

Proof. □

Classification of transitive G-sets

- Since any G-set is a union of transitive G-sets, classifying G-sets reduces to classifying transitive G-sets.
- Coset spaces are examples of transitive G-sets; we now show every transitive G-set is isomorphic to one.

Proposition 3.4. *If X is a transitive G-set, then $X \cong G/G_x$ as G-sets for any $x \in X$.*

Proof. □

Corollary 3.5 (Orbit-Stabilizer)**.** *Let X be a G-set. Then $Gx \cong G/G_x$ as G-sets for any $x \in X$; in particular, if G is finite, then $|Gx| = |G : G_x|$.*

Proof. □

When are transitive G-sets isomorphic?

- Having decomposed G-sets into transitive ones and identified each with a coset space, the remaining question is when two transitive G-sets are isomorphic.

Lemma 3.6. *Let $\varphi : X \to Y$ be a homomorphism of G-sets, and let $x \in X$. Then $G_x \leq G_{\varphi(x)}$, and if φ is an isomorphism, then $G_x = G_{\varphi(x)}$.*

Proof. □

Proposition 3.7. *If H and K are subgroups of G, then the G-sets G/H and G/K are isomorphic iff H and K are conjugate in G.*

Proof. □

Doubly transitive actions and maximality

- Let X be a G-set. We say X is *doubly transitive* (and G acts *doubly transitively* on X) if whenever (x_1, x_2) and (y_1, y_2) are elements of $X \times X$ with $x_1 \neq x_2$ and $y_1 \neq y_2$, there exists some $g \in G$ such that $gx_1 = y_1$ and $gx_2 = y_2$.
- The natural action of S_n on $\{1, \ldots, n\}$ for $n \geq 2$ is doubly transitive. Any doubly transitive G-set is transitive.
- A proper subgroup H of G is *maximal* if there is no proper subgroup of G that properly contains H. Any subgroup of prime index is necessarily maximal by Theorem 1.6.

Proposition 3.8. *Let G be a group, let X be a doubly transitive G-set, and let $x \in X$. Then G_x is a maximal subgroup of G.*

Proof. □

Blocks and primitivity

- A non-empty subset B of a transitive G-set X is a *block* if B and $gB = \{gx \mid x \in B\}$ are either equal or disjoint for every $g \in G$.
- X itself is always a block, and any one-element subset of X is always a block; the transitive G-set X is *primitive* if these are the only blocks.

Proposition 3.9. *There is a natural bijective correspondence between the set of blocks of a transitive G-set X which contain a given element x and the set of subgroups of G which contain G_x.*

Proof. □

Corollary 3.10. *Let X be a transitive G-set. Then X is primitive iff G_x is a maximal subgroup of G for every $x \in X$.*

Proof. □

Consequences for primitivity

- If X is a transitive G-set, all stabilizers are conjugate by Lemma 3.2; so if G_x is maximal for some x, it is maximal for every x, and Corollary 3.10 can be modified accordingly.

Corollary 3.11. *Any doubly transitive G-set is primitive.*

Proof. □

Counting via group actions

- In the remainder of this section, we apply the theory of group actions to elementary counting results. As before, G denotes an arbitrary group.

Proposition 3.12. *If G is finite and $H, K \leq G$, then*

$$|HK| = \frac{|H||K|}{|H \cap K|}.$$

Proof. □

Centralizers and conjugacy classes

- For $x \in G$, the *centralizer* of x in G is $C_G(x) = \{g \in G \mid gx = xg\} \leq G$.
- For $S \subseteq G$, $C_G(S) = \{g \in G \mid gx = xg \text{ for all } x \in S\} = \bigcap_{x \in S} C_G(x)$, called the centralizer of S in G.
- $Z(G) = C_G(G)$, and $x \in Z(G)$ iff $C_G(x) = G$.
- The *conjugacy class* of $x \in G$ is $\{gxg^{-1} \mid g \in G\}$. By Proposition 1.10, the conjugacy class of $\rho \in S_n$ consists of all elements of S_n having the same cycle structure as ρ.

Proposition 3.13. *The conjugacy classes of G form a partition of G, and if G is finite then an element $x \in G$ has $|G : C_G(x)|$ conjugates in G.*

Proof. □

Normalizers and conjugates of subgroups

- A subgroup of G is normal iff it is a union of conjugacy classes; such a union is in fact disjoint.
- The order of a normal subgroup of a finite group G must be a sum of orders of conjugacy classes of G.
- For $H \leq G$, the *normalizer* of H in G is $N_G(H) = \{g \in G \mid gHg^{-1} = H\}$. We have $N_G(H) \leq G$ and $H \trianglelefteq N_G(H)$; $N_G(H)$ is the largest subgroup of G in which H is normal, so $N_G(H) = G$ iff $H \trianglelefteq G$.

Proposition 3.14. *A subgroup H of a finite group G has exactly $|G : N_G(H)|$ conjugates in G. In particular, the number of conjugates of H in G divides $|G : H|$ and is equal to 1 iff $H \trianglelefteq G$.*

Proof. □

Double cosets

- Let G be a group and let H, K be subgroups of G. A *double coset* of H and K in G is a set $HxK = \{hxk \mid h \in H, k \in K\}$ for some $x \in G$.
- Any two double cosets are either disjoint or equal, as is the case with ordinary cosets.

Proposition 3.15. *If G is finite and $H, K \leq G$, then for any $x \in G$ we have*

$$|HxK| = \frac{|H||K|}{|H \cap xKx^{-1}|} = \frac{|H||K|}{|x^{-1}Hx \cap K|}.$$

Proof. □

Exercises

Exercise 3.1. Show that a finite simple group whose order is at least $r!$ cannot have a proper subgroup of index r.

Solution. □

Exercise 3.2. Show that a group G acts doubly transitively on a set X iff G_x acts transitively on $X - \{x\}$ for every $x \in X$. (Here we must assume that X has more than two elements.)

Solution. □

Exercise 3.3. Show directly from the definitions (that is, without reference to Propositions 3.8 and 3.9) that a doubly transitive G-set is primitive.

Solution. □

Exercise 3.4. Let G be the subgroup of S_5 generated by $(1\ 2\ 3\ 4\ 5)$, and let G act on $X = \{1, 2, 3, 4, 5\}$ in the canonical way. Show that this action is primitive, but not doubly transitive.

Solution. □

Exercise 3.5. Let $N \trianglelefteq G$, and let $y \in N$. Show that the conjugacy class of y in G is a union of conjugacy classes of the group N. Show further that there is a bijective correspondence between the conjugacy classes of N which comprise the conjugacy class of y in G and the cosets of $NC_G(y)$ in G.

Solution. □

Exercise 3.6. Let G be a finite group, and let r be the number of conjugacy classes of G. Show that $|\{(a, b) \in G \times G \mid ab = ba\}| = r|G|$.

Solution. □

Exercise 3.7. Show that $C_G(gxg^{-1}) = gC_G(x)g^{-1}$ for any elements g and x of a group G.

Solution. □

Exercise 3.8. Let $n \geq 5$, and assume that A_n is simple. (A proof of this fact is outlined in Exercise 7.8.) Use Exercise 3.1 to show that S_n has no proper subgroup of index less than n other than A_n.

Solution. □

Further Exercises

Let X be a transitive G-set. A *system of imprimitivity* of X is a partition of X which is permuted by the action of G. Note that a system of imprimitivity of a G-set is itself a G-set.

Exercise 3.9. Show that there is a bijective correspondence between the set of blocks which contain a given element of X and the set of systems of imprimitivity of X. (Observe that when X is finite, this implies that any two elements of X lie in the same number of blocks.)

Solution. □

Exercise 3.10. Suppose that the G-set Y is an epimorphic image of a G-set X. Show that there is a G-set isomorphism between Y and some system of imprimitivity of X.

Solution. □

If X is a G-set, we use $[X]$ to denote the isomorphism class of X.

Exercise 3.11. Let G be a finite group, and let $S(G)$ be the set of isomorphism classes of finite G-sets. Show that we can define sum and product operations on $S(G)$ by $[X]+[Y] = [X \cup Y]$ and $[X][Y] = [X \times Y]$.

Solution. □

Exercise 3.12. (cont.) Let $B(G)$ be the set obtained from $S(G)$ by adjoining formal additive inverses of isomorphism classes, in the same way that $\mathbb{Z}$ is obtained from $\mathbb{N}$ by adjoining the additive inverses of positive integers. (The additive identity here will be the isomorphism class of the empty set.) Show that the operations defined above on $S(G)$ extend to give a commutative ring structure on $B(G)$. This ring $B(G)$ is called the *Burnside ring* of G.

Solution. □

Exercise 3.13. (cont.) Show that any element of $B(G)$ can be written uniquely as a $\mathbb{Z}$-linear combination of isomorphism classes of transitive G-sets.

Solution. □

Exercise 3.14. (cont.) Let $H \leq G$. Show that there is a unique ring homomorphism from $B(G)$ to $\mathbb{Z}$ which sends an isomorphism class $[X]$ to the number of elements of X fixed by H.

Solution. □

Exercise 3.15. (cont.) Show that any non-zero homomorphism from $B(G)$ to $\mathbb{Z}$ arises from the construction in Exercise 3.14 and that the intersection of the kernels of all such homomorphisms is zero.

Solution. □

An element e of a ring R is called an *idempotent* if $e^2 = e$.

Exercise 3.16. (cont.) Show that if G has no self-normalizing proper subgroup, then $B(G)$ has no idempotents other than the additive and multiplicative identities. (A subgroup H of G is called *self-normalizing* if $N_G(H) = H$. We shall see in Section 11 that there is an important class of groups, namely the nilpotent groups, that have no self-normalizing proper subgroups.)

Solution. □

Chapter 2

The General Linear Group

4 Basic Structure

Setup and notation

- F a field, $n \in \mathbb{N}$. $\mathcal{M}_n(F)$ denotes the set of all $n \times n$ matrices with entries in F; we write $M = (m_{ij})$ with $m_{ij} \in F$ the (i, j)-entry (row i, column j).
- The *general linear group* $\mathrm{GL}(n, F)$ is the subset of $\mathcal{M}_n(F)$ of invertible matrices, equivalently those with non-zero determinant. It forms a group under matrix multiplication; I denotes the identity.
- More generally, for a finite-dimensional F-vector space V, $\mathrm{GL}(V)$ is the group of invertible linear transformations of V under composition. Taking $V = F^n$ recovers $\mathrm{GL}(n, F)$ up to obvious isomorphism, and any n-dimensional F-vector space is isomorphic to F^n, so we lose nothing by working with $\mathrm{GL}(n, F)$.
- If F is finite, then $|F| = p^a$ for a prime p and $a \in \mathbb{N}$ (Moore's theorem, 1893). For q a prime power we write $\mathrm{GL}(n, q)$ for $\mathrm{GL}(n, F)$ where F is the unique field of order q.

Proposition 4.1. *Let E be a finite abelian group of exponent p, where p is prime. Then* $\mathrm{Aut}(E) \cong \mathrm{GL}(n, p)$*, where $n \in \mathbb{N}$ is such that $|E| = p^n$.*

(Recall that the exponent of a group is the least common multiple of the orders of its elements.)

Proof. □

Elementary abelian p-groups

- If E is as in Proposition 4.1 with basis $\{x_1, \ldots, x_n\}$ over the field of p elements, then as groups $E = \langle x_1 \rangle \times \cdots \times \langle x_n \rangle$ with each $\langle x_i \rangle$ cyclic of order p.
- Hence any finite abelian group of prime exponent p is isomorphic with a direct product of copies of $\mathbb{Z}_p$. Such groups are called *elementary abelian p-groups.*
- The *rank* of an elementary abelian p-group E is n, where $|E| = p^n$.
- In particular $\mathrm{Aut}(\mathbb{Z}_p) \cong \mathrm{GL}(1, p) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ by Proposition 4.1; this was previously established in Proposition 2.1.

Proposition 4.2. *Let $n \in \mathbb{N}$, and let q be a prime power. Then*

$$|\mathrm{GL}(n, q)| = \prod_{k=1}^{n} (q^n - q^{k-1}) = q^{\frac{n(n-1)}{2}} (q^n - 1) \cdots (q - 1).$$

Proof. □

Fixing notation for the section

- We now fix a field F and some $n \in \mathbb{N}$, and write G instead of $\mathrm{GL}(n, F)$.
- For $M = (m_{ij}) \in \mathcal{M}_n(F)$: the *main diagonal* of M consists of the entries m_{ii} for $1 \leq i \leq n$.
- M is *diagonal* if its only non-zero entries lie on the main diagonal.
- M is *upper triangular* if all entries below the main diagonal are zero, i.e. $m_{ij} = 0$ whenever $i > j$.

Proposition 4.3. *The set B consisting of all invertible upper triangular matrices is a subgroup of G, called the* standard Borel subgroup*.*

Proof. □

More generally, a *Borel subgroup* of G is any conjugate of the standard Borel subgroup B.

Permutation matrices

- A *permutation matrix* is a matrix in which every row and column has a unique non-zero entry, and all non-zero entries equal 1.
- The identity matrix is a permutation matrix; every permutation matrix is obtained from I by switching columns (or rows).
- Every permutation matrix is orthogonal, so its inverse (= transpose) is again a permutation matrix. Hence all $n \times n$ permutation matrices lie in G.

Proposition 4.4. *The set W consisting of all permutation matrices is a subgroup of G, called the* Weyl subgroup*.*

Proof. □

Action on the standard basis

- Let $V_n(F)$ denote the vector space of n-dimensional column vectors with entries in F, and let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be the standard basis.
- Multiplying $\mathbf{v}_i$ on the left by an $n \times n$ matrix M produces the ith column of M; we say that M *sends* $\mathbf{v}_i$ to the ith column of M.

Proposition 4.5. $W \cong S_n$.

Proof. □

Permutation matrices as permutations

- We often implicitly regard a permutation matrix w as the element of S_n sending i to j, where $\mathbf{v}_j$ is the ith column of w.
- For example, the matrix
$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$
corresponds to $(1\ 3)(2\ 4) \in S_4$, and so if w is this matrix then we may write $w(1) = 3$, and so forth.
- Useful observation: if $w(i) = j$, then for any $M \in \mathcal{M}_n(F)$, the jth row of wM equals the ith row of M, and the ith column of Mw equals the jth column of M.

Transvections

- For distinct $1 \leq i, j \leq n$ and $\alpha \in F$, define $X_{ij}(\alpha)$ to be the $n \times n$ matrix whose (k, l)-entry equals α if $(k, l) = (i, j)$ and equals δ_{kl} for all other (k, l).
- For example, $X_{23}(\alpha) \in \mathcal{M}_3(F)$ is the matrix

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{pmatrix}.$$

- These matrices $X_{ij}(\alpha)$, and their conjugates by elements of G, are called *transvections.*

Lemma 4.6. *Let $\alpha, \beta \in F$, and let i and j be distinct.*

(i) $X_{ij}(\alpha)$ *has determinant* 1 *and hence lies in* G.

(ii) If $\alpha \neq 0$, *then* $X_{ij}(\alpha) \in B$ *iff* $i < j$.

(iii) $X_{ij}(\alpha)X_{ij}(\beta) = X_{ij}(\alpha + \beta)$, *and hence* $X_{ij}(\alpha)^{-1} = X_{ij}(-\alpha)$.

(iv) $[X_{ij}(\alpha), X_{jk}(\beta)] = X_{ik}(\alpha\beta)$ *whenever* i, j, k *are distinct.*

(v) If $w \in W$, *then* $wX_{ij}(\alpha)w^{-1} = X_{w(i)\,w(j)}(\alpha)$.

(vi) $X_{ij}(\alpha)$ *sends* $\mathbf{v}_j$ *to* $\mathbf{v}_j + \alpha\mathbf{v}_i$ *and fixes* $\mathbf{v}_k$ *whenever* $k \neq j$.

(vii) If $M \in \mathcal{M}_n(F)$, *then the* i*th row of* $X_{ij}(\alpha)M$ *is equal to the sum of the* i*th row of* M *and* α *times the* j*th row of* M, *and for* $k \neq i$ *the* k*th row of* $X_{ij}(\alpha)M$ *is equal to the* k*th row of* M.

Proof. □

Root subgroups

- For distinct i and j, define $X_{ij} = \{X_{ij}(\alpha) \mid \alpha \in F\}$; by parts (i) and (iii) of Lemma 4.6, this is a subgroup of G.
- The subgroups X_{ij} are called *root subgroups* of G. (Terminology from the theory of Lie algebras.)
- These subgroups feed into the Bruhat Decomposition Theorem, the main result of the section.

Lemma 4.7. *Let* $M \in G$. *Then there is a product* b *of upper triangular transvections such that the following property holds: For each* $1 \leq i \leq n$, bM *has exactly one row, say the* k_i*th row, whose entries in the first* $i - 1$ *columns are zero and which has a non-zero entry in the* i*th column.*

Proof. □

Theorem (Bruhat Decomposition Theorem)**.** $G = BWB$.

Proof. □

We have now expressed G as a union of the double cosets BwB, as w ranges through W. We will now show that this union is disjoint. Again we first need a lemma.

Lemma 4.8. *If* $w_1, w_2 \in W$ *and* $b \in B$ *are such that* $w_1bw_2 \in B$, *then* $w_2 = w_1^{-1}$.

Proof. □

Corollary 4.9. *If* w *and* w' *are distinct elements of* W, *then* BwB *and* $Bw'B$ *are disjoint; consequently,* G *is the disjoint union as* w *runs through* W *of the* $n!$ *double cosets* BwB.

Proof. □

The following somewhat technical observation, which we shall need in the next section, is an immediate consequence of what we have done.

Corollary 4.10. *Let $M \in G$, and let w be the unique element of W such that $M \in BwB$. Then w sends $\mathbf{v}_i$ to $\mathbf{v}_{k_i}$ for each i, where the numbers k_i are as defined in the statement of Lemma 4.7. In particular, if M sends $\mathbf{v}_1$ to $\alpha_1\mathbf{v}_1 + \cdots + \alpha_k\mathbf{v}_k$ where $\alpha_k \neq 0$, then w sends $\mathbf{v}_1$ to $\mathbf{v}_k$.*

Proof. □

We close this section with a result giving a smaller generating set for G than that given by the Bruhat decomposition. Once again, we start with a lemma.

Lemma 4.11. *Let $b \in B$. Then there exists a product t of transvections such that tb is a diagonal matrix having the same main diagonal entries as b.*

Proof. □

Theorem 4.12. *G is generated by the set consisting of all invertible diagonal matrices and all transvections.*

Proof. □

Exercises

Exercise 4.1. Show that $\mathrm{GL}(2,2) \cong S_3$.

Solution. □

Exercise 4.2. (cont.) Construct a monomorphism $\varphi\colon \mathrm{GL}(1,4) \to \mathrm{GL}(2,2)$ that corresponds to the inclusion of A_3 in S_3. (Recall that the field $\mathbb{F}_4$ of 4 elements can be written as $\{0,1,\alpha,\alpha^2\}$, where $\alpha + 1 = \alpha^2$ and $\lambda + \lambda = 0$ for all $\lambda \in \mathbb{F}_4$.) Show further that the extension of φ to a map from $\mathbb{F}_4$ to $\mathcal{M}_2(\mathbb{F}_2)$, where $\mathbb{F}_2 = \{0,1\}$ is the field of 2 elements, is a monomorphism of rings.

Solution. □

Exercise 4.3. (cont.) Construct an explicit monomorphism from $\mathrm{GL}(n,4)$ to $\mathrm{GL}(2n,2)$ for any $n \in \mathbb{N}$.

Solution. □

Exercise 4.4. (cont.) More generally, show for any $n \in \mathbb{N}$ and any prime power q that $\mathrm{GL}(2n,q)$ has a subgroup that is isomorphic with $\mathrm{GL}(n,q^2)$. Will $\mathrm{GL}(n,q^m)$ always have a subgroup isomorphic with $\mathrm{GL}(mn,q)$ for any $m,n \in \mathbb{N}$ and any prime power q?

Solution. □

Exercise 4.5. Show that $\mathrm{GL}(4,2) \cong A_8$.

Solution. □

Exercise 4.6. Let β be a non-trivial automorphism of the field F. Use β to construct an outer automorphism of $\mathrm{GL}(n,F)$. Do all outer automorphisms of $\mathrm{GL}(n,F)$ arise in this way?

Solution. □

Further Exercises

Exercise 4.7. A matrix is said to be *monomial* if each row and column has exactly one non-zero entry. Let N be the subset of $G = \mathrm{GL}(n, F)$ consisting of all monomial matrices. Show that $N \leq G$, that $T = B \cap N$ is the subgroup of G consisting of all diagonal matrices, that $N = N_G(T)$, and that $N = T \rtimes W$.

Solution. □

Let G be a group. Suppose that G has subgroups B and N of G satisfying the following conditions:

- G is generated by B and N.
- $T = B \cap N$ is a normal subgroup of N.
- $W = N/T$ is generated by a finite set S of involutions (elements of order 2). In addition, if for each $w \in W$ we choose some $\dot{w} \in N$ such that $\dot{w}T = w$, then we must have
  - $\dot{s}B\dot{w} \subseteq B\dot{w}B \cup B\dot{s}\dot{w}B$ for any $s \in S$ and $w \in W$,
  - $\dot{s}B\dot{s} \nsubseteq B$ for any $s \in S$.

In this case we say that B and N form a *BN-pair* of G, or that (G, B, N, S) is a *Tits system* (after Jacques Tits). We call B the *Borel subgroup* of G, and $W = N/B \cap N$ the *Weyl group* associated with the Tits system. The *rank* of the Tits system is defined to be $|S|$.

Exercise 4.8. (cont.) Let $G = \mathrm{GL}(n, F)$, let B be the standard Borel subgroup of G, let N be the subgroup of G consisting of all monomial matrices, and let $T = B \cap N$. By Exercise 4.7 above, we know that we can regard $W = N/T$ as being imbedded in G as the group of permutation matrices. (By making this identification, we can replace $\dot{w}$ by w in the above formulas.) Let S be the subset of W consisting of those permutation matrices that are obtained from the identity matrix by switching two adjacent columns. (In other words, if we identify W with S_n as in Proposition 4.5, then S corresponds to the set $\{(1\ 2), (2\ 3), \ldots, (n-1\ n)\}$.) Show that (G, B, N, S) is a Tits system of rank $n - 1$.

Solution. □

Exercise 4.9. (cont.) Let G be a group with a BN-pair. Verify that, for $w \in W$, the set $B\dot{w}B$ is independent of the choice of $\dot{w} \in N$ such that $\dot{w}T = w$. Show that we have a Bruhat decomposition

$$G = \bigcup_{w \in W} BwB$$

in which the union is disjoint, where BwB is taken to mean $B\dot{w}B$ for any $\dot{w} \in N$ with $\dot{w}T = w$.

Solution. □

5 Parabolic Subgroups

Setup and Complete Flags

- Throughout, $G = \mathrm{GL}(n, F)$ for some field F and some $n \in \mathbb{N}$, with $V_n(F)$ the space of column vectors of length n over F.
- Elements of G are viewed as matrices, with respect to the standard basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$, of invertible linear transformations of $V_n(F)$.
- A *complete flag* on $V_n(F)$ is a sequence of subspaces

$$0 \subset V_1 \subset V_2 \subset \cdots \subset V_{n-1} \subset V_n = V_n(F),$$

  denoted $(V_1, \ldots, V_n)$. Since $\subset$ is proper containment, $\dim_F V_i = i$.
- The *standard flag* is given by $V_i = F\mathbf{v}_1 \oplus \cdots \oplus F\mathbf{v}_i = V_{i-1} \oplus F\mathbf{v}_i$ (with $V_0 = 0$).

The G-Action on Complete Flags

- G acts on complete flags by $g(V_1, \ldots, V_n) = (gV_1, \ldots, gV_n)$.
- This action is transitive: any complete flag $(W_1, \ldots, W_n)$ is obtained from the standard flag by the matrix whose ith column is $\mathbf{w}_i \in W_i - W_{i-1}$.
- The stabilizer of the standard flag is precisely the standard Borel subgroup B, so by Lemma 3.2 the Borel subgroups of G are exactly the stabilizers of complete flags on $V_n(F)$.

Upper Unitriangular Matrices and the Decomposition of B

- An upper triangular matrix is *upper unitriangular* if all its main-diagonal entries equal 1, equivalently if it is upper triangular with only eigenvalue 1.
- The set U of all invertible upper unitriangular matrices is a subgroup of B (imitate the proof of Proposition 4.3).
- Let T be the set of diagonal matrices in G. Then $T \leq B$ and $U \cap T = 1$.

Proposition 5.1. $B = U \rtimes T$.

Proof. □

Action of U and T on the Standard Flag

- An element of U sends each $\mathbf{v}_i$ to $\mathbf{v}_i$ plus some element of V_{i-1}; conversely, any matrix with this property lies in U.
- Hence U consists of the matrices stabilizing each V_i and inducing the identity on each quotient V_i/V_{i-1}.
- T consists exactly of the matrices stabilizing each of $F\mathbf{v}_1, \ldots, F\mathbf{v}_n$; we say T is the *common stabilizer* of the $F\mathbf{v}_i$.

Unipotent Elements and Subgroups

- An element of G is *unipotent* if its characteristic polynomial is $(X-1)^n$.
- By Jordan form, any unipotent element of G is conjugate to an element of U.
- A subgroup of G is *unipotent* if all its elements are unipotent; U is unipotent and comprises all unipotent elements of B.

Theorem (Kolchin's Theorem)**.** *Any unipotent subgroup of G is conjugate with a subgroup of U.*

Proof. □

General Flags and Parabolic Subgroups

- A *flag* on $V_n(F)$ is a nested sequence of non-zero subspaces of $V_n(F)$ which terminates in $V_n(F)$ and has no repeated terms; that is, a sequence $(W_1, \dots, W_r)$ with
$$0 \subset W_1 \subset W_2 \subset \cdots \subset W_{r-1} \subset W_r = V_n(F).$$
Since $\subset$ is proper, $r \leq n$; a complete flag is the case $r = n$.
- The G-set of all flags is not transitive: two flags $(W_1, \dots, W_r)$ and $(W_1', \dots, W_s')$ lie in the same orbit iff $r = s$ and $\dim_F W_i = \dim_F W_i'$ for all i (same *dimension sequence*).
- A *parabolic subgroup* of G is the stabilizer of some flag on $V_n(F)$.
- Given a flag $(W_1, \dots, W_r)$ with parabolic stabilizer P, choose subspaces Y_i with $W_i = W_{i-1} \oplus Y_i$. The *unipotent radical* of P is the subgroup $U_P \leq P$ of matrices inducing the identity on each W_i/W_{i-1}; e.g. $U_B = U$.
- A *Levi complement* of U_P is the subgroup $L_P \leq P$ which is the common stabilizer of the Y_i.
- A Levi complement is isomorphic with $\mathrm{GL}(y_1, F) \times \cdots \times \mathrm{GL}(y_r, F)$, where $y_i = \dim_F Y_i = \dim_F W_i - \dim_F W_{i-1}$. (In particular, any two Levi complements of U_P are isomorphic.)
- Taking $Y_i = F\mathbf{v}_i$ gives $L_B = T$, isomorphic with a direct product of n copies of $\mathrm{GL}(1, F) \cong F^\times$.

Proposition 5.2. *If P is a parabolic subgroup of G, then we have $P = U_P \rtimes L_P$, where U_P is the unipotent radical of P and L_P is a Levi complement of U_P. Furthermore, $P = N_G(U_P)$.*

Proof. □

Subflags and Staircase Groups

- A *subflag* of the standard flag $(V_1, \dots, V_n)$ is a flag $(W_1, \dots, W_r)$ in which each $W_i = V_j$ for some j; there are 2^{n-1} such subflags.
- A *staircase group* is a parabolic subgroup of G that stabilizes some subflag of the standard flag.
- Example ($n = 6$, subflag $0 \subset V_2 \subset V_3 \subset V_6$): the staircase group consists of all matrices in $G = \mathrm{GL}(6, F)$ of the form
$$\begin{pmatrix} * & * & * & * & * & * \\ * & * & * & * & * & * \\ 0 & 0 & * & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{pmatrix}.$$
Imagine a "staircase" separating the zero entries from the arbitrary entries.
- Its unipotent radical consists of all matrices in G of the form
$$\begin{pmatrix} 1 & 0 & * & * & * & * \\ 0 & 1 & * & * & * & * \\ 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$
and the Levi complement (with canonical choices $Y_2 = F\mathbf{v}_3$ and $Y_3 = F\mathbf{v}_4 \oplus F\mathbf{v}_5 \oplus F\mathbf{v}_6$) consists of all matrices in G of the form
$$\begin{pmatrix} * & * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 & 0 \\ 0 & 0 & * & 0 & 0 & 0 \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \\ 0 & 0 & 0 & * & * & * \end{pmatrix},$$
isomorphic with $\mathrm{GL}(2, F) \times \mathrm{GL}(1, F) \times \mathrm{GL}(3, F)$.

- In general, the unipotent radical of any staircase group is contained in B.
- The G-set of all flags is partitioned into the orbits of the subflags of the standard flag, since any flag has the same dimension sequence as some subflag of the standard flag. By Lemma 3.2, the parabolic subgroups of G are exactly the conjugates of the staircase groups.

Subgroups Containing B

- Goal: there are exactly 2^{n-1} subgroups of G containing B, namely the staircase groups.

Lemma 5.3. *The only non-zero subspaces of $V_n(F)$ left invariant by B are the subspaces $V_1, \ldots, V_n$ that appear in the standard flag.*

Proof. □

Theorem 5.4. *The only subgroups of G that contain B are the staircase groups.*

Proof. □

Exercises

Let $G = \mathrm{GL}(n, F)$, where $n \geq 2$.

Exercise 5.1. Suppose that $g \in G$ fixes some $0 \neq \mathbf{v} \in V_n(F)$ and induces the identity transformation on $V_n(F)/F\mathbf{v}$. Show that g is conjugate with an element of the root subgroup X_{12}.

Solution. □

Exercise 5.2. We say a basis $\mathcal{B}$ of $V_n(F)$ *belongs* to a complete flag $(V_1, \ldots, V_n)$ if each V_i contains exactly one element of $\mathcal{B}$ that is not in V_{i-1}.

(a) Show that if $(V_1, \ldots, V_n)$ and $(W_1, \ldots, W_n)$ are complete flags on $V_n(F)$, then there is a basis of $V_n(F)$ that belongs to both flags.

(b) Use part (a) to give a different proof of the Bruhat Decomposition Theorem.

Solution. □

Exercise 5.3. Let P and Q be distinct parabolic subgroups of G containing the standard Borel subgroup B of G (so that, by Theorem 5.4, P and Q are staircase groups).

(a) Show that P and Q are not conjugate in G.

(b) If L_P and L_Q are corresponding Levi complements, find conditions on P and Q that determine when L_P and L_Q will be conjugate in G.

Solution. □

Exercise 5.4. Prove Proposition 5.2.

Solution. □

Exercise 5.5. Prove the following generalization of Lemma 5.3: If P is a parabolic subgroup of G which is the stabilizer of the flag $(W_1, \ldots, W_r)$, then the W_i are the only subspaces of $V_n(F)$ left invariant by P.

Solution. □

Exercise 5.6. (cont.) Using Exercise 5.5, show that any parabolic subgroup of G is self-normalizing.

Solution. □

Exercise 5.7. Complete the outlined proof of Theorem 5.4.

Solution. □

Exercise 5.8. Assume that the case $r = 1$ of Theorem 5.4 holds; we sketch an alternate proof of the general case of Theorem 5.4. We use induction on n, where $G = \mathrm{GL}(n, F)$. Let H be a subgroup of G which contains B, and suppose that H stabilizes V_m, where $1 \leq m < n$. Let S be the stabilizer of V_m; observe that $H \leq S$. We can write $S = U \rtimes (K_1 \times K_2)$, where the elements of $K_1 \cong \mathrm{GL}(m, F)$ have their only non-zero entries in the first m rows and columns, and where K_2 is a direct product of general linear groups whose elements have their non-zero entries concentrated in the last $n - m$ rows and columns. Since $U \leq B \leq H$, by Exercise 2.10 we have $H = U \rtimes Q$ for some $Q \leq K_1 \times K_2$. Use Goursat's theorem (Exercise 2.8), Exercise 5.6, and the induction hypothesis to analyze Q and thereby conclude that H is a staircase group.

Solution. □

6 The Special Linear Group

Definition

- The *special linear group* is the subgroup $\mathrm{SL}(n,F)$ of $\mathrm{GL}(n,F)$ consisting of all matrices having determinant 1.
- In other words, $\mathrm{SL}(n,F)$ is the kernel of the homomorphism $\det\colon \mathrm{GL}(n,F) \to F^\times$, and hence $\mathrm{SL}(n,F) \trianglelefteq \mathrm{GL}(n,F)$.

Proposition 6.1. *Let $n \in \mathbb{N}$, and let q be a prime power. Then*

$$|\mathrm{SL}(n,q)| = \prod_{k=1}^{n-1}(q^{n+1} - q^k) = q^{\frac{n(n-1)}{2}}(q^n-1)\cdots(q^2-1).$$

Proof. □

Generation by root subgroups

- We showed in Theorem 4.12 that $\mathrm{GL}(n,F)$ is generated by the root subgroups and the diagonal subgroup. We now establish the analogous result for $\mathrm{SL}(n,F)$.

Proposition 6.2. $\mathrm{SL}(n,F)$ *is generated by the root subgroups X_{ij}.*

Proof. □

Proposition 6.3. *The subgroups X_{ij} are conjugate in* $\mathrm{SL}(n,F)$.

Proof. □

Scalar matrices and the center

- Let Z be the subgroup of $\mathrm{GL}(n,F)$ consisting of the multiples of the identity matrix by elements of $F^\times$. The elements of Z are called the (non-zero) *scalar* matrices. We clearly have $Z \cong F^\times$.

Proposition 6.4. *The center of* $\mathrm{GL}(n,F)$ *is Z, and the center of* $\mathrm{SL}(n,F)$ *is* $Z \cap \mathrm{SL}(n,F)$.

Proof. □

Projective groups

- The group $\mathrm{GL}(n,F)/Z$ is called the *projective general linear group*, and the group $\mathrm{SL}(n,F)/Z \cap \mathrm{SL}(n,F)$ is called the *projective special linear group.*
- We denote these groups by $\mathrm{PGL}(n,F)$ and $\mathrm{PSL}(n,F)$, respectively. (Observe that the first isomorphism theorem implies that $\mathrm{PSL}(n,F) \cong Z\,\mathrm{SL}(n,F)/Z \leq \mathrm{GL}(n,F)/Z = \mathrm{PGL}(n,F)$.)
- As $Z \cong F^\times$, we have $|\mathrm{PGL}(n,q)| = |\mathrm{SL}(n,q)|$ for any prime power q; however, to determine $|\mathrm{PSL}(n,q)|$ we must be able to count the number of nth roots of unity in the field of q elements.

Proposition 6.5. *Let q be a prime power. Then*

$$|\mathrm{PSL}(2,q)| = \begin{cases} q^3 - q & \text{if } 2 \mid q, \\ \frac{1}{2}(q^3-q) & \text{if } 2 \nmid q. \end{cases}$$

Proof. □

Simplicity of $\mathrm{PSL}(n, F)$

- Our main objective in this section is to show that $\mathrm{PSL}(n, F)$ is simple whenever $n \geq 2$, except when $n = 2$ and $|F| = 2$ or 3.
- This result dates back to L. E. Dickson, another of the early architects of the mathematical tradition of the University of Chicago, who established it in his 1896 Ph.D. thesis for F a finite field; in 1893 E. H. Moore had established the result for F a finite field and $n = 2$, while in 1870 Camille Jordan had established the result for F a finite field of prime order and n arbitrary.
- We first need two lemmas.

Lemma 6.6. *If $n \geq 2$, then every transvection $X_{ij}(\alpha)$ is a commutator of elements of $\mathrm{SL}(n, F)$, except when $n = 2$ and $|F| = 2$ or 3.*

Proof. □

Consequence

- Observe that Lemma 6.6 and Proposition 6.2 together imply that, except when $n = 2$ and $|F| = 2$ or 3, $\mathrm{SL}(n, F)$ is its own derived group and is consequently also the derived group of $\mathrm{GL}(n, F)$.

Lemma 6.7. *If $n \geq 2$, then the action of $\mathrm{SL}(n, F)$ on the one-dimensional subspaces of $V_n(F)$ is doubly transitive.*

Proof. □

Theorem 6.8. *If $n \geq 2$, then $\mathrm{PSL}(n, F)$ is simple, except when $n = 2$ and $|F| = 2$ or 3.*

Proof. □

Exercises

Exercise 6.1. Let B be the standard Borel subgroup of $\mathrm{GL}(n, F)$. Determine all subgroups of $\mathrm{SL}(n, F)$ which contain $B \cap \mathrm{SL}(n, F)$.

Solution. □

Exercise 6.2. Show that $\mathrm{PSL}(2, 2) \cong S_3$ and $\mathrm{PSL}(2, 3) \cong A_4$, and that these groups are not simple.

Solution. □

Exercise 6.3. Show that $\mathrm{PSL}(2, 4)$ and $\mathrm{PSL}(2, 5)$ are both isomorphic with A_5.

Solution. □

Exercise 6.4. Show that $\mathrm{PSL}(2, 7) \cong \mathrm{GL}(3, 2)$.

Solution. □

Exercise 6.5. Show that $\mathrm{PSL}(2, 9) \cong A_6$.

Solution. □

Exercise 6.6. By Exercise 6.3 and Theorem 6.8, the group A_5 is simple. Attempt to prove this fact directly by mimicking the proof of Theorem 6.8, with A_5 in place of S, A_4 in place of P, and the Klein four-group in place of K.

Solution. □

Further Exercises

Let F be a field, and let $V = F^2$. We define an equivalence relation on $V - \{0\}$ by $\mathbf{v} \sim \mathbf{w}$ iff $\mathbf{v} = \alpha\mathbf{w}$ for some $\alpha \in F^\times$. The set of equivalence classes under this relation is called the *one-dimensional projective space* (or *projective line*) over F and is denoted by $\mathbb{P}^1(F)$. We write elements of V as column vectors, and the element of $\mathbb{P}^1(F)$ that is the equivalence class of $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ shall be written as $\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right]$.

Exercise 6.7. Show that there is a well-defined action of $\mathrm{PSL}(2, F)$ on $\mathbb{P}^1(F)$ given by

$$\overline{\begin{pmatrix} a & b \\ c & d \end{pmatrix}} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

and that this action is doubly transitive. (Here $\overline{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)}$ denotes the image in $\mathrm{PSL}(2, F)$ of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}(2, F)$.)

Solution. □

Exercise 6.8. (cont.) Determine the proper definition of n-dimensional projective space $\mathbb{P}^n(F)$ for arbitrary n, and show that $\mathrm{PSL}(n + 1, F)$ acts doubly transitively on $\mathbb{P}^n(F)$.

Solution. □

Exercise 6.9. Let F be the field of 7 elements. For $0 \le i \le 6$, let i represent $\left[\begin{smallmatrix} 1 \\ i \end{smallmatrix}\right] \in \mathbb{P}^1(F)$; denote the remaining element $\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$ of $\mathbb{P}^1(F)$ by ∞. View S_8 as being the group of permutations of $\{\infty, 0, 1, 2, 3, 4, 5, 6\}$, and let $\varphi\colon \mathrm{PSL}(2, 7) \to S_8$ be the monomorphism arising (as in Proposition 3.1) from the action of $\mathrm{PSL}(2, 7)$ on $\mathbb{P}^1(F)$. Show that the image of φ is the subgroup of A_8 generated by $(0\ 1\ 2\ 3\ 4\ 5\ 6)$, $(1\ 4\ 2)(3\ 5\ 6)$, and $(\infty\ 0)(1\ 3)(2\ 5)(4\ 6)$. (Hint: Show first that $\mathrm{PSL}(2, 7)$ is generated by the images under the natural map of the elements $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 4 & 0 \\ 0 & 2 \end{smallmatrix}\right)$, and $\left(\begin{smallmatrix} 0 & 3 \\ 2 & 0 \end{smallmatrix}\right)$ of $\mathrm{SL}(2, 7)$.) For the relevance of this exercise, see Exercises 7.13–7.17.

Solution. □

Recall the definition of a BN-pair from the further exercises to Section 4.

Exercise 6.10. Let $G = \mathrm{GL}(n, F)$. Let B be the standard Borel subgroup of G, let N be the subgroup of G of monomial matrices, and let $T = B \cap N$ be the subgroup of diagonal matrices. Let $B_0 = B \cap \mathrm{SL}(n, F)$, let $N_0 = N \cap \mathrm{SL}(n, F)$, and let $T_0 = T \cap \mathrm{SL}(n, F) = B_0 \cap N_0$. Show that B_0 and N_0 form a BN-pair of $\mathrm{SL}(n, F)$, with the associated Weyl group $W_0 = N_0/T_0$ being isomorphic with S_n.

Solution. □

Exercise 6.11. Let $P = \mathrm{PSL}(n, F)$, and let $\overline{B_0}$, $\overline{N_0}$, and $\overline{T_0}$ be the images in P of the subgroups B_0, N_0, and T_0 of $\mathrm{SL}(n, F)$. Show that $\overline{B_0}$ and $\overline{N_0}$ form a BN-pair of P (with again the Weyl group $\overline{W_0} = \overline{N_0}/\overline{T_0}$ being isomorphic with S_n).

Solution. □

Chapter 3

Local Structure

7 Sylow's Theorem

Setup and notation

- Throughout the section, G is a finite group and p a prime divisor of $|G|$.
- $|G|_p$ denotes the highest power of p dividing $|G|$, so $|G|_p = p^n$ with $p^n \mid |G|$ but $p^{n+1} \nmid |G|$.

p-elements, p-groups, p-subgroups

- $g \in G$ is a *p-element* if its order is a power of p.
- G is a *p-group* if $|G|$ is a power of p; $H \leq G$ is a *p-subgroup* if $|H|$ is a power of p.
- Every element of a p-group is a p-element. (For infinite groups, p-group means every element is a p-element.)
- $H \leq G$ is a *Sylow p-subgroup* of G if H is a p-subgroup of order $|G|_p$, the maximal possible order of a p-subgroup of G.
- A subgroup of G is a *Sylow subgroup* if it is a Sylow p-subgroup for some prime divisor p of $|G|$.

Motivating example: $\mathrm{GL}(n, p)$

- For $G = \mathrm{GL}(n, p)$, Proposition 4.2 gives $|G| = p^{\frac{n(n-1)}{2}}(p^n - 1)\cdots(p-1)$, so $|G|_p = p^{\frac{n(n-1)}{2}}$.
- The subgroup U of upper uni-triangular matrices satisfies $|U| = p^{n-1}p^{n-2}\cdots p^2 p = p^{\frac{n(n-1)}{2}} = |G|_p$, hence U is a Sylow p-subgroup.
- Every p-element of $\mathrm{GL}(n, p)$ is unipotent (its characteristic polynomial is $(X-1)^n$), and by Kolchin's Theorem any p-subgroup is conjugate with a subgroup of U.
- In particular: $\mathrm{GL}(n, p)$ has a Sylow p-subgroup, all are conjugate, and every p-subgroup is contained in one. Sylow's Theorem extends this to arbitrary finite G.

Theorem (Sylow's Theorem)**.** *(i) G has at least one Sylow p-subgroup.*
(ii) All the Sylow p-subgroups of G are conjugate.
(iii) Any p-subgroup of G is contained in a Sylow p-subgroup.
(iv) The number of Sylow p-subgroups of G is congruent to 1 *modulo p.*

Proof. □

Corollary 7.1. *The number of Sylow p-subgroups of G divides $|G|/|G|_p$.*

Proof. □

Theorem (Cauchy's Theorem)**.** *G has an element of order p.*

Proof. □

(Other proofs of Sylow's Theorem often assume that Cauchy's Theorem is already known, as was the case historically.)

Sylow subgroups of normal subgroups and quotients

- Sylow p-subgroups behave well with respect to a normal subgroup $N \trianglelefteq G$: both PN/N and $P \cap N$ are Sylow p-subgroups (of G/N and N respectively).
- For a non-normal subgroup $H \leq G$, $P \cap H$ need not be Sylow in H, but some conjugate $gPg^{-1} \cap H$ is.

Proposition 7.2. *Let $N \trianglelefteq G$, and let P be a Sylow p-subgroup of G. Then PN/N is a Sylow p-subgroup of G/N, and $P \cap N$ is a Sylow p-subgroup of N.*

Proof. □

Applications: groups of order pq and simplicity of A_5

- Sylow's theorem classifies groups of order pq (distinct primes $p > q$).
- It also gives a clean count of p-elements via the orbit count of Sylow p-subgroups, used to prove A_5 is simple.

Proposition 7.3. *Let p and q be distinct primes, with $p > q$. If $p \not\equiv 1 \pmod q$, then any group of order pq is isomorphic with $\mathbb{Z}_{pq}$. If $p \equiv 1 \pmod q$, then any abelian group of order pq is isomorphic with $\mathbb{Z}_{pq}$, and there is exactly one isomorphism class of non-abelian groups of order pq.*

Proof. □

Theorem 7.4. *A_5 is simple.*

Proof. □

Theorem 7.5. *Any simple group of order 60 is isomorphic with A_5.*

Proof. □

Corollary 7.6. $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5) \cong A_5$.

Proof. □

It is a relatively easy exercise to show that the only simple groups of order less than 60 are the cyclic groups of prime order, and hence that A_5 is the non-abelian finite simple group having smallest order. The non-abelian finite simple group of next smallest order is $\mathrm{PSL}(2,7)$, which has order 168.

Exercises

Throughout these exercises, G is a finite group and p is a prime divisor of $|G|$.

Exercise 7.1. Let $H \leq G$ and let P be a Sylow p-subgroup of G. Prove, without reference to Sylow's Theorem, that there is some conjugate of P whose intersection with H is a Sylow p-subgroup of H.

Solution. □

Exercise 7.2. (cont.) Use Exercise 7.1 to give an alternate proof of part (i) of Sylow's Theorem.

Solution. □

Exercise 7.3. Give an alternate proof of parts (ii) and (iii) of Sylow's Theorem by considering the action of an arbitrary p-subgroup Q of G on the coset space G/P, where P is a Sylow p-subgroup of G.

Solution. □

Exercise 7.4. Prove the following generalization of part (iv) of Sylow's Theorem: If $|G|$ is divisible by p^b, and $H \leq G$ has order p^a where $a \leq b$, then the number of subgroups of G that both contain H and have order p^b is congruent to 1 modulo p.

Solution. □

Exercise 7.5. Show that if P is a Sylow p-subgroup of G, then $N_G(P)$ is a self-normalizing subgroup of G, meaning that $N_G(N_G(P)) = N_G(P)$. (Any subgroup of G that can be written as $N_G(P)$ for some non-trivial p-subgroup P of G is said to be a *p-local subgroup* of G.)

Solution. □

Exercise 7.6. Given a prime p, find an example of a finite group having exactly $1+p$ Sylow p-subgroups. Can this be done for $1+2p$? $1+3p$?

Solution. □

Exercise 7.7. Show that if G has exactly $1+kp$ Sylow p-subgroups for some $k \in \mathbb{N}$, then there is a subgroup of S_{1+kp} having exactly $1+kp$ Sylow p-subgroups.

Solution. □

Exercise 7.8. Let $n \geq 6$, and assume by induction that A_{n-1} is simple. Let N be a non-trivial proper normal subgroup of A_n.

(a) Show that $A_n = N \rtimes A_{n-1}$.

(b) Let $N^\#$ be the set of non-identity elements of N, and consider the action of A_{n-1} on $N^\#$ given by the conjugation homomorphism from A_{n-1} to $\mathrm{Aut}(N)$. Show that $N^\#$ is isomorphic as an A_{n-1}-set to $\{1, \ldots, n-1\}$.

(c) Show that A_{n-1} acts triply transitively on $N^\#$. (An action of a group G on a set X is *triply transitive* if for every pair of triples (x_1, x_2, x_3) and (y_1, y_2, y_3), where $x_i, y_i \in X$ for all i and $x_i \neq x_j$ (resp., $y_i \neq y_j$) when $i \neq j$, there is some $g \in G$ such that $gx_i = y_i$ for all i.)

(d) Derive a contradiction, and conclude that A_n is simple.

Solution. □

The remaining exercises develop a proof that if G is a finite simple group with $|G| \leq 200$, then either $G \cong \mathbb{Z}_p$ for some prime p, or $G \cong A_5$, or $G \cong \mathrm{PSL}(2,7)$. This was established by Hölder in 1892. We will need to assume two facts, which will be proved in later sections:

A. If $|G| = p^n$ where $n > 1$, then G is not simple (Section 8).

B. If G is a p-group and $H < G$, then $H < N_G(H)$ (Section 11).

Exercise 7.9. Show that G is not simple whenever one of the following statements is true, where p is an odd prime:

(a) $|G| = 2^m p^n$, where $2^k \not\equiv 1 \pmod{p}$ for any $1 \leq k \leq m$.

(b) $|G| = p^n q$, where $q \neq p$ is prime and $q \not\equiv 1 \pmod{p}$.

(c) $|G| = ap$, where $p \nmid a$ and $kp+1 \nmid a$ for any $k \in \mathbb{N}$.

Solution. □

Exercise 7.10. Show that G is not simple whenever one of the following statments is true:

(a) $|G| = p^a(p+1)$, where $a > 1$.

(b) $|G| = p^a(p+3)$, where $a > 3$ if $p = 2$ and $a > 1$ if $p > 3$.

(c) $|G| = p^a(p^2-1)$, where $a > 1$ and p is odd.

Solution. □

Exercise 7.11. Show that if $|G| \in \{30, 56, 105, 132\}$, then G is not simple.

Solution. □

Exercise 7.12. Show that G is not simple in the following cases:

(a) $|G| = 90$. (HINT: Show that if Q is a Sylow 3-subgroup of G and $1 \neq x \in Q$, then $C_G(x) = Q$.)

(b) $|G| = 112$. (HINT: Use fact B above to show that any two Sylow 2-subgroups of G intersect trivially.)

(c) $|G| = 120$. (HINT: Show that G imbeds in S_6, and contradict Exercise 3.8.)

(d) $|G| = 144$. (HINT: Argue as in part (i) above.)

(e) $|G| = 180$. (HINT: Show that if Q is a Sylow 3-subgroup of G and $1 \neq x \in Q$, then $|C_G(x) : Q| \leq 2$.)

Solution. □

From fact A above and Exercises 9–12, we can conclude that if G is a simple group with $|G| \leq 200$, then $|G|$ either is prime or is equal to either 60 or 168. (Verify this.) In light of Theorem 7.5 and Theorem 6.8, all we need now show is that any two simple groups of order $168 = 2^3 \cdot 3 \cdot 7$ are isomorphic. We accomplish this in Exercises 13–17 below by showing that any simple group of order 168 is isomorphic with a specific subgroup of A_8. (Compare with Exercise 6.9.) Let G be a simple group of order 168, let $P = \langle x \rangle$ be a Sylow 7-subgroup of G, and let $H = N_G(P)$.

Exercise 7.13. Show that $H = \{g \in G \mid gxg^{-1} \in P\}$ is a non-abelian group of order 21 which is generated by x and y, where y has order 3 and $yxy^{-1} = x^4$.

Solution. □

Exercise 7.14. (cont.) Show that $N_G(\langle y \rangle)$ is isomorphic with S_3 and is generated by y and z, where z has order 2 and $yzy = z$.

Solution. □

Exercise 7.15. (cont.) Show that $G/H = \{H, zH, xzH, \ldots, x^6zH\}$. Conclude that $G = \langle x, y, z \rangle$.

Solution. □

Let $\varphi\colon G \to S_8$ be the monomorphism corresponding to the action of G on the coset space G/H. Observe that $\varphi(G) = \varphi(G)' \leq S_8' \leq A_8$. View S_8 as the group of permutations of the set $\{\infty, 0, 1, 2, 3, 4, 5, 6\}$; for each $0 \leq i \leq 6$, associate i with the coset $x^i zH$, and associate ∞ with the coset H.

Exercise 7.16. (cont.) Show that $\varphi(x) = (0\,1\,2\,3\,4\,5\,6)$, that $\varphi(y) = (1\,4\,2)(3\,5\,6)$, and that $\varphi(z) = z_i$ for some i, where $z_1 = (\infty\,0)(1\,3)(2\,5)(4\,6)$, $z_2 = (\infty\,0)(1\,5)(2\,6)(3\,4)$, and $z_3 = (\infty\,0)(1\,6)(2\,3)(4\,5)$.

Solution. □

Exercise 7.17. (cont.) Let $L_i = \langle \varphi(x), \varphi(y), z_i \rangle \leq A_8$ for $i = 1, 2, 3$. Show that $L_1 = L_2 = L_3$. Conclude that G is uniquely determined up to isomorphism.

Solution. □

8 Finite p-groups

Why study finite p-groups?

- Sylow's Theorem reduces many questions about a finite group to questions about its subgroups of prime-power order.
- Finite p-groups form a rich subject in their own right; this section only scratches the surface.

The center and maximal subgroups

- The center of a non-trivial finite p-group is non-trivial (Theorem 8.1); more generally, every non-trivial normal subgroup meets $Z(P)$ non-trivially.
- Consequently, the only finite simple p-groups are the groups $\mathbb{Z}_p$ of prime order.
- For non-abelian G, the quotient $G/Z(G)$ is never cyclic (Lemma 8.2); in particular $|P : Z(P)| \neq p$.
- Every maximal subgroup of a finite p-group is normal of index p (Proposition 8.3, Corollary 8.4).

Abelian p-groups and the basis theorem

- Every finite abelian p-group is a direct product of cyclic p-groups (Theorem 8.6).
- The non-generators of a cyclic p-group are exactly its pth powers (Lemma 8.5).
- Combining Theorems 8.6 and 8.7 yields the *basis theorem*: every finite abelian group is a direct product of cyclic p-groups (Corollary 8.8).

Nilpotent groups and the Frattini argument

- The Frattini Argument states that $G = N_G(P)N$ whenever P is a Sylow subgroup of a normal subgroup $N \trianglelefteq G$.
- A finite group G is *nilpotent* if every Sylow subgroup of G is normal; equivalently, G is the direct product of its Sylow subgroups, equivalently every maximal subgroup is normal (Theorem 8.7).
- Finite abelian groups and finite p-groups are nilpotent; an alternative definition of nilpotence which extends to infinite groups is taken up in Section 11.

Groups of small order

- Any group of order p is isomorphic with $\mathbb{Z}_p$.
- Any group of order p^2 is abelian (by Theorem 8.1 and Lemma 8.2) and is isomorphic with either $\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.
- Any abelian group of order p^3 is isomorphic with one of $\mathbb{Z}_{p^3}$, $\mathbb{Z}_{p^2} \times \mathbb{Z}_p$, or $\mathbb{Z}_p \times \mathbb{Z}_p \times \mathbb{Z}_p$ (by Theorem 8.6).
- For odd primes p, there are exactly two isomorphism classes of non-abelian groups of order p^3: one having an element of order p^2 (Proposition 8.10) and one with no such element (Proposition 8.11).
- Lemma 8.9 provides the technical commutator identities supporting the classification.
- For $p = 2$, the non-abelian groups of order 8 are the dihedral group D_8 and the *quaternion group*; the latter is the unique group of order p^3 that cannot be written non-trivially as a semidirect product.

Theorem 8.1. *If P is a non-trivial finite p-group, then $Z(P)$ is also non-trivial. Moreover, if $1 \neq N \trianglelefteq P$, then $N \cap Z(P) \neq 1$.*

Proof. □

Lemma 8.2. *If G is a non-abelian group, then $G/Z(G)$ cannot be cyclic. In particular, if P is a finite p-group, then $|P : Z(P)| \neq p$.*

Proof. □

Proposition 8.3. *If P is a finite p-group, then every maximal subgroup of P is normal in P.*

Proof. □

Corollary 8.4. *Any maximal subgroup of a finite p-group is of index p.*

Proof. □

Lemma 8.5. *Every non-generator of a cyclic p-group P is a pth power in P.*

Proof. □

Theorem 8.6. *A finite abelian p-group is a direct product of cyclic p-groups.*

Proof. □

Theorem (Frattini Argument)**.** *Let G be a finite group, let $N \trianglelefteq G$, and let P be a Sylow subgroup of N. Then $G = N_G(P)N$.*

Proof. □

Theorem 8.7. *The following statements about a finite group G are equivalent:*

(1) Every Sylow subgroup of G is normal in G.

(2) G is the direct product of its Sylow subgroups.

(3) Every maximal subgroup of G is normal in G.

Proof. □

Corollary 8.8. *A finite abelian group is a direct product of cyclic p-groups.*

Proof. □

Lemma 8.9. *Let G be a group and let $x, y \in G$. If $[x, y] \in Z(G)$, then $[x^n, y] = [x, y]^n$ and $x^n y^n = (xy)^n [x, y]^{\frac{n(n-1)}{2}}$ for any $n \in \mathbb{N}$.*

Proof. □

Proposition 8.10. *If p is an odd prime, then there is exactly one isomorphism class of non-abelian groups of order p^3 having elements of order p^2.*

Proof. □

Proposition 8.11. *If p is an odd prime, then there is exactly one isomorphism class of non-abelian groups of order p^3 having no elements of order p^2.*

Proof. □

Exercises

Exercise 8.1. Show that if P is a non-cyclic finite p-group, then P has a normal subgroup N such that $P/N \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

Solution. □

Exercise 8.2. Let P be a group of order p^n. Show that P has a normal subgroup N_a of order p^a for every $0 \leq a \leq n$, and that these subgroups can be chosen so that N_a is contained in N_b whenever $a \leq b$.

Solution. □

Exercise 8.3. Let $G = \mathrm{GL}(n, p)$, and let P be a Sylow p-subgroup of G. What is the order of $Z(P)$? What is the order of $Z(P/Z(P))$? If we let $Z_2(P) \leq P$ be such that $Z_2(P)/Z(P) = Z(P/Z(P))$ and continue in this way, what happens?

Solution. □

Exercise 8.4. Let U be the subgroup of $\mathrm{GL}(n, p)$ consisting of the upper unitriangular matrices, and let Q be the subgroup of U consisting of matrices whose (i, j)-entry is zero whenever $1 < i < j < n$. Determine $Z(Q)$, and show that $Q/Z(Q)$ is abelian.

Solution. □

Exercise 8.5. Show that subgroups and quotient groups of finite nilpotent groups are nilpotent, and that direct products of finite nilpotent groups are nilpotent.

Solution. □

Exercise 8.6. Let U be the subgroup of $\mathrm{GL}(3, p)$ consisting of the upper unitriangular matrices. Show that if p is an odd prime, then U is a non-abelian group of order p^3 having no elements of order p^2. If $p = 2$, with which group of order 8 is U isomorphic?

Solution. □

Further Exercises

Exercise 8.7. Show that a finite group G has a largest nilpotent normal subgroup, in the sense that it contains all nilpotent normal subgroups of G. (This subgroup is called the *Fitting subgroup* of G.)

Solution. □

The intersection of all maximal subgroups of a finite group G is called the *Frattini subgroup* of G and is denoted by $\Phi(G)$.

Exercise 8.8. Show that $\Phi(G)$ is a nilpotent normal subgroup of G.

Solution. □

Exercise 8.9. Show that $g \in \Phi(G)$ iff whenever $G = \langle S \rangle$ and $g \in S$, then $G = \langle S - \{g\} \rangle$.

Solution. □

Exercise 8.10. Show that if P is a finite p-group, then $P/\Phi(P)$ is an elementary abelian p-group.

Solution. □

9 The Schur-Zassenhaus Theorem

Setting and terminology

- This section is devoted to another of the basic results of finite group theory, Schur–Zassenhaus Theorem.
- A *complement* to a normal subgroup N of a group G is a subgroup H of G such that $G = N \rtimes H$.
- A subgroup H of a finite group G is called a *Hall subgroup* if $|H|$ and $|G : H|$ are coprime.

Theorem (Schur–Zassenhaus Theorem)**.** *Any normal Hall subgroup of a finite group has a complement.*

Proof. □

Cohomological framework

- We now wish to fit some of the ideas developed in the proof of Schur–Zassenhaus Theorem into a more general framework.
- Let H be an arbitrary group, and let A be an abelian group written additively. Suppose that we have an action of H as automorphisms of A, or equivalently a homomorphism from H to $\mathrm{Aut}(A)$.
- A function $f : H \times H \to A$ satisfying the cocycle identity given in the above proof is called a 2-*cocycle* of the pair (H, A).
- The set of all 2-cocycles of (H, A) is denoted by $Z^2(H, A)$, and if we define $(f + g)(h_1, h_2) = f(h_1, h_2) + g(h_1, h_2)$ for $f, g \in Z^2(H, A)$ and $h_1, h_2 \in H$, then $Z^2(H, A)$ becomes an abelian group.
- A 2-cocycle is called a 2-*coboundary* if there is a function $c : H \to A$ such that $f(h_1, h_2) = c(h_1) + {}^{h_1}c(h_2) - c(h_1h_2)$ for all $h_1, h_2 \in H$.
- The set of 2-coboundaries of (H, A) is denoted by $B^2(H, A)$ and is a subgroup of $Z^2(H, A)$.
- The quotient group $Z^2(H, A)/B^2(H, A)$ is called the *second cohomology group* of (H, A) and is denoted by $H^2(H, A)$. (This terminology comes from algebraic topology. We shall define the nth cohomology group of (H, A) for any $n \in \mathbb{N}$ in the further exercises to Section 12.)
- Now suppose that A is an abelian normal Hall subgroup of a finite group G. We showed during the proof of Schur–Zassenhaus Theorem that any 2-cocycle f of $(G/A, A)$ was a 2-coboundary, or equivalently that $H^2(G/A, A) = 0$. (The difference of sign between the definition of 2-coboundary given above and what was established in the proof is irrelevant.) Furthermore, we showed that this fact implied the existence of a complement to A in G. We attempt to make this connection between the second cohomology group of $(G/A, A)$ and the existence of complements to A in G more transparent in the further exercises below.
- A stronger version of Schur–Zassenhaus Theorem asserts that if N is a normal Hall subgroup of a finite group G, then not only does N have a complement in G, but all such complements are conjugate in G. To prove this, one needs the concept of solvable groups, which we shall introduce in Section 11; see [22, pp. 246–8] for further details.

Exercises

Exercise 9.1. Suppose that G is a finite group such that $G = N \rtimes H$, where H is abelian. Show that if $|N|$ and $|H|$ are relatively coprime, then all complements to N in G are conjugate.

Solution. □

Exercise 9.2. Repeat Exercise 9.1 under the assumption that H is nilpotent rather than abelian.

Solution. □

Further Exercises

These exercises are a continuation of the further exercises to Section 2. Let N and H be groups. A *factor pair* of N by H is a pair (f, φ) of set maps $f : H \times H \to N$ and $\varphi : H \to \mathrm{Aut}(N)$ satisfying properties **1**, **2**, and **3** listed on page 27. Let $\mathcal{E}$ be the set of extensions of N by H, and let $\mathcal{F}$ be the set of factor pairs of N by H. In what follows, we will always use "extension" to mean an element of $\mathcal{E}$, and "factor pair" to mean an element of $\mathcal{F}$.

Exercise 9.3. Let (f, φ) be a factor pair, and define

$$(x, \alpha) \cdot (y, \beta) = (x\varphi(\alpha)(y)f(\alpha, \beta), \alpha\beta)$$

for $(x, \alpha), (y, \beta) \in N \times H$. Show that this gives a group structure on $N \times H$; call this group $E_{f,\varphi}$. Show further that $(E_{f,\varphi}, i, \pi)$ is an extension, where $i(x) = (x, 1)$ and $\pi(x, \alpha) = \alpha$, and that (f, φ) is the factor pair arising from some normalized section of $E_{f,\varphi}$. (Observe that this construction generalizes the notion of external semidirect product.)

Solution. □

We have seen in Exercise 2.13 that an extension gives rise to a factor pair via a choice of normalized section, and we have just given an explicit construction of an extension from a given factor pair. We view these processes as giving maps between $\mathcal{E}$ and $\mathcal{F}$, and we now investigate the relationship between these maps. We must first consider the relation between factor pairs arising from different normalized sections of the same extension.

Exercise 9.4. (cont.) Suppose that t and u are normalized sections of an extension E, and let (f, φ) and (g, ρ) be the factor pairs arising from t and u, respectively. Let $c : H \to N$ be the set map such that $u(\alpha) = c(\alpha)t(\alpha)$ for every $\alpha \in H$. Show that the following properties hold:

4 $\rho(\alpha) = \Psi(c(\alpha))\varphi(\alpha)$ for $\alpha \in H$, where $\Psi(c(\alpha))$ is the inner automorphism of N corresponding to $c(\alpha)$.

5 $g(\alpha, \beta) = c(\alpha)\varphi(\alpha)(c(\beta))f(\alpha, \beta)c(\alpha\beta)^{-1}$ for $\alpha, \beta \in H$.

Solution. □

The above exercise motivates the following definition: We say that two factor pairs (f, φ) and (g, ρ) are *equivalent* if there is a map $c : N \to H$ such that properties **4** and **5** hold. (Verify that this is an equivalence relation on $\mathcal{F}$.) Let $\overline{\mathcal{F}}$ denote the set of equivalence classes of factor pairs; we will use $[f, \varphi]$ to denote the class of the factor pair (f, φ). We have a well-defined map from $\mathcal{E}$ to $\overline{\mathcal{F}}$ which sends an extension to the class of a factor pair arising from any normalized section. We must now consider what happens when we pass from $\overline{\mathcal{F}}$ back to $\mathcal{E}$ via the construction in Exercise 9.3. Here we will need to recall the exact definition of an extension.

Exercise 9.5. (cont.) Let (f, φ) and (g, ρ) be factor pairs, with $[f, \varphi] = [g, \rho]$. Let $i : N \to E_{f,\varphi}$ and $j : N \to E_{g,\rho}$ be the natural inclusions (of N into the underlying set $N \times H$), and let $\pi : E_{f,\varphi} \to H$ and $\tau : E_{g,\rho} \to H$ be the natural projections (of the underlying set $N \times H$ onto H). Show that there is an isomorphism $\xi : E_{f,\varphi} \to E_{g,\rho}$ such that $\xi \circ i = j$ and $\tau \circ \xi = \pi$.

Solution. □

Motivated by the above exercise, we say that two extensions (E, i, π) and (F, j, τ) are *equivalent* if there is an isomorphism $\xi : E \to F$ such that $\xi \circ i = j$ and $\tau \circ \xi = \pi$. (Verify that this gives an equivalence relation on $\mathcal{E}$.) We let $\overline{\mathcal{E}}$ denote the set of equivalence classes of extensions, and we let $[E]$ denote the class of an extension E. In this context, Exercise 9.5 asserts that there is a well-defined map from $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}}$, sending $[f, \varphi]$ to $[E_{f,\varphi}]$.

Exercise 9.6. (cont.) If p is an odd prime, show that $\mathbb{Z}_{p^2}$ can be realized in $p - 1$ nonequivalent ways as an extension of $\mathbb{Z}_p$ by $\mathbb{Z}_p$.

Solution. □

Exercise 9.7. (cont.) We have already obtained a map from $\mathcal{E}$ to $\overline{\mathcal{F}}$, sending an extension E to the class of the factor pair arising from any normalized section of E. Show that this map induces a map from $\overline{\mathcal{E}}$ to $\overline{\mathcal{F}}$.

Solution. □

Exercise 9.8. (cont.) Show that the map from $\overline{\mathcal{E}}$ to $\overline{\mathcal{F}}$ obtained in Exercise 9.7 is inverse to the map from $\overline{\mathcal{F}}$ to $\overline{\mathcal{E}}$ sending $[f,\varphi]$ to $[E_{f,\varphi}]$. Conclude that there is a bijective correspondence between the set of equivalence classes of extensions and the set of equivalence classes of factor pairs.

Solution. □

Exercise 9.8 implies that in order to study extensions up to equivalence, it suffices to study equivalence classes of factor pairs. The next two exercises give a slight refinement of the correspondence just obtained.

Exercise 9.9. (cont.) Let $\eta : \mathrm{Aut}(N) \to \mathrm{Out}(N)$ be the natural map. Show that if (f,φ) and (g,ρ) are any two factor pairs arising from an extension E, then $\eta \circ \varphi = \eta \circ \rho$, and this map from H to $\mathrm{Out}(N)$ (which we denote by ψ_E) is a homomorphism. Conclude that there is a well-defined map from $\mathcal{E}$ to the set of homomorphisms from H to $\mathrm{Out}(N)$, sending E to ψ_E.

Solution. □

Exercise 9.10. (cont.) Let $\psi : H \to \mathrm{Out}(N)$ be a given homomorphism. Show that there is a bijective correspondence between the set of classes $[E]$ of $\overline{\mathcal{E}}$ for which $\psi_E = \psi$ and the set of classes $[f,\varphi]$ of $\overline{\mathcal{F}}$ for which $\eta \circ \varphi = \psi$, where $\eta : \mathrm{Aut}(N) \to \mathrm{Out}(N)$ is the natural map.

Solution. □

We now consider the case where the group N is abelian; we write A instead of N, and we will use additive notation for A. Observe that $\mathrm{Out}(A) = \mathrm{Aut}(A)$. We fix a homomorphism $\varphi : H \to \mathrm{Aut}(A)$, and we write ${}^x a$ in lieu of $\varphi(x)(a)$ for $x \in H$ and $a \in A$. We would like to study those equivalence classes $[E]$ of extensions of N by H for which $\psi_E = \varphi$; we say that such extensions *respect* the action of H on A. By Exercise 9.10, it suffices to study equivalence classes $[f,\varphi]$ of factor pairs of A by H. We suppress φ in our notation, so that we are studying functions $f : H \times H \to A$ such that $f(x,1) = f(1,x) = 0$ for all $x \in H$ and which in addition satisfy

$$f(x,y) + f(xy,z) = {}^x f(y,z) + f(x,yz)$$

for all $x,y,z \in H$, with two such functions f and g being equivalent if there is a map $c : H \to A$ such that $c(1) = 0$ and

$$g(x,y) = f(x,y) + c(x) + {}^x c(y) - c(xy)$$

for all $x,y \in H$. As discussed in the section, the set $H^2(H,A)$ of equivalence classes of such functions forms an abelian group that is called the second cohomology group of (H,A). It now follows from Exercise 9.10 that there is a bijective correspondence between the group $H^2(H,A)$ and the set of equivalence classes of extensions of A by H which respect the action of H on A. In particular, if $H^2(H,A) = 0$, then every extension of A by H is split.

Exercise 9.11. (cont.) Suppose that H and A are both finite. Show that the order of each element of $H^2(H,A)$ divides both $|H|$ and the exponent of A. (This implies that $H^2(G/A,A) = 0$ when A is an abelian normal Hall subgroup of a finite group G, which we established in proving Schur–Zassenhaus Theorem.)

Solution. □

Chapter 4

Normal Structure

10 Composition Series

Composition Series

- A series of subgroups $G = G_0 > G_1 > \ldots > G_r = 1$ of a group G is a *composition series* of G if $G_{i+1} \trianglelefteq G_i$ for every i and if each *successive quotient* G_i/G_{i+1} is simple.
- The composition series above is said to have *length* r.
- The successive quotients of a composition series are called the *composition factors* of the series.
- More generally, a group is said to be a *composition factor* of G if it is isomorphic with one of the composition factors in some composition series of G.
- Example: $S_5 > A_5 > 1$ is the only composition series of S_5 (since A_5 is simple by Theorem 7.4 and is the only non-trivial proper normal subgroup of S_5 by Theorem 7.5), with composition factors $\mathbb{Z}_2$ and A_5.

Maximal Normal Subgroups

- N is a *maximal normal subgroup* of G if $N \trianglelefteq G$ and there is no proper normal subgroup of G that properly contains N.
- By Correspondence Theorem, N is a maximal normal subgroup of G iff G/N is simple.
- A maximal subgroup that is also normal is clearly a maximal normal subgroup (and must be of prime index), but a maximal normal subgroup need not be a maximal subgroup, for it could be properly contained in a proper subgroup that is not normal.
- For example, $A_5 \times \mathbb{Z}_2$ has $1 \times \mathbb{Z}_2$ as a maximal normal subgroup that is not maximal.

Proposition 10.1. *Finite groups have composition series.*

Proof. $\square$

- Infinite groups need not have composition series. For example, using Theorem 1.5 we see that every non-trivial subgroup of the infinite cyclic group $\mathbb{Z}$ is isomorphic with $\mathbb{Z}$; as $\mathbb{Z}$ is not simple, $\mathbb{Z}$ has no simple subgroups, and hence we cannot construct a composition series of $\mathbb{Z}$, since the last non-trivial term of such a series must be a simple subgroup of $\mathbb{Z}$.

Lemma 10.2. *Let G be a group that has a composition series, and let $N \trianglelefteq G$. Then N has a composition series.*

Proof. $\square$

Equivalence of Composition Series

- Let $G = G_0 > G_1 > \ldots > G_r = 1$ be a composition series, and suppose $G = H_0 > H_1 > \ldots > H_r = 1$ is another composition series of the same length r. These series are *equivalent* if there is some $\rho \in S_r$ such that $G_{i-1}/G_i \cong H_{\rho(i)-1}/H_{\rho(i)}$ for every i.
- Example: $G = \langle x \rangle \cong \mathbb{Z}_6$ with $G_1 = \langle x^2 \rangle$, $H_1 = \langle x^3 \rangle$ gives equivalent composition series $G > G_1 > 1$ and $G > H_1 > 1$, since $G/G_1 \cong H_1/1 \cong \mathbb{Z}_2$ and $G_1/1 \cong G/H_1 \cong \mathbb{Z}_3$. (Here $\rho = (1\ 2) \in S_2$.)
- Up to equivalence, a group has at most one composition series, so a group having a composition series has a well-defined collection of composition factors.
- Since composition factors are simple groups, classifying finite simple groups (completed in 1980) gives comprehensive understanding of finite groups.

Theorem (Jordan–Hölder Theorem)**.** *Suppose that G is a group that has a composition series. Then any two composition series of G have the same length and are equivalent.*

Proof. □

Subnormal, Normal, and Chief Series

- A series of subgroups $G = G_0 \geq G_1 \geq \ldots \geq G_r = 1$ of a group G is called a *subnormal series* of G if $G_{i+1} \trianglelefteq G_i$ for every i.
- A subnormal series is called a *normal series* if $G_i \trianglelefteq G$ for every i.
- Some authors use "normal series" for what we call a subnormal series, and "invariant series" for what we call a normal series.
- Composition series are examples of subnormal series, but subnormal series can have successive quotients that are trivial or that are non-trivial but not simple.
- Two subnormal series of the same length are said to be *equivalent* if they satisfy the condition stated as the definition of equivalence of composition series.
- A *refinement* of a subnormal series is one obtained by interposing additional terms; it is *proper* if at least one of the interposed terms was not already present.
- A composition series is a subnormal series that has no repeated terms and does not admit a proper refinement.
- A *chief series* of G is a normal series of G with no repeated terms and with the additional property that no normal subgroup of G is contained properly between any two terms of the series.
- Analogy: A composition series is a subnormal series having no subnormal series as a proper refinement, and a chief series is a normal series having no normal series as a proper refinement.
- The successive quotients of a chief series are its *chief factors*; a group is a *chief factor* of G if it is isomorphic with a chief factor in some chief series of G.
- The analogue of the Jordan–Hölder Theorem for chief series holds—namely, any two chief series of a group have the same length and are equivalent, with a proof virtually identical to that for composition series. This is a special case of the Jordan–Hölder theorem for groups with operators (see [26, Section 2.3]).

Minimal Normal Subgroups

- N is a *minimal normal subgroup* of G if $1 \neq N \trianglelefteq G$ and there is no non-trivial normal subgroup of G that is properly contained in N.
- Any non-trivial finite group has minimal normal subgroups.
- A simple group has a unique minimal normal subgroup, namely itself.

Proposition 10.3. *Finite groups have chief series.*

Proof. □

- By definition, the composition factors of finite groups are simple groups. We now turn to determining the nature of chief factors of finite groups; the first step is to rephrase the problem in different terms.

Lemma 10.4. *Let G be a group having a chief series. Then every chief factor of G is a minimal normal subgroup of a quotient group of G.*

Proof. □

Theorem 10.5. *A minimal normal subgroup of a finite group is a direct product of mutually isomorphic simple groups.*

Proof. □

Corollary 10.6. *A chief factor of a finite group is a direct product of mutually isomorphic simple groups.*

Proof. □

Exercises

Exercise 10.1. Show that an abelian group has a composition series iff it is finite.

Solution. □

Exercise 10.2. Show that $\mathrm{GL}(n, F)$ has a composition series iff F is finite.

Solution. □

Exercise 10.3. Using Third Isomorphism Theorem (Exercise 2.14), prove the *Schreier refinement theorem*: Any two subnormal series of a given group have equivalent refinements.

Solution. □

Exercise 10.4. (cont.) Use the Schreier refinement theorem to give an alternate proof of Jordan–Hölder Theorem.

Solution. □

Exercise 10.5. Determine all composition series and all chief series of S_n for $n \geq 2$. (Hint: Use Exercise 3.8.)

Solution. □

Exercise 10.6. Show that any group having a composition series has a chief series.

Solution. □

Exercise 10.7. (cont.) Show that any chief factor of a group having a composition series is a direct product of mutually isomorphic simple groups.

Solution. □

Exercise 10.8. Show that any finite group that has no proper non-trivial characteristic subgroups is a direct product of mutually isomorphic simple groups. Use this to give a new proof of Theorem 10.5.

Solution. □

11 Solvable Groups

The derived series and solvability

- We define a series $G^{(k)}$ of subgroups of a group G by setting $G^{(0)} = G$ and taking $G^{(k)}$ to be the derived group of $G^{(k-1)}$ for $k \in \mathbb{N}$. This series is called the *derived series* of G.
- Since $G^{(k+1)}$ is a characteristic subgroup of $G^{(k)}$ for each k by Lemma 2.5, we see via Lemma 2.4 that the derived series is a normal series of G. Moreover, we see from Proposition 2.6 that each successive quotient $G^{(k)}/G^{(k+1)}$ of the derived series is abelian.
- We say that a group is *solvable* if its derived series terminates in the identity. (Authors from British Commonwealth countries often write "soluble" in lieu of "solvable" and "insoluble" in lieu of "non-solvable.")
- Since a group is abelian iff its derived group is trivial, we see that abelian groups are solvable. However, not all groups are solvable; for example, A_5 is non-abelian and simple by Theorem 7.4, so $A_5^{(k)} = A_5$ for all k.
- A group whose derived group is itself is said to be *perfect*; any non-abelian simple group is perfect, and in particular is not solvable. However, a perfect group need not be simple: $\mathrm{SL}(n, F)$ is perfect but has non-trivial center, except when $n = 2$ and $|F| = 3$.
- Solvable groups are to be thought of as the opposite of simple groups: simple groups have very few normal subgroups while solvable groups are rife with them.

Proposition 11.1. *A simple solvable group has prime order.*

Proof. □

Proposition 11.2. *The following statements about a group G are equivalent:*

(1) G is solvable.

(2) G has a normal series in which every successive quotient is abelian.

(3) G has a subnormal series in which every successive quotient is abelian.

Proof. □

Proposition 11.3. *(i) If G is solvable and $H \leq G$, then H is solvable.*

(ii) If G is solvable and $N \trianglelefteq G$, then G/N is solvable.

(iii) If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable.

(iv) If G and H are solvable, then $G \times H$ is solvable.

Proof. □

Proposition 11.4. *A group having a composition series is solvable iff all of its composition factors have prime order. (In particular, such groups are finite.)*

Proof. □

Examples and consequences

- There are groups, such as S_5, which are neither simple nor solvable, and there are non-abelian groups, such as S_3 (which has the composition series $S_3 > A_3 > 1$), which are solvable.
- Proposition 11.4 also implies that an infinite abelian group, being solvable, cannot possess a composition series (which was Exercise 10.1).

Corollary 11.5. *Finite p-groups are solvable.*

Proof. □

Proposition 11.6. *A group having a composition series is solvable iff all of its chief factors are elementary abelian.*

(Any group having a composition series also has a chief series by Exercise 10.6. Alternately, a solvable group having a composition series is finite by Proposition 11.4 and hence has a chief series by Proposition 10.3.)

Proof. □

Corollary 11.7. *A group having a composition series is solvable iff it has a normal series in which every successive quotient is a p-group.*

Proof. □

Supersolvable groups and Hall's theorems

- Those finite groups whose chief factors all have prime order are said to be *supersolvable*; finite supersolvable groups are solvable by Proposition 11.6.

- It follows from Exercise 8.2 that finite p-groups are supersolvable. However, not all finite solvable groups are supersolvable. The series $S_4 > A_4 > K > 1$, where K is the Klein four-group, is a chief series of S_4, and we have $S_4/A_4 \cong \mathbb{Z}_2$, $A_4/K \cong \mathbb{Z}_3$, and $K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, which by Proposition 11.6 shows that S_4 is solvable but not supersolvable.

- There is an equivalent definition of supersolvability which extends to infinite groups; see Exercise 11.6.

- Solvable groups possess a number of properties beyond those which hold for arbitrary groups. The following theorem, due to Philip Hall, is a generalization for finite solvable groups of part (i) of Sylow's Theorem.

Theorem 11.8. *Let G be a finite solvable group of order mn, where m and n are coprime. Then G has a subgroup of order m.*

(In other words, this theorem asserts that a finite solvable group possesses Hall subgroups of all possible orders.)

Proof. □

More on solvability criteria

- Theorem 11.8 does not hold for all finite groups; for instance, we observed in the proof of Theorem 7.5 that the group A_5 of order $60 = 20 \cdot 3$ has no subgroup of order 20.

- P. Hall also proved the following generalization for finite solvable groups of parts (ii) and (iii) of Sylow's Theorem, which the reader is asked to prove in the exercises.

Theorem 11.9. *Let G be a finite solvable group of order mn, where m and n are coprime. Then any two subgroups of G of order m are conjugate, and any subgroup of G whose order divides m is contained in a subgroup of order m.*

Proof. □

Classical theorems on solvability

- We now state, without proof, a number of well-known theorems giving conditions under which a finite group is solvable. We start with a classical result that generalizes Corollary 11.5.

Theorem (Burnside's Theorem)**.** *If p and q are primes, then any group of order $p^a q^b$ is solvable.*

Proof. □

- Burnside's Theorem is the best possible result, in the sense that a finite group whose order has exactly three prime divisors need not be solvable, with A_5 providing a counterexample.

- Burnside proved this theorem in 1904 using the methods of character theory, which will be discussed in Chapter 6; we present a character-theoretic proof of Burnside's theorem in the Appendix (Burnside's Theorem). Until the 1960s, there was no proof of this theorem that did not involve character theory.

Theorem (Feit–Thompson Theorem)**.** *All finite groups of odd order are solvable.*

Proof. □

- This result, also known as the "odd order theorem," had first been conjectured by Burnside in 1911. As a corollary, we see that the only simple groups of odd order are cyclic groups of prime order.

- Once it had finally been established that all non-abelian finite simple groups are of even order, the movement toward a classification of all finite simple groups gathered steam. Feit and Thompson completed their proof of this theorem during a special "Finite Group Theory Year," held at the University of Chicago in the school year 1960–61. Their original proof [12] was 255 pages long, and even at that length the proof requires too much background knowledge for it to be readily comprehensible to anyone but the specialists at the time at which the proof was published. In recent years there has been a program, of which [8] represents a major part, to produce a more accessible proof of this fundamental result.

- The following result is a converse to Theorem 11.8 and is also due to P. Hall. While this theorem generalizes Burnside's Theorem, its proof relies on the fact that groups of order $p^a q^b$ are solvable.

Theorem 11.10. *Let G be a finite group. If G has a subgroup of order m whenever m and n are coprime numbers such that $|G| = mn$, then G is solvable.*

Proof. □

- This next result was conjectured by P. Hall in the same paper [14] in which he proved Theorem 11.10; it was later proved by Thompson in [27].

Theorem 11.11. *A finite group G is not solvable iff there exist non-trivial elements x, y, z of G of pairwise coprime orders a, b, c such that $xy = z$.*

Proof. □

- For example, in A_5 this criterion for non-solvability is satisfied by taking $a = 5$, $b = 2$, $c = 3$ and $x = (1\,2\,3\,4\,5)$, $y = (1\,2)(3\,4)$, $z = (1\,3\,5)$. (Compare with Exercise 1.5.)

Galois's theorem and the affine group

- The concept of solvable groups originated with Galois around 1830 in his work on the solution by radicals of polynomial equations; this in fact is the origin of the term "solvable" for this class of groups.

- Let V be a vector space over a field F. For $v \in V$, let T_v be the self-map of V corresponding to translation by v, and let $\mathcal{T}(V)$ be the set of all such translations; this is a group isomorphic with the abelian group V, and if we regard $\mathcal{T}(V)$ and $\mathrm{GL}(V)$ as subgroups of the group of all invertible self-maps of V, then $\mathrm{GL}(V) \cap \mathcal{T}(V)$ is trivial.

- We easily verify that $ST_vS^{-1} = T_{S(v)}$ for any $S \in \mathrm{GL}(V)$ and $v \in V$. This gives rise to a homomorphism from $\mathrm{GL}(V)$ to $\mathrm{Aut}(\mathcal{T}(V))$, and hence we have an (internal) semidirect product $\mathcal{T}(V) \rtimes \mathrm{GL}(V)$. This group is called the *affine group* of V and is denoted $\mathrm{Aff}(V)$; its elements are called the *affine transformations* of V.

Theorem 11.12. *Let G be a finite, solvable, primitive permutation group on a set X. Then X can be given the structure of a vector space over the field $\mathbb{Z}/p\mathbb{Z}$ for some prime p in such a way that G is isomorphic with a subgroup of* $\mathrm{Aff}(X)$ *that contains $\mathcal{T}(X)$.*

Proof. □

Nilpotent groups

- A normal series $G = G_0 \geq G_1 \geq \ldots \geq G_r = 1$ of a group G is said to be a *central series* of G if, for each i, G_i/G_{i+1} is contained in the center of G/G_{i+1}.
- A group G is said to be *nilpotent* if it has a central series. An abelian group G has the central series $G > 1$, and hence abelian groups are nilpotent.

Proposition 11.13. *Nilpotent groups are solvable.*

Proof. □

- There are solvable groups that are not nilpotent. For example, the group S_3 cannot have a central series, as the penultimate term of such a series would have to be a non-trivial subgroup of $Z(S_3) = 1$.

Lemma 11.14. *Finite p-groups are nilpotent.*

Proof. □

- If H and K are subgroups of a group G, we define a new subgroup $[H, K]$ of G by $[H, K] = \langle \{[h, k] \mid h \in H, k \in K\} \rangle$. Observe that $G' = [G, G]$.
- We now reconcile the above definition of nilpotence with that made in Section 8 for finite groups. Recall that a subgroup H of a group G is said to be *self-normalizing* if $N_G(H) = H$.

Theorem 11.15. *The following statements about a finite group G are equivalent:*

(1) G is nilpotent.

(2) G has no proper self-normalizing subgroups.

(3) Every Sylow subgroup of G is normal in G.

(4) G is the direct product of its Sylow subgroups.

(5) Every maximal subgroup of G is normal in G.

Proof. □

Exercises

Exercise 11.1. Show that any finite group has a largest solvable normal subgroup.

Solution. □

Exercise 11.2. Let N be a solvable normal Hall subgroup of a finite group G. Show that any two complements to N in G are conjugate. (The strong version of Schur–Zassenhaus Theorem asserts that this is true for any normal Hall subgroup N, whether solvable or not. The proof uses Feit–Thompson Theorem to argue that if N is not solvable, then G/N must be solvable.)

Solution. □

Exercise 11.3. Complete the following sketch, which gives a proof of Theorem 11.8 that is independent of Schur–Zassenhaus Theorem. Here G is a finite solvable group of order mn, where m and n are coprime, and we are to show that G has a subgroup of order m.

(a) Using induction, reduce to the case where G has a unique minimal normal subgroup N of order n.

(b) Let M/N be a minimal normal subgroup of G/N; since G/N is solvable by Proposition 11.3, it follows from Proposition 11.6 that $|M/N|$ is a power of some prime divisor p of m. If P is a Sylow p-subgroup of M, show that $|N_G(P)| = m$.

Solution. □

Exercise 11.4. Prove Theorem 11.9. If your proof involves Schur–Zassenhaus Theorem, then construct a second proof that is independent of Schur-Zassenhaus by mimicking Exercise 11.3 above.

Solution. □

Exercise 11.5. Let V be a finite-dimensional vector space over the field of p elements. Suppose that G is a solvable subgroup of $\mathrm{Aff}(V)$ which contains $\mathcal{T}(V)$ and that the stabilizer of the origin under the action of G on V does not leave any non-zero proper subspace of V invariant. Show that G is a primitive permutation group on V.

Solution. □

Exercise 11.6. Show that a finite group is supersolvable iff it has a normal series having cyclic successive quotients. (An arbitrary group is called supersolvable if this latter condition holds.)

Solution. □

Exercise 11.7. Show that finite nilpotent groups are supersolvable. (If we define supersolvability for arbitrary groups as in Exercise 11.6, is an infinite nilpotent group necessarily supersolvable?)

Solution. □

Exercise 11.8. (cont.) Give an example of a finite supersolvable group that is not nilpotent. (Hence we have a proper ascending chain of classes of finite groups, starting with cyclic groups, then abelian, then nilpotent, then supersolvable, and ending with solvable.)

Solution. □

Further Exercises

Let G be a group. Let $\Gamma_1 = G$, and for $n \in \mathbb{N}$ let $\Gamma_{n+1} = [\Gamma_n, G] \leq G$. Let $\mathcal{Z}_0 = 1$, and for $n \in \mathbb{N}$ let $\mathcal{Z}_n$ be the unique subgroup of G such that $\mathcal{Z}_n/\mathcal{Z}_{n-1} = Z(G/\mathcal{Z}_{n-1})$. (Observe that $\Gamma_2 = G'$ and $\mathcal{Z}_1 = Z(G)$.) We call the series $G = \Gamma_1 \geq \Gamma_2 \geq \Gamma_3 \geq \ldots$ the *lower central series* of G and the series $1 = \mathcal{Z}_0 \leq \mathcal{Z}_1 \leq \mathcal{Z}_2 \leq \ldots$ the *upper central series* of G.

Exercise 11.9. Show that each Γ_n and $\mathcal{Z}_n$ is a characteristic subgroup of G.

Solution. □

Exercise 11.10. (cont.) Show that G is nilpotent iff $\Gamma_r = 1$ for some r iff $\mathcal{Z}_s = G$ for some s.

Solution. □

Exercise 11.11. (cont.) Suppose that G is nilpotent. Show that the lower and upper central series of G have the same length, say c, and that no central series of G can have length less than c. (This number c is called the *nilpotency class* of G. The groups of nilpotency class 1 are exactly the abelian groups.)

Solution. □

Chapter 5

Semisimple Algebras

12 Modules and Representations

From Group Actions to Linear Actions

- In Section 3 we studied actions of groups on sets; if a group acts on a set carrying additional algebraic structure, it is the structure-respecting actions that are most interesting.
- Here we focus on actions of groups on vector spaces that respect the vector space structure.
- Let F be a field and let G be a group acting on an F-vector space V. The action is *linear* if
  - $g(v + w) = gv + gw$ for all $g \in G$ and $v, w \in V$;
  - $g(\alpha v) = \alpha(gv)$ for all $g \in G$, $\alpha \in F$, and $v \in V$.
- Proposition 3.1 gave the bijection between group actions on a set and homomorphisms to symmetric groups; we now deduce its analogue for linear actions.

Proposition 12.1. *There is a bijective correspondence between the set of linear actions of a group G on an F-vector space V and the set of homomorphisms from G to* $\mathrm{GL}(V)$.

Proof. □

Linear Representations and Modules

- A homomorphism $\rho\colon G \to \mathrm{GL}(V)$, where G is a group and V a vector space, is called a *linear representation* of G in V.
- By Proposition 12.1, the study of linear representations is equivalent to the study of linear actions; with emphasis on finite groups and finite-dimensional vector spaces, this area originated in the late nineteenth century.
- The modern approach to the representation theory of finite groups involves yet another equivalent concept, that of finitely generated modules over group algebras. It is therefore necessary at this juncture to review some elementary module theory.
- As in Section 1, we will omit most proofs on the assumption that the reader will have seen this material previously; however, our account requires only a mild knowledge of rings, fields, and vector spaces.
- Let R be a ring with unit, and let M be an abelian group written additively. Then M is a *left R-module* if there is a map $R \times M \to M$, $(r, m) \mapsto rm$, satisfying:
  - $1m = m$ for all $m \in M$;
  - $r(m + n) = rm + rn$ for all $r \in R$ and $m, n \in M$;
  - $(r + s)m = rm + sm$ for all $r, s \in R$ and $m \in M$;
  - $r(sm) = (rs)m$ for all $r, s \in R$ and $m \in M$.

- If $R = \mathbb{Z}$, then $\mathbb{Z}$-modules are exactly abelian groups; if F is a field, then F-modules are precisely F-vector spaces. A module is the natural generalisation of a vector space to an arbitrary ring.
- One similarly defines right R-modules; if R is commutative, every left module is canonically a right module. In general, a left R-module is a right module over the *opposite ring* R^{op}.
- Let S be another ring with unit. An abelian group M that is both a left R-module and a right S-module is an (R, S)*-bimodule* provided $r(ms) = (rm)s$ for all $r \in R$, $m \in M$, $s \in S$. Any left R-module is an $(R, \mathbb{Z})$-bimodule, and R itself is an (R, R)-bimodule.
- An R-module M is *finitely generated* if every element is an R-linear combination of elements from some finite subset of M. Unlike vector spaces, minimal generating sets need not have a common size.
- If N is a subgroup of M closed under the R-action, N is an R*-submodule*. The submodules of the regular module R are exactly the left ideals. A nonzero module with no submodules other than 0 and itself is *simple*.
- Quotient modules M/N are defined by $r(m + N) = rm + N$.
- Sums, intersections, internal and external direct sums of submodules are defined as for groups; N is a *direct summand* of M if $M = N \oplus N'$ for some submodule N'. We write nM or M^n for $\underbrace{M \oplus \cdots \oplus M}_{n}$.
- A *composition series* of M is a descending series of submodules ending at 0 in which each successive quotient is simple. Modules need not have one (any infinite $\mathbb{Z}$-module is an example), but when they do, Jordan–Hölder Theorem holds.
- Every composition factor of any submodule or quotient module of an R-module M must also be a composition factor of M, as any composition series of a submodule (resp., quotient module) can be extended (resp., lifted) to give a composition series of M.
- An R*-module homomorphism* $\varphi \colon M \to N$ is a group homomorphism with $\varphi(rm) = r\varphi(m)$ for all $r \in R$, $m \in M$. Mono-, epi-, iso-, endo-, automorphisms are defined as for groups; the kernel is a submodule, the image is a submodule, and Fundamental Theorem on Homomorphisms, Correspondence Theorem, and the two isomorphism theorems (First Isomorphism Theorem and Second Isomorphism Theorem) all carry over.

Lemma (Schur's Lemma)**.** *Any non-zero homomorphism between simple R-modules is an isomorphism.*

Proof. □

Hom-Modules, Balanced Maps, and Tensor Products

- Given R-modules M and N, we write $\mathrm{Hom}_R(M, N)$ for the set of R-module homomorphisms $M \to N$, and $\mathrm{End}_R(M)$ for $\mathrm{Hom}_R(M, M)$. Pointwise addition makes $\mathrm{Hom}_R(M, N)$ an abelian group.
- If S is another ring, M an (R, S)-bimodule, and N an R-module, then $\mathrm{Hom}_R(M, N)$ is an S-module via $(s\varphi)(m) = \varphi(ms)$. In particular, when F is a field and U, V are F-vector spaces, $\mathrm{Hom}_F(U, V)$ is an F-vector space.
- The map $\mathrm{Hom}_R(R, M) \to M$ sending φ to $\varphi(1)$ is an R-module isomorphism.
- Let M be an (R, S)-bimodule and N a left S-module. A map $f \colon M \times N \to U$ to an R-module U is *balanced* if
  - $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$;
  - $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$;
  - $f(ms, n) = f(m, sn)$;
  - $f(rm, n) = rf(m, n)$.

- The *tensor product* $M \otimes_S N$ is the R-module equipped with a balanced map $\eta\colon M \times N \to M \otimes_S N$ such that any balanced $f\colon M \times N \to U$ factors uniquely through η: there is a unique R-module homomorphism $\alpha\colon M \otimes_S N \to U$ with $f = \alpha \circ \eta$. We write $m \otimes n = \eta(m, n)$.
- Concretely, $M \otimes_S N$ is generated by the symbols $\{m \otimes n \mid m \in M, n \in N\}$ subject to the relations
  - $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$;
  - $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$;
  - $(ms) \otimes n = m \otimes (sn)$;
  - $(rm) \otimes n = r(m \otimes n)$.

  A general element of $M \otimes_S N$ is a *sum* of elementary tensors, not just one.
- Special case: if F is a field and U, V are finite-dimensional with bases $\{u_1, \ldots, u_r\}$ and $\{v_1, \ldots, v_s\}$, then $U \otimes_F V$ has basis $\{u_i \otimes v_j\}$ and dimension rs. For $u = \sum_i a_i u_i$ and $v = \sum_j b_j v_j$, $u \otimes v = \sum_{i,j} a_i b_j (u_i \otimes v_j)$.
- (An understanding of this special case will, strictly speaking, suffice in reading the remainder of the book; however, an understanding of the general case will be beneficial in Section 16.)

Proposition 12.2. *Let R be a ring with unit, and let M be an R-module. Then M and $R \otimes_R M$ are isomorphic R-modules.*

Proof. □

Proposition 12.3. *Let R and S be rings with unit, let $M_1, \ldots, M_r$ be (R, S)-bimodules, and let N be an S-module. Then $(\bigoplus_i M_i) \otimes_S N$ and $\bigoplus_i M_i \otimes_S N$ are isomorphic R-modules.*

Proof. □

Proposition 12.4. *Let F be a field, and let U and V be F-vector spaces, with $\dim_F(U) < \infty$. Let $U^* = \mathrm{Hom}_F(U, F)$. Then the map $\Gamma\colon U^* \otimes_F V \to \mathrm{Hom}_F(U, V)$ defined by setting $\Gamma(\varphi \otimes v)(u) = \varphi(u)v$ and extending linearly is an isomorphism of F-vector spaces.*

Proof. □

Group Rings and Group Algebras

- Let R be a ring and G a group. The *group ring* of G over R, denoted RG, consists of all finite formal R-linear combinations of elements of G, with the obvious addition and with multiplication extending that in G:
$$\Big(\sum_{x \in G} \alpha_x x\Big)\Big(\sum_{y \in G} \beta_y y\Big) = \sum_{x \in G}\sum_{y \in G} \alpha_x \beta_y \, xy = \sum_{x \in G}\Big(\sum_{g \in G} \alpha_g \beta_{g^{-1}x}\Big) x.$$
- The unit element is the identity of G.
- When $R = F$ is a field and G is finite, FG is also an F-vector space with basis G, hence has finite dimension $|G|$. In this case FG is called the *group algebra*.
- An *algebra* over F (or *F-algebra*) is a set A carrying both a ring structure and an F-vector space structure that share the same addition, with $(\lambda a)b = a(\lambda b) = \lambda(ab)$ for $\lambda \in F$, $a, b \in A$. We do *not* assume an algebra has a ring unit.
- An algebra is *finite-dimensional* if it has finite dimension as an F-vector space; the matrix ring $\mathcal{M}_n(F)$ is a finite-dimensional F-algebra, and FG is finite-dimensional whenever G is finite.
- A homomorphism of F-algebras is a ring homomorphism that is also F-linear.

Lemma 12.5. *If F is a field and G a finite group, then an FG-module is finitely generated iff it has finite dimension as an F-vector space.*

Proof. □

Proposition 12.6. *If F is a field and G a finite group, then there is a bijective correspondence between finitely generated FG-modules and linear actions of G on finite-dimensional F-vector spaces.*

Proof. □

Examples and Constructions of FG-Modules

- In what follows G is a finite group, F a field, and all F-vector spaces are finite-dimensional; all FG-modules will be finitely generated and hence finite-dimensional by Lemma 12.5.
- It is enough to specify the action of G on V; the FG-action then extends linearly.
- *Trivial module*: F regarded as an FG-module via $g\lambda = \lambda$.
- *Permutation module*: if G acts on a finite set $X = \{x_1, \ldots, x_n\}$, the F-vector space FX with basis X is an FG-module via $g\big(\sum_i c_i x_i\big) = \sum_i c_i (gx_i)$.
- Direct sums of FG-modules: $g(u, v) = (gu, gv)$.
- *Tensor product* $U \otimes_F V$ with the diagonal action $g(u \otimes v) = gu \otimes gv$.
- $\mathrm{Hom}_F(U, V)$ with action $(g\varphi)(u) = g(\varphi(g^{-1}u))$; one checks $(g_1 g_2)\varphi = g_1(g_2\varphi)$.
- Taking $V = F$ trivial gives the *dual module* $U^* = \mathrm{Hom}_F(U, F)$ with $(g\varphi)(u) = \varphi(g^{-1}u)$.

Proposition 12.7. *Let U and V be FG-modules. Then $U^* \otimes_F V$ and $\mathrm{Hom}_F(U, V)$ are isomorphic FG-modules.*

Proof. □

Maschke's Theorem and Semisimplicity

- We now reach the most basic result of the representation theory of finite groups, discovered in 1898 by Heinrich Maschke at the University of Chicago.

Theorem (Maschke's Theorem)**.** *Let G be a group, and suppose that the characteristic of F is either zero or coprime to $|G|$. If U is an FG-module and V is an FG-submodule of U, then V is a direct summand of U as FG-modules.*

Proof. □

- We summarise the hypothesis by saying that the characteristic of F *does not divide* $|G|$, even though this is a slight abuse.
- A module is *semisimple* if it is a direct sum of simple modules.

Corollary 12.8. *Let G be a group, and let F be a field whose characteristic does not divide $|G|$. Then every non-zero FG-module is semisimple.*

Proof. □

- In Chapter 6 we shall be concerned only with the case $F = \mathbb{C}$, called *ordinary* representation theory; since $\mathbb{C}$ has characteristic zero, by Corollary 12.8 every non-zero $\mathbb{C}G$-module is semisimple.
- The next section concentrates on algebras with this property. The case where the characteristic of F divides $|G|$, in which FG-modules need not be semisimple, is the subject of *modular* representation theory (originated by Dickson in 1907; mostly developed by Brauer from the 1930s onward).

Exercises

Throughout these exercises, let F be a field and let G be a finite group.

Exercise 12.1. Let U and V be FG-modules having the same dimension n, and let $\rho\colon G \to \mathrm{GL}(U)$ and $\tau\colon G \to \mathrm{GL}(V)$ be the corresponding representations. By fixing F-bases for U and V, we consider ρ and τ as homomorphisms from G to $\mathrm{GL}(n, F)$. Show that $U \cong V$ iff there exists some $M \in \mathrm{GL}(n, F)$ such that $\rho(g)M = M\tau(g)$ for every $g \in G$. (Two representations are said to be *equivalent* if this latter condition holds; hence two representations are equivalent iff their corresponding modules over the group algebra are isomorphic.)

Solution. □

Exercise 12.2. Let $\sigma = \sum_{g \in G} g \in FG$. Show that the subspace $F\sigma$ of FG is the unique submodule of FG that is isomorphic with the trivial module.

Solution. □

Exercise 12.3. (cont.) Let $\epsilon\colon FG \to F$ be the FG-module epimorphism defined by $\epsilon(g) = 1$ for all $g \in G$, and let $\Delta = \ker \epsilon$. (We call Δ the *augmentation ideal* of FG.) Show that Δ is the unique submodule of FG whose quotient is isomorphic with the trivial module.

Solution. □

Exercise 12.4. (cont.) Suppose that the characteristic of F divides $|G|$. Show that $F\sigma \subseteq \Delta$. Conclude that Δ is not a direct summand of FG and hence that the FG-module FG is not semisimple. (This shows that the converse of Corollary 12.8 is true.)

Solution. □

Further Exercises

In Section 9 we defined the second cohomology group of a pair (H, A), where H is a group and A an abelian group, with respect to a given homomorphism from H to $\mathrm{Aut}(A)$. It is easy to verify that, for an abelian group A, specifying a homomorphism from a group H to $\mathrm{Aut}(A)$ is exactly the same as specifying a $\mathbb{Z}H$-module structure on A. Therefore, it is customary to talk about the second cohomology group of a pair (H, A) where H is a group and A a $\mathbb{Z}H$-module. (We anticipated this development by writing xa instead of $\varphi(x)(a)$ on page 86.) We will now generalize the second cohomology group.

Let H be a group, let H^n be the set of n-tuples of elements of H for some $n \in \mathbb{N}$, and let A be a $\mathbb{Z}H$-module. A *normalized n-cochain* of (H, A) is a set function $f\colon H^n \to A$ such that $f(h_1, \dots, h_n) = 0$ whenever $h_i = 1$ for some i; the set of normalized n-cochains of (H, A) is denoted $C^n(H, A)$. We give $C^n(H, A)$ an abelian group structure by defining

$$(f + g)(h_1, \dots, h_n) = f(h_1, \dots, h_n) + g(h_1, \dots, h_n).$$

We define a homomorphism $\delta^n\colon C^{n-1}(H, A) \to C^n(H, A)$ by

$$(\delta^n f)(h_1, \dots, h_n) = h_1 f(h_2, \dots, h_n) + \sum_{i=1}^{n-1} (-1)^i f(h_1, \dots, h_i h_{i+1}, \dots, h_n) + (-1)^n f(h_1, \dots, h_{n-1}).$$

The image of δ^n is called the set of *n-coboundaries* and is denoted $B^n(H, A)$; the kernel of δ^{n+1} is called the set of *n-cocycles* and is denoted $Z^n(H, A)$. For example, $\ker \delta^3$ consists of all normalized 2-cochains f satisfying $h_1 f(h_2, h_3) - f(h_1 h_2, h_3) + f(h_1, h_2 h_3) - f(h_1, h_2) = 0$ for all $h_1, h_2, h_3 \in H$, which agrees with our definition of a 2-cocycle in Section 9. Both $B^n(H, A)$ and $Z^n(H, A)$ are abelian subgroups of $C^n(H, A)$. We find that $\delta^{n+1} \circ \delta^n$ is the zero map, or equivalently that $B^n(H, A) \subseteq Z^n(H, A)$. (Verify this.) The quotient group $Z^n(H, A)/B^n(H, A)$ is denoted $H^n(H, A)$ and is called the *nth cohomology group* of (H, A).

While $H^3(H, A)$ plays some role in the theory of extensions of a non-abelian group N by H, where $Z(N) = A$ (see [20, pp. 124–131]), the groups $H^n(H, A)$ for $n > 3$ have no known group-theoretic meaning. Nonetheless, higher cohomology groups are studied for their own sake, and in recent years connections between the cohomology of groups and the representation theory of groups have attracted much attention. (As the title of one of the first author's papers [6] proclaims, "Cohomology *is* representation theory.") We developed the group-theoretic meaning of $H^2(H, A)$ in the further exercises to Section 9, and in what follows we develop the meaning of $H^1(H, A)$.

Exercise 12.5. (cont.) Define a homomorphism $\varphi\colon H \to \mathrm{Aut}(A)$ by $\varphi(x)(a) = xa$. Show that there is a bijective correspondence between the set of splitting maps of $A \rtimes_\varphi H$ and $Z^1(H, A)$. (Recall that a homomorphism $t\colon H \to A \rtimes_\varphi H$ is called a splitting map if $\eta \circ t$ is the identity map on H, where $\eta\colon A \rtimes_\varphi H \to H$ is the natural map. The natural inclusion of H into $A \rtimes_\varphi H$ is a splitting map, but there may be others.)

Solution. □

Exercise 12.6. (cont.) We say that two splitting maps t and u of $A \rtimes_{\varphi} H$ are *conjugate* if there is some $a \in A$ such that $u(x) = at(x)a^{-1}$ for all $x \in H$. (Verify that this is an equivalence relation on the set of splitting maps.) Show that there is a bijective correspondence between the set of conjugacy classes of splitting maps of $A \rtimes_{\varphi} H$ and $H^1(H, A)$.

Solution. □

13 Wedderburn Theory

Setting and Goal

- Maschke (Corollary 12.8) shows that if $\mathrm{char}(F) \nmid |G|$ then every finitely generated FG-module decomposes as a direct sum of simple modules.
- This section identifies the finite-dimensional F-algebras with this property and uses the resulting structure theorems to finish Kolchin's theorem on unipotent subgroups.
- Throughout, all algebras are finite-dimensional F-algebras (with unit unless stated), and all modules are finitely generated, equivalently finite-dimensional as F-vector spaces (Lemma 12.5).
- The classification is due to J. H. M. Wedderburn.

Semisimplicity for Modules

- Three equivalent formulations of semisimplicity for an A-module: every submodule is a direct summand; the module is a direct sum of simple submodules; the module is a sum of simple submodules.
- Semisimplicity is preserved under submodules and quotient modules.
- An algebra A is *semisimple* if every non-zero A-module is semisimple; equivalently, A is semisimple as an A-module over itself.

Lemma 13.1. *The following statements about an A-module M are equivalent:*

1. *Any submodule of M is a direct summand of M.*
2. *M is semisimple.*
3. *M is a sum (but not, a priori, a direct sum) of simple submodules.*

Proof. □

Lemma 13.2. *Submodules and quotient modules of semisimple modules are semisimple.*

Proof. □

Lemma 13.3. *The algebra A is semisimple iff the A-module A is semisimple.*

Proof. □

Counting Simple Summands

- Decomposing $A \cong S_1 \oplus \cdots \oplus S_r$ into simple summands captures all isomorphism types of simple A-modules.
- Multiplicities of simple summands in a semisimple module are determined uniquely (Jordan–Hölder Theorem).

Proposition 13.4. *Let A be a semisimple algebra, and suppose that as A-modules we have $A \cong S_1 \oplus \cdots \oplus S_r$, where the S_i are simple submodules of A. Then any simple A-module is isomorphic with some S_i.*

Proof. □

Proposition 13.5. *Suppose that A is a semisimple algebra, and let $S_1, \ldots, S_r$ be a collection of simple A-modules such that every simple A-module is isomorphic with exactly one S_i. Let M be an A-module, and write $M \cong n_1S_1 \oplus \cdots \oplus n_rS_r$ for some non-negative integers n_i. Then the n_i are uniquely determined.*

Proof. □

Matrix Algebras over Division Algebras

- For a finite-dimensional F-algebra D, the matrix algebra $\mathcal{M}_n(D)$ has dimension $n^2 \dim_F D$.
- $E_{ij}(\alpha)$ denotes the matrix with sole non-zero entry α in position (i, j).
- D^n is the natural left $\mathcal{M}_n(D)$-module of column vectors.
- A *division algebra* is an algebra whose non-zero elements form a group under multiplication; field extensions of F are division algebras, but non-commutative examples also exist.

Theorem 13.6. *Let D be a division algebra, and let $n \in \mathbb{N}$. Then any simple $\mathcal{M}_n(D)$-module is isomorphic with D^n, and $\mathcal{M}_n(D)$ is isomorphic as $\mathcal{M}_n(D)$-modules with the direct sum of n copies of D^n. In particular, $\mathcal{M}_n(D)$ is a semisimple algebra.*

Proof. □

Simple Algebras

- An algebra is *simple* if its only two-sided ideals are itself and the zero ideal.
- Simple algebras are semisimple, and matrix algebras over division algebras are simple.

Lemma 13.7. *Simple algebras are semisimple.*

Proof. □

Theorem 13.8. *Let D be a division algebra, and let $n \in \mathbb{N}$. Then $\mathcal{M}_n(D)$ is a simple algebra.*

Proof. □

Direct Sums of Algebras

- The *external direct sum* $B = B_1 \oplus \cdots \oplus B_r$ has the Cartesian product of underlying sets with componentwise operations; each B_i embeds as an ideal.
- A B-module M becomes a B_i-module via $b_i m = (0, \ldots, b_i, \ldots, 0)m$, and is simple/semisimple as a B_i-module iff it is simple/semisimple as a B-module.
- An *internal direct sum* of ideals $B_1, \ldots, B_r$ (vector-space direct sum) is isomorphic to the external direct sum.
- Two-sided ideals of a finite direct sum decompose componentwise.

Lemma 13.9. *Let $B = B_1 \oplus \cdots \oplus B_n$ be a direct sum of algebras. Then the two-sided ideals of B are exactly the sets of the form $J_1 \oplus \cdots \oplus J_n$, where J_i is a two-sided ideal of B_i for each i.*

Proof. □

Theorem 13.10. *Let $r \in \mathbb{N}$. For each $1 \leq i \leq r$, let D_i be a division algebra over F, let $n_i \in \mathbb{N}$, and let $B_i = \mathcal{M}_{n_i}(D_i)$. Let B be the external direct sum of the B_i. Then B is a semisimple algebra having exactly r isomorphism classes of simple modules and exactly 2^r two-sided ideals, namely every sum of the form $\bigoplus_{j \in J} B_j$, where J is a subset of $\{1, \ldots, n\}$.*

Proof. □

Endomorphism and Opposite Algebras

- For an A-module M, $\mathrm{End}_A(M)$ is an F-algebra under composition; we call it the *endomorphism algebra* of M.
- The *opposite algebra* B^{op} has the same underlying additive structure but reversed multiplication: $a \cdot b := ba$. Note $(B^{op})^{op} = B$, and the opposite of a division algebra is again a division algebra.
- These two notions are linked by $B^{op} \cong \mathrm{End}_B(B)$: every B-endomorphism of B is right multiplication by some element.
- Endomorphism algebras decompose along A-module direct sums into block form, and over a single simple module they yield a matrix algebra.

Lemma 13.11. *Let B be an algebra. Then $B^{op} \cong \mathrm{End}_B(B)$.*

Proof. □

Lemma 13.12. *Let $S_1, \ldots, S_r$ be the distinct simple A-modules; for each i, let U_i be a direct sum of copies of S_i, and let $U = U_1 \oplus \cdots \oplus U_r$. Then we have $\mathrm{End}_A(U) \cong \mathrm{End}_A(U_1) \oplus \cdots \oplus \mathrm{End}_A(U_r)$.*

Proof. □

Lemma 13.13. *If S is a simple A-module, then for any $n \in \mathbb{N}$ we have $\mathrm{End}_A(nS) \cong \mathcal{M}_n(\mathrm{End}_A(S))$.*

Proof. □

Lemma 13.14. *Suppose that F is algebraically closed, and let S be a simple A-module. Then $\mathrm{End}_A(S) \cong F$.*

Proof. □

Lemma 13.15. *Let B be an algebra. Then $\mathcal{M}_n(B)^{op} \cong \mathcal{M}_n(B^{op})$ for any $n \in \mathbb{N}$.*

Proof. □

Wedderburn's Structure Theorems

- Wedderburn's main structure theorem (Theorem 13.16): a finite-dimensional algebra is semisimple iff it is isomorphic to a direct sum of matrix algebras over division algebras.
- Equivalently, semisimple iff a direct sum of simple algebras, confirming the consistency of terminology.
- An algebra is simple iff it is isomorphic to a single matrix algebra over a division algebra (Theorem 13.17).
- Over an algebraically closed field, every semisimple algebra is a direct sum of full matrix algebras over F itself (Theorem 13.18).

Theorem 13.16. *The algebra A is semisimple iff it is isomorphic with a direct sum of matrix algebras over division algebras.*

Proof. □

Theorem 13.17. *The algebra A is simple iff it is isomorphic with a matrix algebra over a division algebra.*

Proof. □

Theorem 13.18. *Suppose that the field F is algebraically closed. Then any semisimple algebra is isomorphic with a direct sum of matrix algebras over F.*

Proof. □

Nilpotent Ideals and the Radical

- In this part, A is an algebra possibly without unit. An element $x \in A$ is *nilpotent* if $x^n = 0$ for some $n \in \mathbb{N}$.
- An ideal I is *nilpotent* if $I^n = 0$ for some n, where I^n is the ideal spanned by all products of n elements; an algebra is nilpotent if some such n exists for $I = A$.
- A nilpotent algebra cannot have a unit.
- The sum of two nilpotent ideals is nilpotent, and (using finite-dimensionality) a unique largest nilpotent ideal exists.
- Submodules with semisimple quotient are closed under finite intersection, so a smallest such submodule of A exists.
- For an A-module M, $\mathrm{Ann}(M) = \{a \in A \mid aM = 0\}$ is the *annihilator* of M, an ideal of A; the *common annihilator* of a family $\{M_t\}_{t\in\mathcal{T}}$ is $\bigcap_{t\in\mathcal{T}} \mathrm{Ann}(M_t)$.
- Wedderburn (Theorem 13.23): the largest nilpotent ideal, the common annihilator of all simple A-modules, and the smallest submodule of A with semisimple quotient all coincide. This common object is the *radical* $\mathrm{rad}(A)$, also called the Jacobson radical $J(A)$.

Lemma 13.19. *Let I and J be nilpotent ideals of A. Then $I + J$ is also a nilpotent ideal of A.*

Proof. □

Corollary 13.20. *A has a largest nilpotent ideal.*

Proof. □

Lemma 13.21. *Let M and N be submodules of A such that A/M and A/N are semisimple A-modules. Then $A/M \cap N$ is a semisimple A-module.*

Proof. □

Corollary 13.22. *A has a smallest submodule having semisimple quotient.*

Proof. □

Theorem 13.23. *Let A be an algebra with unit. Then the following are equal:*

- *The largest nilpotent ideal of A.*
- *The common annihilator of all simple A-modules.*
- *The smallest submodule of A having semisimple quotient.*

Proof. □

Corollary 13.24. *Let A be an algebra with unit. Then A is semisimple iff A has no non-zero nilpotent ideals.*

Proof. □

Algebras Without Unit; Nilpotence

- Any algebra (possibly without unit) embeds as an ideal in an algebra with unit, with quotient F.
- Consequence: an algebra without unit having radical zero cannot exist—no such algebras occur.
- Wedderburn: an algebra generated as an F-vector space by nilpotent elements is itself nilpotent. This finishes Kolchin's Theorem.

Lemma 13.25. *Let A be an algebra, possibly without unit. Then there exists an algebra B with unit and an ideal I of B such that, as algebras, we have $I \cong A$ and $B/I \cong F$.*

Proof. □

Theorem 13.26. *Let A be a non-zero algebra, possibly without unit. If A has no non-zero nilpotent ideals, then A is an algebra with unit.*

Proof. □

Theorem 13.27. *Suppose that the algebra A can be generated as an F-vector space by a set consisting of nilpotent elements. Then A is nilpotent.*

Proof. □

Proposition 13.28. *Let $V_n(F)$ be the space of column vectors of length $n \in \mathbb{N}$ with entries from a field F, and let H be a unipotent subgroup of $\mathrm{GL}(n, F)$. Then there exists some $0 \neq \mathbf{v} \in V_n(F)$ such that $x\mathbf{v} = \mathbf{v}$ for all $x \in H$.*

Proof. □

Exercises

Throughout these exercises, A denotes a finite-dimensional algebra with unit over a field F, and all A-modules are finitely generated.

Exercise 13.1. Let $n \in \mathbb{N}$. Let V be the n-dimensional subspace of A^n spanned by the identity elements of the summands. Show that if M is an A-module and $T : V \to M$ is a linear transformation, then there is a unique A-module homomorphism from A^n to M which extends T.

Solution. □

Exercise 13.2. (cont.) Suppose that B is an A-module having an n-dimensional subspace U such that whenever M is an A-module and $T : U \to M$ is a linear transformation, there is a unique extension of T to an A-module homomorphism from B to M. Show that B and A^n are isomorphic A-modules. (Modules of the form A^n for some $n \in \mathbb{N}$ are called *free* A-modules; hence this exercise provides an alternate characterization of freeness, which we shall generalize for group algebras in the exercises to Section 16.)

Solution. □

Exercise 13.3. Let $n \in \mathbb{N}$, and let $T_n(F)$ be the algebra of upper triangular $n \times n$ matrices. Show that the set $V_n(F)$ of column vectors over F of length n is a $T_n(F)$-module that has a unique composition series in which every simple $T_n(F)$-module appears exactly once as a composition factor.

Solution. □

Exercise 13.4. (cont.) Show that the $T_n(F)$-module $T_n(F)$ is isomorphic with the direct sum of all non-zero submodules of $V_n(F)$.

Solution. □

Exercise 13.5. (cont.) Find the largest nilpotent ideal of $T_n(F)$, and show without reference to Theorem 13.23 that it is also the common annihilator of all simple $T_n(F)$-modules and the smallest submodule of $T_n(F)$ having semisimple quotient.

Solution. □

Exercise 13.6. Let U be an A-module, let $n \in \mathbb{N}$, and let U^n be the set of column vectors of length n with entries from U, considered in the obvious way as an $\mathcal{M}_n(A)$-module. Show that U is a simple A-module iff U^n is a simple $\mathcal{M}_n(A)$-module.

Solution. □

Exercise 13.7. (cont.) Show that $\mathrm{Hom}_A(U, V) \cong \mathrm{Hom}_{\mathcal{M}_n(A)}(U^n, V^n)$ for any A-modules U and V.

Solution. □

Exercise 13.8. (cont.) Show that every $\mathcal{M}_n(A)$-module is isomorphic with a module of the form U^n for some A-module U. (What Exercises 13.6–13.8 show is that the module theory of the algebras A and $\mathcal{M}_n(A)$ is, in a sense that can be made precise, the same. In the language of category theory, we say that the categories of A-modules and $\mathcal{M}_n(A)$-modules are *Morita equivalent.*)

Solution. □

Exercise 13.9. Show that A is a simple A-module iff A is a division algebra.

Solution. □

Exercise 13.10. We can clearly regard an $A/\operatorname{rad} A$-module as being an A-module; on the other hand, a simple A-module is annihilated by $\operatorname{rad}(A)$ and hence can be considered as an $A/\operatorname{rad} A$-module. Show that an A-module is simple iff it is a simple $A/\operatorname{rad}(A)$-module. (Since $A/\operatorname{rad}(A)$ is a semisimple algebra, this implies that the determination of simple modules over arbitrary algebras reduces to the case of semisimple algebras.)

Solution. □

Exercise 13.11. Let B be a finite-dimensional F-algebra, possibly without unit. Show that if B is a simple algebra whose multiplication is not identically zero, then B is isomorphic with a matrix algebra over a division algebra.

Solution. □

Further Exercises

Let A be as above, and let M be a finitely generated A-module.

Exercise 13.12. Show that the following objects (exist and) are equal: the intersection of all maximal submodules of M; the smallest submodule of M having semisimple quotient; and $\operatorname{rad}(A)M$. This submodule of M is called the *radical* of M and is denoted $\operatorname{rad}(M)$. (Observe that our two definitions of the symbol $\operatorname{rad}(A)$ agree.)

We define a descending series $\operatorname{rad}^n(M)$ of submodules of M by setting $\operatorname{rad}^0(M) = M$ and $\operatorname{rad}^n(M) = \operatorname{rad}(\operatorname{rad}^{n-1}(M))$ for $n \in \mathbb{N}$. It follows from Exercise 13.12 that $\operatorname{rad}^n(M) = (\operatorname{rad}(A))^n M$ for any $n \in \mathbb{N}$. This series is called the *radical series* of M. Observe that $\operatorname{rad}^n(M) = 0$ for some n, since $\operatorname{rad}(A)$ is a nilpotent ideal, and that each successive quotient of the radical series is a semisimple module, with M being semisimple iff $\operatorname{rad}(M) = 0$.

Solution. □

Exercise 13.13. (cont.) Show that M has a unique largest semisimple submodule and that this submodule is equal to $\{m \in M \mid \operatorname{rad}(A)m = 0\}$. We call this submodule the *socle* of M, and we denote it by $\operatorname{soc}(M)$.

Solution. □

Exercise 13.14. (cont.) We define an ascending series $\operatorname{soc}^n(M)$ of submodules of M as follows: We let $\operatorname{soc}^0(M) = 0$ and $\operatorname{soc}^1(M) = \operatorname{soc}(M)$, and for $n > 1$ we let $\operatorname{soc}^n(M)$ be the submodule of M such that $\operatorname{soc}^n(M)/\operatorname{soc}^{n-1}(M) = \operatorname{soc}(M/\operatorname{soc}^{n-1}(M))$. Show that we have $\operatorname{soc}^n(M) = \{m \in M \mid \operatorname{rad}(A)^n m = 0\}$ for any $n \in \mathbb{N}$. (This series is called the *socle series* of M. As with the radical series, we observe that $\operatorname{soc}^n(M) = M$ for some n, since $\operatorname{rad}(A)$ is nilpotent, and that each successive quotient of the socle series is semisimple, with M being semisimple iff $\operatorname{soc}(M) = M$.)

Solution. □

Exercise 13.15. (cont.) Show that $\operatorname{rad}^n(M) = 0$ iff $\operatorname{soc}^n(M) = M$; conclude that the radical and socle series of M have the same finite length. If this common length is r, then show that $\operatorname{rad}^n(M) \subseteq \operatorname{soc}^{r-n}(M)$ for every $0 \le n \le r$.

Solution. □

Chapter 6

Group Representations

14 Characters

Standing assumptions

- In this chapter G denotes a finite group.
- All $\mathbb{C}G$-modules are finitely generated, equivalently, finite-dimensional as $\mathbb{C}$-vector spaces.
- $\mathbb{C}$ is the field of complex numbers.

Wedderburn structure of $\mathbb{C}G$

- Since $\mathbb{C}$ is algebraically closed of characteristic zero, Wedderburn theory pins down the structure of $\mathbb{C}G$ precisely.
- The simple summands $\mathcal{M}_{f_i}(\mathbb{C})$ correspond bijectively to isomorphism classes of simple $\mathbb{C}G$-modules S_i, with $\dim_{\mathbb{C}} S_i = f_i$.

Theorem 14.1. *There is some $r \in \mathbb{N}$ and some $f_1, \ldots, f_r \in \mathbb{N}$ such that $\mathbb{C}G \cong \mathcal{M}_{f_1}(\mathbb{C}) \oplus \ldots \oplus \mathcal{M}_{f_r}(\mathbb{C})$ as $\mathbb{C}$-algebras. Furthermore, there are exactly r isomorphism classes of simple $\mathbb{C}G$-modules, and if we let $S_1, \ldots, S_r$ be representatives of these r classes, then we can order the S_i so that $\mathbb{C}G \cong f_1 S_1 \oplus \ldots \oplus f_r S_r$ as $\mathbb{C}G$-modules, where $\dim_{\mathbb{C}} S_i = f_i$ for each i. Any $\mathbb{C}G$-module can be written uniquely in the form $a_1 S_1 \oplus \ldots \oplus a_r S_r$, where the a_i are non-negative integers.*

Proof. □

Degrees of G

- The $\mathbb{C}$-dimensions $f_1, \ldots, f_r$ of the r simple $\mathbb{C}G$-modules are called the *degrees* of G.
- The trivial $\mathbb{C}G$-module $\mathbb{C}$ is one-dimensional and hence simple, so G always has at least one degree equal to 1; by convention we set $f_1 = 1$.

Corollary 14.2. *We have $\sum_{i=1}^{r} f_i^2 = |G|$.*

Proof. □

Degrees and conjugacy classes

- In fact the degrees of G must divide $|G|$; we shall not use this fact, but we provide a proof of it in the Appendix.
- The next theorem links the number of simple $\mathbb{C}G$-modules to the conjugacy structure of G via the center of $\mathbb{C}G$.

Theorem 14.3. *The number r of simple $\mathbb{C}G$-modules is equal to the number of conjugacy classes of G.*

Proof. □

Defining characters

- In general there is no natural bijective correspondence between the conjugacy classes of G and the simple $\mathbb{C}G$-modules. However, if G is a symmetric group, then there is such a correspondence, although we shall not develop it here; for an overview, see [13, Section 4.1].
- If U is a $\mathbb{C}G$-module, then each $g \in G$ defines an invertible linear transformation of U that sends $u \in U$ to gu. The *character* of U is the function $\chi_U : G \to \mathbb{C}$, where $\chi_U(g)$ is the trace of this linear transformation of U defined by g.
- For example, $\chi_U(1) = \dim_{\mathbb{C}} U$, since the identity element of G induces the identity transformation on U.
- If $\rho\colon G \to \mathrm{GL}(U)$ is the representation corresponding to U, then $\chi_U(g)$ is just the trace of the map $\rho(g)$.
- Isomorphic $\mathbb{C}G$-modules have equal characters.
- For any $g, h \in G$ the linear transformations of U defined by g and hgh^{-1} are similar and hence have the same trace; so any character of G is constant on each conjugacy class of G.
- For example, let $U = \mathbb{C}G$ and let $g \in G$. With respect to the basis G of $\mathbb{C}G$, $\chi_U(g)$ equals the number of elements $x \in G$ for which $gx = x$. Therefore $\chi_U(1) = |G|$ and $\chi_U(g) = 0$ for every $1 \neq g \in G$. This character is called the *regular character* of G.
- The theory of characters was developed by Frobenius and others, starting in 1896. At first nothing was known about linear representations, let alone modules over the group algebra; Frobenius defined characters as being functions from G to $\mathbb{C}$ satisfying certain properties, but it turned out that his characters were exactly the trace functions of finitely generated $\mathbb{C}G$-modules.
- We will denote the characters of the r simple $\mathbb{C}G$-modules by $\chi_1, \dots, \chi_r$; these characters will be referred to as the *irreducible characters* of G. Whenever we say that $S_1, \dots, S_r$ are the distinct simple $\mathbb{C}G$-modules, we implicitly order them so that $\chi_{S_i} = \chi_i$ for each i.
- In keeping with the convention that f_1 denotes the degree of the trivial representation, we let χ_1 be the character of the trivial representation; we call χ_1 the *principal character* of G, and we have $\chi_1(g) = 1$ for all $g \in G$.
- A character of a one-dimensional $\mathbb{C}G$-module is called a *linear character*. Since one-dimensional modules are simple, all linear characters are irreducible. If χ arises from a one-dimensional $\mathbb{C}G$-module U, then for $g, h \in G$ and $u \in U$ we have $gu = \chi(g)u$ and $hu = \chi(h)u$, so $\chi(gh)u = (gh)u = \chi(g)\chi(h)u$; hence χ is a homomorphism from G to $\mathbb{C}^\times$. Conversely, given a homomorphism $\varphi\colon G \to \mathbb{C}^\times$, we can define a one-dimensional $\mathbb{C}G$-module U by $gu = \varphi(g)u$ for $g \in G$ and $u \in U$, and we then have $\chi_U = \varphi$. Therefore, linear characters of G are exactly the same as group homomorphisms from G to $\mathbb{C}^\times$.

Proposition 14.4. *Let U be a $\mathbb{C}G$-module, let $\rho\colon G \to \mathrm{GL}(U)$ be the representation corresponding to U, and let $g \in G$ be of order n. Then:*

(i) *$\rho(g)$ is diagonalizable.*

(ii) *$\chi_U(g)$ equals the sum (with multiplicities) of the eigenvalues of $\rho(g)$.*

(iii) *$\chi_U(g)$ is a sum of $\chi_U(1)$ nth roots of unity.*

(iv) *$\chi_U(g^{-1}) = \overline{\chi_U(g)}$. (Here $\overline{z}$ denotes the complex conjugate of $z \in \mathbb{C}$.)*

(v) *$|\chi_U(g)| \leq \chi_U(1)$.*

(vi) *$\{x \in G \mid \chi_U(x) = \chi_U(1)\}$ is a normal subgroup of G.*

Proof. □

Algebra of characters

- If χ and ψ are characters of G, we can define new functions $\chi + \psi$ and $\chi\psi$ from G to $\mathbb{C}$ by $(\chi + \psi)(g) = \chi(g) + \psi(g)$ and $(\chi\psi)(g) = \chi(g)\psi(g)$ for $g \in G$.
- These new functions obtained from characters are not, a priori, characters themselves.
- Given a scalar $\lambda \in \mathbb{C}$, we can also define a new function $\lambda\chi\colon G \to \mathbb{C}$ by $(\lambda\chi)(g) = \lambda\chi(g)$, so we view characters of G as elements of a $\mathbb{C}$-vector space of functions from G to $\mathbb{C}$.

Proposition 14.5. *The irreducible characters of G are, as functions from G to $\mathbb{C}$, linearly independent over $\mathbb{C}$.*

Proof. □

Lemma 14.6. $\chi_{U\oplus V} = \chi_U + \chi_V$ *for any $\mathbb{C}G$-modules U and V.*

Proof. □

Characters distinguish modules

- Characters of $\mathbb{C}G$-modules suffice to distinguish $\mathbb{C}G$-modules up to isomorphism.

Theorem 14.7. *If $S_1, \ldots, S_r$ are the distinct simple $\mathbb{C}G$-modules, then the character of the $\mathbb{C}G$-module $a_1S_1 \oplus \ldots \oplus a_rS_r$ (where the a_i are non-negative integers) is $a_1\chi_1 + \ldots + a_r\chi_r$. Consequently, two $\mathbb{C}G$-modules are isomorphic iff their characters are equal.*

Proof. □

Tensor products, duals, and Hom

- Although each character determines a $\mathbb{C}G$-module up to isomorphism by Theorem 14.7, there is no generic way to construct a module from its corresponding character, and so in some sense information is lost by studying characters in lieu of modules.
- Characters turn out to be an efficient means of translating information about the ordinary representation theory of G to information about G itself; from now on we focus on characters rather than modules.
- Recall from Section 12 that if U and V are $\mathbb{C}G$-modules, then $U \otimes_{\mathbb{C}} V$ and $\mathrm{Hom}_{\mathbb{C}}(U, V)$, written here as $U \otimes V$ and $\mathrm{Hom}(U, V)$, admit natural $\mathbb{C}G$-module structures.

Proposition 14.8. *Let U and V be $\mathbb{C}G$-modules. Then:*

(i) $\chi_{U\otimes V} = \chi_U\chi_V$.

(ii) $\chi_{U^*} = \overline{\chi_U}$. *(Recall that $U^* = \mathrm{Hom}(U, F)$.)*

(iii) $\chi_{\mathrm{Hom}(U,V)} = \overline{\chi_U}\chi_V$.

Proof. □

Virtual characters and class functions

- A *virtual character* of G is a $\mathbb{Z}$-linear combination of the irreducible characters of G. (Some authors prefer the term "generalized character.") Characters are virtual characters by Theorem 14.7.

Corollary 14.9. *The virtual characters of G form a ring.*

Proof. □

Class functions

- A *class function* on G is a function from G to $\mathbb{C}$ whose value within any conjugacy class is constant.
- Characters of $\mathbb{C}G$-modules are class functions.
- The set of class functions on G forms a $\mathbb{C}$-vector space of dimension r, where r is the number of conjugacy classes of G; an obvious basis is the set of functions taking the value 1 on a single conjugacy class and 0 on all other classes.

Proposition 14.10. *The irreducible characters of G form a basis for the space of class functions on G.*

Proof. □

Inner product on class functions

- If α and β are class functions on G, then their *inner product* is the complex number

$$(\alpha, \beta) = \frac{1}{|G|} \sum_{g \in G} \alpha(g)\overline{\beta(g)}.$$

- This function $(\,\cdot\,,\,\cdot\,)$ is, as the name suggests, an inner product on the space of class functions:
  - $(\alpha, \alpha) \geq 0$ for all α, and $(\alpha, \alpha) = 0$ iff $\alpha = 0$.
  - $(\alpha, \beta) = \overline{(\beta, \alpha)}$ for all α, β.
  - $(\lambda\alpha, \beta) = \lambda(\alpha, \beta)$ for all α, β and all $\lambda \in \mathbb{C}$.
  - $(\alpha_1 + \alpha_2, \beta) = (\alpha_1, \beta) + (\alpha_2, \beta)$ for all $\alpha_1, \alpha_2, \beta$.
- The following are easy consequences of the above properties:
  - $(\alpha, \lambda\beta) = \overline{\lambda}(\alpha, \beta)$ for all α, β and all $\lambda \in \mathbb{C}$.
  - $(\alpha, \beta_1 + \beta_2) = (\alpha, \beta_1) + (\alpha, \beta_2)$ for all α, β_1, β_2.
- We need a small piece of notation: if U is a $\mathbb{C}G$-module, the set of elements of U on which G acts trivially is a $\mathbb{C}G$-submodule, denoted $U^G = \{u \in U \mid gu = u \text{ for all } g \in G\}$.

Lemma 14.11. *If U is a $\mathbb{C}G$-module, then* $\dim_{\mathbb{C}} U^G = \frac{1}{|G|} \sum_{g \in G} \chi_U(g)$.

Proof. □

Theorem 14.12. *We have* $(\chi_U, \chi_V) = \dim_{\mathbb{C}} \operatorname{Hom}_{\mathbb{C}G}(U, V)$ *for any $\mathbb{C}G$-modules U and V.*

Proof. □

Exercises

Throughout these exercises, G denotes a finite group, and all modules are finitely generated.

Exercise 14.1. Let A be a semisimple $\mathbb{C}$-algebra. Let $[A, A]$ be the subspace of A spanned by the set $\{ab - ba \mid a, b \in A\}$. Show that the codimension of $[A, A]$ in A (which is just the dimension of $A/[A, A]$) is equal to the number of isomorphism classes of simple A-modules.

Solution. □

Exercise 14.2. (cont.) Let $A = \mathbb{C}G$, and let $g_1, \ldots, g_r$ be representatives of the r conjugacy classes of G. Show that $\{g_i + [A, A] \mid 1 \leq i \leq r\}$ is a basis of $A/[A, A]$.

Solution. □

Exercise 14.3. Show that if S is a simple $\mathbb{C}G$-module and U is a one-dimensional $\mathbb{C}G$-module, then $S \otimes U$ is simple.

Solution. □

Exercise 14.4. Show that the set of isomorphism classes of the one-dimensional $\mathbb{C}G$-modules forms a group under the taking of tensor products.

Solution. □

Exercise 14.5. Let χ be an irreducible character of G. Show that if λ is any $|G|$th root of unity, then $\{x \in G \mid \chi(x) = \lambda\chi(1)\} \trianglelefteq G$.

Solution. □

Exercise 14.6. Show that a $\mathbb{C}G$-module U is simple iff its dual U^* is simple. Conclude that the complex conjugate of an irreducible character is again an irreducible character.

Solution. □

If F is any field and U is any FG-module, then we define the character of U to be an F-valued function on U whose value at $g \in G$ is the trace of the F-endomorphism of U induced by g, exactly as was done in this section in the case $F = \mathbb{C}$.

Exercise 14.7. Suppose that F has characteristic $p > 0$. Give an example to demonstrate that two FG-modules that have the same character need not be isomorphic.

Solution. □

Exercise 14.8. Suppose that F has characteristic $p > 0$. Show that if U is an FG-module then we have $\chi_U(g) = \chi_U(g_{p'})$ for any $g \in G$, where $g_{p'}$ is the p'-part of g as defined in Exercise 1.4.

Solution. □

15 The Character Table

Definition of the Character Table

- Any character of G is a $\mathbb{Z}$-linear combination of the r irreducible characters $\chi_1, \ldots, \chi_r$, where r is the number of conjugacy classes of G (Theorem 14.3).
- Each irreducible character is determined by its values on the r conjugacy classes, so the irreducible characters are completely determined by an $r \times r$ array.
- The *character table* of G is this $r \times r$ array $\mathcal{X} = (\chi_i(g_j))_{1 \le i,j \le r}$, where $g_1, \ldots, g_r$ are conjugacy class representatives.
- The character table is well-defined only up to reordering of rows and columns.
- By convention, $g_1 = 1$, so the first column consists of the degrees of G.
- We write $\mathcal{X}$ in the form

$$\begin{array}{c|cccc} & 1 & k_2 & \cdots & k_r \\ & 1 & g_2 & \cdots & g_r \\ \hline \chi_1 & 1 & 1 & \cdots & 1 \\ \chi_2 & f_2 & \chi_2(g_2) & \cdots & \chi_2(g_r) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \chi_r & f_r & \chi_r(g_2) & \cdots & \chi_r(g_r) \end{array}$$

  where the f_i are the degrees of G, and $k_i = |G : C_G(g_i)|$ is the order of the conjugacy class of g_i (Proposition 3.13).
- By convention χ_1 is always the principal character.

Orthogonality of Rows

Theorem (Row Orthogonality Theorem). *$(\chi_i, \chi_j) = \delta_{ij}$ for any i and j.*

Proof. □

In other words, this theorem asserts that

$$\delta_{ij} = \frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_j(g)} = \frac{1}{|G|} \sum_{t=1}^{r} k_t \chi_i(g_t)\overline{\chi_j(g_t)}$$

for any i and j, where the g_t are conjugacy class representatives and the k_t are the orders of the conjugacy classes. The rows of the character table, considered as vectors in $\mathbb{C}^r$, are orthogonal with respect to an inner product that differs slightly from the standard one.

Corollary 15.1. *The irreducible characters of G form an orthonormal basis for the vector space of class functions on G.*

Proof. □

Corollary 15.2. *If $\alpha = \sum_i a_i \chi_i$ and $\beta = \sum_j b_j \chi_j$ are virtual characters of G, then $(\alpha, \beta) = \sum_i a_i b_i$.*

Proof. □

Corollary 15.3. *If α is a character of G and $n \in \{1, 2, 3\}$, then $(\alpha, \alpha) = n$ iff α is a sum of n irreducible characters.*

Proof. □

Corollary 15.4. *If α is a virtual character of G, then each χ_j appears with coefficient (α, χ_j) in the unique expression of α as a $\mathbb{Z}$-linear combination of the irreducible characters of G.*

Proof. □

Proposition 15.5. *If α is a linear character of G and χ is an irreducible character of G, then $\alpha\chi$ is an irreducible character of G.*

Proof. □

Orthogonality of Columns

Theorem (Column Orthogonality Theorem)**.** *If $g_1, \ldots, g_r$ are a set of conjugacy class representatives of G, and $k_1, \ldots, k_r$ are the orders of the conjugacy classes, then for any $1 \leq i, j \leq r$ we have*

$$\sum_{t=1}^{r} \chi_t(g_i)\overline{\chi_t(g_j)} = \frac{|G|}{k_i}\delta_{ij} = |C_G(g_i)|\delta_{ij}.$$

Proof. □

Characters and Quotient Groups

Lemma 15.6. *Let $N \trianglelefteq G$, and let U be a $\mathbb{C}(G/N)$-module. Then U admits a canonical $\mathbb{C}G$-module structure, with a subspace of U being a $\mathbb{C}G$-submodule iff it is a $\mathbb{C}(G/N)$-submodule. If ψ is the character of the $\mathbb{C}(G/N)$-module U, then the character of the $\mathbb{C}G$-module U is $\psi \circ \eta$, where $\eta\colon G \to G/N$ is the natural map.*

Proof. □

We define $K_\chi = \{x \in G \mid \chi(x) = \chi(1)\}$ for any character χ of G; we have $K_\chi \trianglelefteq G$ by part (vi) of Proposition 14.4. We call K_χ the *kernel* of χ, as it is the kernel of the corresponding representation. We write K_i instead of K_{χ_i}.

Proposition 15.7. *The normal subgroups of G are exactly the sets of the form $\bigcap_{i\in I} K_i$ for some $I \subseteq \{1, \ldots, r\}$.*

Proof. □

Corollary 15.8. *G is simple iff the only irreducible character χ_i for which $\chi_i(g) = \chi_i(1)$ for some $1 \neq g \in G$ is the principal character χ_1.*

Proof. □

Corollary 15.9. *The character table of G can be used to determine whether or not G is solvable.*

Proof. □

We define $Z_\chi = \{x \in G \mid |\chi(x)| = \chi(1)\}$ for any character χ of G. Again we write Z_i instead of Z_{χ_i}.

Lemma 15.10. *$Z_\chi \leq G$ for any character χ of G, and if in addition χ is irreducible, then $Z_\chi/K_\chi = Z(G/K_\chi)$.*

Proof. □

Corollary 15.11. *If G is non-abelian and simple, then $Z_i = 1$ for all $i > 1$.*

Proof. □

Proposition 15.12. *$Z(G) = \bigcap_{i=1}^{r} Z_i$.*

Proof. □

Lemma 15.13. *If $N \trianglelefteq G$, then the irreducible characters of G/N can be determined from those of G.*

Proof. □

Corollary 15.14. *The character table of G can be used to determine whether or not G is nilpotent.*

Proof. □

Observe that in Corollary 15.9 we assumed that we knew both the irreducible characters of G and the orders of the conjugacy classes, whereas in Corollary 15.14 we did not need to know the orders of the classes. As evidenced by Lemma 15.13, there may be circumstances in which we can construct a character table without knowing the orders of the conjugacy classes. There is a theorem of G. Higman (see [15, p. 136]) which asserts that if we know the irreducible characters of G, but not the orders of the conjugacy classes, then we can at least determine the sets of prime divisors of the orders of the conjugacy classes.

Examples: Constructing Character Tables

The remainder of this section consists of a series of examples in which we develop methods of finding characters and use these methods to construct the character tables of various groups.

Example 15.1. Let $G = \langle g \rangle \cong \mathbb{Z}_n$ for some $n \in \mathbb{N}$, and let λ be a primitive nth root of unity. For each $1 \leq i \leq n$, let V_i be a one-dimensional $\mathbb{C}$-vector space, and let g act on V_i by multiplication by λ^{i-1}; since G is cyclic, this definition completely determines a $\mathbb{C}G$-module structure on V_i. Each V_i, being one-dimensional, is a simple $\mathbb{C}G$-module. If χ_i is the character of V_i, then $\chi_i(g) = \lambda^{i-1}$, and hence $\chi_i(g^a) = \lambda^{a(i-1)}$ for any a. The characters $\chi_1, \ldots, \chi_n$ are n distinct linear characters of G, and since G can have at most $|G| = n$ irreducible characters, we see that the χ_i are precisely the irreducible characters of G. Thus, the character table of G is

$$
\begin{array}{c|cccccc}
 & 1 & 1 & 1 & \cdots & 1 \\
 & 1 & g & g^2 & \cdots & g^{n-1} \\
\hline
\chi_1 & 1 & 1 & 1 & \cdots & 1 \\
\chi_2 & 1 & \lambda & \lambda^2 & \cdots & \lambda^{n-1} \\
\chi_3 & 1 & \lambda^2 & \lambda^4 & \cdots & \lambda^{n-2} \\
\vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\
\chi_n & 1 & \lambda^{n-1} & \lambda^{n-2} & \cdots & \lambda
\end{array}
$$

Observe that the set $\{\chi_1, \ldots, \chi_n\}$ is a cyclic group under multiplication of characters, with generator χ_2; we have $\chi_i = \chi_2^{i-1}$ for any i.

Example 15.2. Let G and H be groups with respective irreducible characters $\chi_1, \ldots, \chi_r$ and $\psi_1, \ldots, \psi_s$. We wish to determine the irreducible characters of $G \times H$. We see that two elements (x, y) and (x', y') of $G \times H$ are conjugate iff x and x' are conjugate in G and y and y' are conjugate in H. Since by Theorem 14.3, G and H have r and s conjugacy classes, respectively, we conclude that $G \times H$ has rs conjugacy classes, with each class of $G \times H$ being the product of a class of G and a class of H; hence by Theorem 14.3, $G \times H$ has rs irreducible characters.

Let $S_1, \ldots, S_r$ and $T_1, \ldots, T_s$ be the distinct simple $\mathbb{C}G$- and $\mathbb{C}H$-modules, respectively. For each i and j, we give $S_i \otimes T_j$ the structure of a $\mathbb{C}(G \times H)$-module by $(g, h)(s \otimes t) = gs \otimes ht$ and linear extension. Let τ_{ij} be the character of $S_i \otimes T_j$ for each i and j. By imitating the proof of part (i) of Proposition 14.8, we find that $\tau_{ij}(g, h) = \chi_i(g)\psi_j(h)$; we adopt the notation $\tau_{ij} = \chi_i \times \psi_j$. We note that the τ_{ij} are distinct and that we can recover the χ_i and ψ_j from the τ_{ij} by appropriate restriction. Now for any i, i', j, j' we have

$$
\begin{aligned}
(\tau_{ij}, \tau_{i'j'}) &= \frac{1}{|G \times H|} \sum_{(g,h) \in G \times H} \tau_{ij}(g, h)\overline{\tau_{i'j'}(g, h)} \\
&= \frac{1}{|G||H|} \sum_{g \in G} \sum_{h \in H} \chi_i(g)\psi_j(h)\overline{\chi_{i'}(g)\psi_{j'}(h)} \\
&= \left(\frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_{i'}(g)} \right) \left(\frac{1}{|H|} \sum_{h \in H} \psi_j(h)\overline{\psi_{j'}(h)} \right) \\
&= (\chi_i, \chi_{i'})(\psi_j, \psi_{j'}) = \delta_{ii'}\delta_{jj'}
\end{aligned}
$$

by Row Orthogonality Theorem. In particular, we have $(\tau_{ij}, \tau_{ij}) = 1$ for each i and j, which by Corollary 15.3 implies that τ_{ij} is an irreducible character. Therefore the τ_{ij} are the rs irreducible characters of $G \times H$; that is, the set of irreducible characters of $G \times H$ is $\{\chi_i \times \psi_j\}_{i,j}$.

Example 15.3. Suppose that G is abelian. Then every conjugacy class of G contains exactly one element, and consequently G has $|G|$ irreducible characters by Theorem 14.3. But $\sum_{i=1}^{|G|} f_i^2 = |G|$ by Corollary 14.2, so we must have $f_i = 1$ for each i. Therefore all irreducible characters of G are linear. By Corollary 8.8, G is a direct product of cyclic p-groups; let these cyclic p-groups be of orders $p_1^{a_1}, \ldots, p_t^{a_t}$ with respective generators $g_1, \ldots, g_t$. We could determine the character table of G from those of the cyclic p-groups using Examples 15.1 and 15.2, but there is an alternate method, which we now describe. A linear character χ of G is just a homomorphism from G to $\mathbb{C}^\times$, and hence to define such a χ it suffices to specify $\chi(g_i)$ for each i, with the only restriction on $\chi(g_i)$ being that it is a $p_i^{a_i}$-th root of unity.

Therefore, there is a bijective correspondence between irreducible characters of G and ordered t-tuples $(\lambda_1, \dots, \lambda_t)$ in which each λ_i is a $p_i^{a_i}$-th root of unity.

As a simple example, let $G = \langle a, b \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Both a and b have order 2, so by the above paragraph we see that the four irreducible characters of G correspond to the ordered pairs $(1, 1), (1, -1), (-1, 1)$, and $(-1, -1)$. Therefore, the character table for G is

	1	1	1	1
	1	a	b	ab
χ_1	1	1	1	1
χ_2	1	1	-1	-1
χ_3	1	-1	1	-1
χ_4	1	-1	-1	1

Here the second and third columns correspond to the ordered pairs given above, and the fourth column is wholly determined by the previous two columns.

Example 15.4. Let X be a finite G-set. We saw in Section 12 that the vector space $\mathbb{C}X$ having basis X is a $\mathbb{C}G$-module. We wish to determine the character π of $\mathbb{C}X$. Consider the matrix, with respect to the basis X, of the linear transformation of $\mathbb{C}X$ defined by $g \in G$. Since $gx \in X$ for every $x \in X$, this matrix is a permutation matrix; but $\pi(g)$ is the sum of the diagonal entries of this matrix, so we conclude that $\pi(g)$ equals the number of elements of X which are fixed by g.

Let χ_1 be the principal character of G. We have

$$\begin{aligned}(\pi, \chi_1) &= \frac{1}{|G|} \sum_{g \in G} \pi(g)\chi_1(g) = \frac{1}{|G|} \sum_{g \in G} \pi(g) \\ &= \frac{1}{|G|} \sum_{g \in G} |\{x \in X \mid gx = x\}| \\ &= \frac{1}{|G|} |\{(g, x) \in G \times X \mid gx = x\}| \\ &= \frac{1}{|G|} \sum_{x \in X} |\{g \in G \mid gx = x\}|.\end{aligned}$$

For each $x \in X$, let G_x be the stabilizer of x; using Corollary 3.5, we now have

$$(\pi, \chi_1) = \sum_{x \in X} \frac{|G_x|}{|G|} = \sum_{\text{orbits } \mathcal{O}} \sum_{x \in \mathcal{O}} \frac{1}{|G : G_x|} = \sum_{\text{orbits } \mathcal{O}} \sum_{x \in \mathcal{O}} \frac{1}{|\mathcal{O}|} = \sum_{\text{orbits } \mathcal{O}} |\mathcal{O}| \cdot \frac{1}{|\mathcal{O}|} = \sum_{\text{orbits } \mathcal{O}} 1,$$

which allows us to conclude that (π, χ_1) is equal to the number of orbits of X under the action of G. (This result is often attributed to Burnside, although as argued in [21] it would be more accurate to call this result the Cauchy-Frobenius lemma.) In particular, if G acts transitively on X, then we see via Corollary 15.4 that the unique expression of the character $\pi - \chi_1$ as a linear combination of the irreducible characters of G does not involve χ_1.

Example 15.5. The group G/G' is abelian by Proposition 2.6, and hence (as seen in Example 15.3) all of its irreducible characters are linear. It follows from Lemma 15.6 that each linear character of G/G' can be lifted to a linear character of G, and that the lifts of distinct characters are distinct. Now let χ be a linear character of G, so that $\chi\colon G \to \mathbb{C}^\times$ is a homomorphism whose kernel is K_χ. Then G/K_χ is abelian, being isomorphic with X, a subgroup of $\mathbb{C}^\times$, and hence $G' \le K_\chi$ by Proposition 2.6. We can now define a linear character ψ of G/G' whose lift is χ by $\psi(gG') = \chi(g)$. We conclude that every linear character of G is the lift of a linear character of G/G'.

Example 15.6. Consider S_3. The conjugacy classes of S_3 are $\{1\}$, $\{(1\,2), (1\,3), (2\,3)\}$, and $\{(1\,2\,3), (1\,3\,2)\}$. (This follows from Proposition 1.10.) We have $S_3/A_3 \cong \mathbb{Z}_2$, so S_3 has two linear characters, which are lifted from the characters of $\mathbb{Z}_2$ as in Lemma 15.6. Letting these characters be χ_1 and χ_2, we have

	1	3	2
	1	$(1\,2)$	$(1\,2\,3)$
χ_1	1	1	1
χ_2	1	-1	1
χ_3			

Since $6 = |S_3| = f_1^2 + f_2^2 + f_3^2 = 2 + f_3^2$ by Corollary 14.2, we have $f_3 = 2$. Column Orthogonality Theorem now gives

$$0 = \sum_{i=1}^{3} f_i\chi_i((1\,2)) = 1 \cdot 1 + 1 \cdot (-1) + 2\chi_3((1\,2))$$

and

$$0 = \sum_{i=1}^{3} f_i\chi_i((1\,2\,3)) = 1 \cdot 1 + 1 \cdot 1 + 2\chi_3((1\,2)),$$

from which we see that the character table of S_3 is

	1	3	2
	1	(1 2)	(1 2 3)
χ_1	1	1	1
χ_2	1	-1	1
χ_3	2	0	-1

Example 15.7. Consider S_4. Recall that S_4 has a normal subgroup of order 4, namely

$$K = \{1,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}.$$

The subgroup S_3 of S_4 intersects K trivially, and $|S_4| = |S_3||K|$; therefore $S_4 = K \rtimes S_3$. In particular, we have $S_4/K \cong S_3$, and thus we can obtain irreducible characters of S_4 from the irreducible characters of S_3 as in Lemma 15.6.

By Proposition 1.10, each conjugacy class of S_4 consists of all elements having a given cycle structure. Hence S_4 has 5 classes, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, $(1\,2)$, and $(1\,2\,3\,4)$; the orders of the classes are 1, 3, 8, 6, and 6, respectively.

Let χ_1, χ_2, and χ_3 be the irreducible characters of S_4 obtained from those of S_3. To determine these characters, we need information about the images in S_4/K of the class representatives. We easily see that the images of 1 and $(1\,2)(3\,4)$ are trivial, that the image of $(1\,2\,3)$ has order 3, and that the images of $(1\,2)$ and $(1\,2\,3\,4)$ have order 2.

Using this observation and the character table for S_3 obtained in Example 15.6, we obtain the following partial character table for S_4:

	1	3	8	6	6
	1	(1 2)(3 4)	(1 2 3)	(1 2)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	2	2	-1	0	0
χ_4					
χ_5					

Now S_4 acts transitively on the set $X = \{1, 2, 3, 4\}$; let π be the character of the $\mathbb{C}G$-module $\mathbb{C}X$. Since $\pi(g)$ equals the number of fixed points under the action of g on X by Example 15.4, we have

	1	3	8	6	6
	1	(1 2)(3 4)	(1 2 3)	(1 2)	(1 2 3 4)
π	4	0	1	2	0
$\pi - \chi_1$	3	-1	0	1	-1

We also have

$$(\pi - \chi_1, \pi - \chi_1) = \frac{1}{|S_4|}\sum_{i=1}^{5} k_i[(\pi - \chi_1)(g_i)]^2 = \frac{1}{24}(1 \cdot 9 + 3 \cdot 1 + 8 \cdot 0 + 6 \cdot 1 + 6 \cdot 1) = 1,$$

and hence $\pi - \chi_1$ is irreducible by Corollary 15.3. Let $\chi_4 = \pi - \chi_1$. Now $\chi_2\chi_4 \neq \chi_4$, and $\chi_2\chi_4$ is irreducible by Proposition 15.5; therefore $\chi_5 = \chi_2\chi_4$, and hence the character table of S_4 is

	1	3	8	6	6
	1	(1 2)(3 4)	(1 2 3)	(1 2)	(1 2 3 4)
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	2	2	-1	0	0
χ_4	3	-1	0	1	-1
χ_5	3	-1	0	-1	1

Example 15.8. Let G be a non-abelian group of order 8. We will show that this information completely determines the character table of G. Since there are two isomorphism classes of non-abelian groups of order 8 (see page 79), this will imply that the character table does not specify a group up to isomorphism.

We have $|Z(G)| = 2$ by Theorem 8.1 and Lemma 8.2, and hence $Z(G) \cong \mathbb{Z}_2$. Since G is non-abelian and $G/Z(G)$ is abelian (being a group of order 4), this forces $G' = Z(G)$ by Proposition 2.6. Now $|G/G'| = 4$, and by Lemma 8.2 G/G' cannot be cyclic since G is non-abelian and $G' = Z(G)$, so we must have $G/G' \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. By Examples 15.3 and 15.5, we now know that G has exactly 4 linear characters. By Corollary 14.2, we have $\sum_{i=1}^{r} f_i^2 = 8$, where $f_i = 1$ for $1 \leq i \leq 4$ and $f_i > 1$ for $i > 4$; this forces $r = 5$ and $f_5 = 2$.

Let x be the generator of G'. As $G' = Z(G)$, we see that 1 and x are the only elements of G lying in single-element conjugacy classes, and hence each of the other three classes must have order 2. Let a, b, and c be representatives of the multi-element classes. Since G/G' also has four conjugacy classes, we see that the images of a, b, and c in G/G' must lie in distinct conjugacy classes. Using the character table for $\mathbb{Z}_2 \times \mathbb{Z}_2$ determined in Example 15.3, we can now obtain a partial character table for G by lifting:

	1	1	2	2	2
	1	x	a	b	c
χ_1	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2				

We see via Column Orthogonality Theorem that $\chi_5(a) = \chi_5(b) = \chi_5(c) = 0$ and that $\chi_5(x) = -2$, completing the character table.

Example 15.9. Consider the group A_5. Using Proposition 1.10, we find that the elements of A_5 form 4 conjugacy classes in S_5, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, and $(1\,2\,3\,4\,5)$ and respective orders 1, 15, 20, and 24. However, elements of A_5 that are conjugate in S_5 may not be conjugate in A_5.

Let $x \in A_5$, and let $\mathcal{K}$ be the conjugacy class in S_5 of x. We see from Proposition 3.13 that $|\mathcal{K}| = |S_5 : C_{S_5}(x)|$ and that the conjugacy class in A_5 of x is contained in $\mathcal{K}$ and has order $|A_5 : C_{A_5}(x)|$. Suppose first that $C_{S_5}(x) \not\leq A_5$. This forces $A_5 C_{S_5}(x) = S_5$ since A_5 is a maximal subgroup of S_5. The first isomorphism theorem gives $S_5/A_5 = A_5 C_{S_5}(x)/A_5 \cong C_{S_5}(x)/C_{S_5}(x) \cap A_5 = C_{S_5}(x)/C_{A_5}(x)$, and hence

$$\begin{aligned} |A_5 : C_{A_5}(x)| &= \frac{|S_5 : C_{A_5}(x)|}{|S_5 : A_5|} = \frac{|S_5 : C_{A_5}(x)|}{|C_{S_5}(x) : C_{A_5}(x)|} = |S_5 : C_{S_5}(x)| \\ &= |\mathcal{K}|. \end{aligned}$$

Thus $\mathcal{K}$ is the conjugacy class in A_5 of x. Now suppose $C_{S_5}(x) \leq A_5$. Then we have $C_{A_5}(x) = C_{S_5}(x) \cap A_5 = C_{S_5}(x)$, and hence

$$|A_5 : C_{A_5}(x)| = |A_5 : C_{S_5}(x)| = \tfrac{1}{2}|S_5 : C_{S_5}(x)| = \tfrac{1}{2}|\mathcal{K}|,$$

from which we see that $\mathcal{K}$ splits into two conjugacy classes in A_5 of equal cardinality.

We easily see that each of the elements 1, $(1\,2)(3\,4)$, and $(1\,2\,3)$ commutes with some odd permutation in S_5; for example, $(4\,5)$ and $(1\,2\,3)$ commute. Hence by the above paragraph, the conjugacy classes in A_5 of these elements are the respective classes in S_5. Since $24 \nmid 60$, we can now conclude that the conjugacy class in S_5 of $(1\,2\,3\,4\,5)$ splits into two classes in A_5, and hence that the conjugacy classes of A_5 have orders 1, 15, 20, 12, and 12. (Using this fact and the remark made after Proposition 3.13, we could now give a slick proof of the simplicity of A_5.) We will now show that $x = (1\,2\,3\,4\,5)$ and $x^2 = (1\,3\,5\,2\,4)$ are not conjugate in A_5, thereby establishing that the conjugacy class in A_5 of x splits in A_5 into the class containing x and the class containing x^2.

Suppose that $g \in A_5$ is such that $gxg^{-1} = x^2$. Then we have $g\langle x\rangle g^{-1} = \langle x^2\rangle = \langle x\rangle$, and hence $g \in N_{A_5}(\langle x\rangle)$. Now $\langle x\rangle$ is a Sylow 5-subgroup of A_5, and we observed in the proof of Theorem 7.4 that A_5 has 6 Sylow 5-subgroups. Thus $|N_{A_5}(\langle x\rangle)| = 10$, and we find that $N_{A_5}(\langle x\rangle) = \langle x\rangle \rtimes \langle t\rangle \cong D_{10}$, where $t = (2\,5)(3\,4)$. But $txt^{-1} = x^{-1} \neq x^2$; from this, we see that there is no element $g \in N_{A_5}(\langle x\rangle)$ such that $gxg^{-1} = x^2$. Contradiction; therefore x and x^2 are not conjugate in A_5.

We now see that A_5 has 5 conjugacy classes, with representatives 1, $(1\,2)(3\,4)$, $(1\,2\,3)$, $(1\,2\,3\,4\,5)$, and $(1\,3\,5\,2\,4)$ and respective orders 1, 15, 20, 12, and 12. Since A_5 is simple, we cannot produce irreducible characters of A_5 by lifting irreducible characters of quotient groups, as we have done in previous examples. However, A_5 is a group of permutations, so we have as a starting point the characters of certain obvious permutation modules.

Let $X = \{1, 2, 3, 4, 5\}$; A_5 acts transitively on X in the obvious way. Let π be the character of the $\mathbb{C}A_5$-module $\mathbb{C}X$. By Example 15.4, we see that $\pi(g)$ equals the number of fixed points of X under the action of $g \in A_5$, and hence we have

	1	15	20	12	12
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2\,3\,4\,5)$	$(1\,3\,5\,2\,4)$
π	5	1	2	0	0
$\pi - \chi_1$	4	0	1	-1	-1

We also have

$$(\pi - \chi_1, \pi - \chi_1) = \frac{1}{|A_5|} \sum_{i=1}^{5} k_i [(\pi - \chi_1)(g_i)]^2 = \frac{1}{60}(1 \cdot 16 + 15 \cdot 0 + 20 \cdot 1 + 12 \cdot 1 + 12 \cdot 1) = 1,$$

and hence $\pi - \chi_1$ is irreducible by Corollary 15.3. Let $\chi_2 = \pi - \chi_1$.

Let Y be the set of two-element subsets of X; this is a transitive A_5-set under the obvious action. Let ψ be the character of the $\mathbb{C}A_5$-module $\mathbb{C}Y$. We have

	1	15	20	12	12
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2\,3\,4\,5)$	$(1\,3\,5\,2\,4)$
ψ	10	2	1	0	0
$\psi - \chi_1$	9	1	0	-1	-1

Now

$$(\psi - \chi_1, \psi - \chi_1) = \frac{1}{|A_5|} \sum_{i=1}^{5} k_i [(\psi - \chi_1)(g_i)]^2 = \frac{1}{60}(1 \cdot 81 + 15 \cdot 1 + 20 \cdot 0 + 12 \cdot 1 + 12 \cdot 1) = 2,$$

so by Corollary 15.3, $\psi - \chi_1$ is the sum of two irreducible characters; as noted in Example 15.4, neither of these two characters is χ_1, since Y is transitive. We have

$$(\psi - \chi_1, \chi_2) = \frac{1}{|A_5|} \sum_{i=1}^{5} k_i (\psi - \chi_1)(g)\chi_2(g) = \frac{1}{60}(1 \cdot 9 \cdot 4 + 15 \cdot 1 \cdot 0 + 20 \cdot 0 \cdot 1 + 24 \cdot (-1)(-1)) = 1,$$

and therefore by Corollary 15.4 we see that $\psi - \chi_1 - \chi_2$ is an irreducible character that is neither χ_1 nor χ_2. Let $\chi_3 = \psi - \chi_1 - \chi_2$, and observe that $f_3 = 5$.

By Corollary 14.2 we have $\sum_{i=1}^{5} f_i^2 = |A_5| = 60$, which gives $f_4^2 + f_5^2 = 18$ and hence $f_4 = f_5 = 3$. Thus we have the following partial character table:

	1	15	20	12	12
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2\,3\,4\,5)$	$(1\,3\,5\,2\,4)$
χ_1	1	1	1	1	1
χ_2	4	0	1	-1	-1
χ_3	5	1	-1	0	0
χ_4	3				
χ_5	3				

Let $a = \chi_4((1\,2)(3\,4))$ and $b = \chi_5((1\,2)(3\,4))$. By Column Orthogonality Theorem, we have

$$0 = \sum_{i=1}^{5} f_i \chi_i((1\,2)(3\,4)) = 1 \cdot 1 + 4 \cdot 0 + 5 \cdot 1 + 3a + 3b,$$

and hence $a + b = -2$. By part (iii) of Proposition 14.4, we see that each of a and b is a sum of three square roots of unity; therefore $a, b \in \{-3, -1, 1, 3\}$. But by Column Orthogonality Theorem we have

$$4 = \frac{60}{15} = \frac{|A_5|}{k_2} = \sum_{i=1}^{5} |\chi_i((1\,2)(3\,4))|^2 = 1 + 0 + 1 + a^2 + b^2,$$

and hence $a^2 + b^2 = 2$. It now follows that $a = b = -1$.

By Column Orthogonality Theorem, we have

$$3 = \frac{60}{20} = \frac{|A_5|}{k_3} = \sum_{i=1}^{5} |\chi_i((1\,2\,3))|^2 = 1 + 1 + 1 + |\chi_4((1\,2\,3))|^2 + |\chi_5((1\,2\,3))|^2;$$

thus $|\chi_4((1\,2\,3))|^2 + |\chi_5((1\,2\,3))|^2 = 0$, and therefore we conclude that $\chi_4((1\,2\,3)) = \chi_5((1\,2\,3)) = 0$. We now have the following partial character table:

	1	15	20	12	12
	1	$(1\,2)(3\,4)$	$(1\,2\,3)$	$(1\,2\,3\,4\,5)$	$(1\,3\,5\,2\,4)$
χ_1	1	1	1	1	1
χ_2	4	0	1	-1	-1
χ_3	5	1	-1	0	0
χ_4	3	-1	0		
χ_5	3	-1	0		

Let $x = (1\,2\,3\,4\,5)$. Recall that in proving that x and x^2 are not conjugate in A_5, we found that x and x^{-1} are conjugate in A_5; hence $\chi_4(x) = \chi_4(x^{-1})$, and thus $\chi_4(x)$ is real-valued by part (iv) of Proposition 14.4. The same argument shows that $\chi_4(x^2)$, $\chi_5(x)$, and $\chi_5(x^2)$ are also real-valued. Let $c = \chi_4(x)$ and $d = \chi_4(x^2)$. Then by Row Orthogonality Theorem, we have

$$0 = \sum_{i=1}^{5} k_i \chi_4(g_i) = 1 \cdot 3 + 15 \cdot (-1) + 20 \cdot 0 + 12c + 12d,$$

which gives $c + d = 1$, and also (since χ_4 is real-valued)

$$60 = \sum_{i=1}^{5} k_i \chi_4(g_i)^2 = 1 \cdot 9 + 15 \cdot 1 + 20 \cdot 0 + 12c^2 + 12d^2,$$

which gives $c^2 + d^2 = 3$. Hence $c^2 - c - 1 = 0$, so without loss of generality we have $c = \frac{1+\sqrt{5}}{2}$ and $d = 1 - c = \frac{1-\sqrt{5}}{2}$. We can similarly show that we must have $\chi_5(x), \chi_5(x^2) \in \{\frac{1+\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2}\}$; we leave it to the reader to verify that $\chi_5(x) = \frac{1-\sqrt{5}}{2}$ and $\chi_5(x^2) = \frac{1+\sqrt{5}}{2}$, which completes the character table of A_5.

Exercises

Throughout these exercises, G denotes a finite group, and $\mathcal{X}$ denotes the character table of G.

Exercise 15.1. What is the determinant of $\mathcal{X}$?

Solution. □

Exercise 15.2. Show that the sum of the entries in any row of $\mathcal{X}$ is a non-negative integer.

Solution. □

Exercise 15.3. Compute the character of the G-set G, where G acts via conjugation, and determine the multiplicities in this character of the irreducible characters.

Solution. □

Exercise 15.4. Observe that the complex conjugate of an irreducible character is an irreducible character. (This is either Exercise 14.6 or an easy consequence of part (ii) of Proposition 14.8 and Corollary 15.3.) Let P be the $r \times r$ permutation matrix such that $P\mathcal{X}$ is the matrix obtained from $\mathcal{X}$ by interchanging two rows if their corresponding characters are conjugate. Let Q be the $r \times r$ permutation matrix such that $\mathcal{X}Q$ is the matrix obtained from $\mathcal{X}$ by interchanging two columns if their corresponding conjugacy classes are inverse, in the sense that one class consists of the inverses of the elements of the other class. Show that $P\mathcal{X} = \mathcal{X}Q$.

Solution. □

Exercise 15.5. Suppose that $|G|$ is odd. Show that the only irreducible character that is real-valued is the principal character.

Solution. $\square$

Exercise 15.6. Let χ be an irreducible character of G, and let $e \in \mathbb{C}G$ be the identity element of the corresponding matrix summand of $\mathbb{C}G$. Show that

$$e = \frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1}) g.$$

Solution. $\square$

Exercise 15.7. If $N \trianglelefteq G$ and $g \in G$, show that $|C_G(g)| \geq |C_{G/N}(gN)|$.

Solution. $\square$

Exercise 15.8. Show that if G acts doubly transitively on X, then the character of $\mathbb{C}X$ is the sum of the principal character and exactly one other irreducible character.

Solution. $\square$

Exercise 15.9. Determine the character table of D_{2n} for any $n \in \mathbb{N}$.

Solution. $\square$

Exercise 15.10. Determine the character table of S_5.

Solution. $\square$

16 Induction

Overview

- Induced modules play a critical role in the representation theory of finite groups.
- Two equivalent definitions of induction are presented: one via tensor products over arbitrary rings (slick but heavier), the other using only tensor products over fields (clumsier but more elementary). A third definition is developed in Exercise 16.1.
- Let $H \leq G$ and let V be an FH-module. Since the group algebra FG is an (FG, FH)-bimodule, we form the FG-module $FG \otimes_{FH} V$, the *induced* FG-module of V (or the *induction* of V to G), denoted $\mathrm{Ind}_H^G V$. Other notations: V^G, $V \uparrow G$, and $V \uparrow_H^G$.
- For an FG-module U, regarding U as an FH-module yields the *restriction* $\mathrm{Res}_H^G U$. Other notations: U_H, $U \downarrow H$, and $U \downarrow_H^G$.

Lemma 16.1. *We have* $\dim_F \mathrm{Ind}_H^G V = |G : H| \dim_F V$ *for* V *an* FH*-module. Moreover, as* F*-vector spaces we have*

$$\mathrm{Ind}_H^G V = \bigoplus_{t \in T} t \otimes V,$$

where T *is a transversal for* H *in* G *and* $t \otimes V = \{t \otimes v \mid v \in V\}$.

Proof. □

Theorem (Frobenius Reciprocity Theorem)**.** *Let* U *be an* FG*-module, let* $H \leq G$*, and let* V *be an* FH*-module. Then as* F*-vector spaces we have* $\mathrm{Hom}_{FH}(V, \mathrm{Res}_H^G U) \cong \mathrm{Hom}_{FG}(\mathrm{Ind}_H^G V, U)$.

Proof. □

Induced Characters

- Restrict attention to $F = \mathbb{C}$. If V is a $\mathbb{C}H$-module with character φ, the character of $\mathrm{Ind}_H^G V$ is denoted φ^G and called an *induced character*; we have $\varphi^G(1) = |G : H|\varphi(1)$.
- If U is a $\mathbb{C}G$-module with character χ, the character of $\mathrm{Res}_H^G U$ is denoted $\chi|_H$, and $\chi|_H(1) = \chi(1)$.
- The form of "Frobenius reciprocity" actually known to Frobenius (Theorem 16.2) follows from Frobenius Reciprocity Theorem and Theorem 14.12.

Theorem 16.2. *Let* $H \leq G$*, let* U *be a* $\mathbb{C}G$*-module having character* χ*, and let* V *be a* $\mathbb{C}H$*-module having character* φ*. Then* $(\varphi, \chi|_H)_H = (\varphi^G, \chi)_G$.

Proof. □

Induction–Restriction Tables

- If $H \leq G$ and $\varphi_1, \ldots, \varphi_r$ and $\chi_1, \ldots, \chi_s$ are the irreducible characters of H and G respectively, the *induction-restriction table* of (G, H) is the $r \times s$ matrix whose (i, j)-entry gives the common value of $(\varphi_i^G, \chi_j)_G$ and $(\varphi_i, \chi_j|_H)_H$.
- By Corollary 15.4, the ith row gives the multiplicities with which the χ_j appear in the decomposition of φ_i^G, while the jth column gives the multiplicities with which the φ_i appear in the decomposition of $\chi_j|_H$.
- One can compute the table by calculating the restrictions of the χ_j and reading off the expressions for the φ_i^G in terms of the χ_j.
- For example, with $G = S_3$ and $H = S_2 \leq G$, inspection gives $\chi_1|_H = \varphi_1$, $\chi_2|_H = \varphi_2$, $\chi_3|_H = \varphi_1 + \varphi_2$, so the induction-restriction table yields $\varphi_1^G = \chi_1 + \chi_3$ and $\varphi_2^G = \chi_2 + \chi_3$.

Proposition 16.3. *Let $H \leq G$, and let V be a $\mathbb{C}H$-module having character χ. If T is a transversal for H in G, then for any $g \in G$ we have*

$$\chi^G(g) = \sum_{\substack{t \in T, \\ t^{-1}gt \in H}} \chi(t^{-1}gt).$$

Proof. □

Corollary 16.4. *Let χ be a character of $H \leq G$, and let $g \in G$. Then*

$$\chi^G(g) = \frac{1}{|H|} \sum_{\substack{x \in G, \\ x^{-1}gx \in H}} \chi(x^{-1}gx).$$

Proof. □

Proposition 16.5. *Let χ be a character of $H \leq G$, and let $g \in G$, and let s be the number of conjugacy classes of H whose members are conjugate in G to g. If $s = 0$, then $\chi^G(g) = 0$. Otherwise, if we let $h_1, \ldots, h_s$ be representatives of these s conjugacy classes of H, then*

$$\chi^G(g) = \sum_{i=1}^{s} \frac{|C_G(g)|}{|C_H(h_i)|} \chi(h_i).$$

Proof. □

Corollary 16.6. *Let χ be a character of $H \leq G$. Let $g \in G$, and suppose that the number s of conjugacy classes of H whose members are conjugate in G to g is positive. Let ℓ be the order of the conjugacy class in G of g; let $h_1, \ldots, h_s$ be representatives of these s conjugacy classes of H, and let $k_1, \ldots, k_s$ be the orders of these classes. Then*

$$\chi^G(g) = \sum_{i=1}^{s} |G : H| \frac{k_i}{\ell} \chi(h_i).$$

Proof. □

Frobenius Groups

- We end the section by presenting an important theorem on finite groups which, quite remarkably, has no known proof that does not use character theory.
- A *Frobenius group* is a group G having a subgroup H such that $H \cap gHg^{-1} = 1$ for every $g \in G - H$. We call H a *Frobenius complement* of G.
- If G is a finite Frobenius group, applying Frobenius' Theorem yields a normal subgroup N, called the *Frobenius kernel* of G. One has $G = N \rtimes H$, $N \cap H = 1$, and $|N||H| = |G|$.
- For $G = N \rtimes H$ a finite Frobenius group with Frobenius kernel N and complement H, the conjugation map $\varphi : H \to \mathrm{Aut}(N)$ has the property that for each $1 \neq h \in H$, $\varphi(h)$ is a *fixed-point-free automorphism* of N: $\varphi(h)(n) = n$ forces $n = 1$. In particular, φ is injective.
- Conversely, given finite groups N and H and a homomorphism $\varphi : H \to \mathrm{Aut}(N)$ for which every $\varphi(h)$, $h \neq 1$, is fixed-point-free, $N \rtimes_\varphi H$ is a Frobenius group.
- Example: if p, q are distinct primes with $p \equiv 1 \pmod q$, then any monomorphism $\varphi : \mathbb{Z}_q \to \mathrm{Aut}(\mathbb{Z}_p)$ is fixed-point-free, and hence the unique non-abelian group of order pq is a Frobenius group.
- A theorem of Thompson (his 1959 Chicago Ph.D. thesis) states that any finite group having a fixed-point-free automorphism of prime order is nilpotent. Hence the Frobenius kernel of a finite Frobenius group is nilpotent.

Theorem (Frobenius' Theorem)**.** *Let G be a transitive permutation group on a finite set X, and suppose that each non-identity element of G fixes at most one element of X. Then the union of the identity element and the set of elements of G that have no fixed points is a normal subgroup of G.*

Proof. □

Exercises

Throughout these exercises, G is a finite group, H is a subgroup of G, F is a field, and all FG-modules are finitely generated.

Exercise 16.1. Let V be an FH-module, and let W be an FG-module containing V. Suppose that W has the property that for any FG-module U and any $\varphi \in \mathrm{Hom}_{FH}(V, U)$, there is a unique FG-module homomorphism from W to U which extends V. (In this case, we say that W is relatively H-free with respect to V; compare with Exercises 13.1–13.2.) Show that $W \cong \mathrm{Ind}_H^G V$.

Solution. □

Exercise 16.2. Let W be an FG-module that is generated by an FH-submodule V, and suppose that $\dim_F W = |G : H| \dim_F V$. Show that we must have $W \cong \mathrm{Ind}_H^G V$.

Solution. □

Exercise 16.3. Show that $\mathrm{Hom}_{FG}(U, \mathrm{Ind}_H^G V) \cong \mathrm{Hom}_{FH}(\mathrm{Res}_H^G U, V)$ as F-vector spaces for any FG-module U and FH-module V. (Hint: Show first that if $\varphi \in \mathrm{Hom}_{FH}(\mathrm{Res}_H^G U, V)$, then the map sending $u \in U$ to $\sum_{t\in T} t \otimes \varphi(t^{-1}u)$, where T is a transversal for H in G, lies in $\mathrm{Hom}_{FG}(U, \mathrm{Ind}_H^G V)$.)

Solution. □

Exercise 16.4. Show that if X is a transitive G-set, then $FX \cong \mathrm{Ind}_{G_x}^G F$ for any $x \in X$.

Solution. □

Exercise 16.5. Compute the induction-restriction table of (A_5, A_4).

Solution. □

Exercise 16.6. Compute the character table of the unique non-abelian group of order pq, where p and q are distinct primes and $p \equiv 1 \pmod q$.

Solution. □

Exercise 16.7. Suppose that G is a Frobenius group having Frobenius kernel N. Show that the irreducible characters of G either are lifts of irreducible characters of G/N or are induced from the non-principal irreducible characters of N. When is it true that the inductions to G of two distinct non-principal irreducible characters of N are equal?

Solution. □

Further Exercises

In this set of exercises, we develop the character table of $G = \mathrm{GL}(2, q)$, where q is any prime power. Recall from Proposition 4.2 that we have $|G| = q(q-1)^2(q+1)$. Our first task is to determine the conjugacy classes of G. Here we sketch a comparatively "hands-on" approach, which uses only some standard topics from linear algebra. (A slicker approach might, for instance, make use of the invariant factor formulation of rational canonical form.) We denote by $m_g(X)$ the minimal polynomial of an element $g \in G$. Our strategy is to exploit the following:

Exercise 16.8. Show that conjugate elements of G have the same minimal polynomial.

Solution. □

Since $m_g(X)$ is a factor of the characteristic polynomial of g, which is the quadratic polynomial $\det(XI - g)$, we see that $m_g(X)$ is either linear, or a product of two like or unlike linear factors, or an irreducible monic quadratic.

Exercise 16.9. (cont.) Show that if $m_g(X) = X - a$ for some $a \in \mathbb{F}_q^\times$, then $g = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \in Z(G)$.

Solution. □

Exercise 16.10. (cont.) Show that if $m_g(X) = (X-a)^2$ for some $a \in \mathbb{F}_q^\times$, then g is conjugate with $\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$, and the conjugacy class of g has order q^2-1.

Solution. □

Exercise 16.11. (cont.) Show that if $m_g(X) = (X-a)(X-b)$ for some distinct $a, b \in \mathbb{F}_q^\times$, then g is conjugate with $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$, and the conjugacy class of g has order q^2+q.

Solution. □

Exercise 16.12. (cont.) Show that if $r, s \in \mathbb{F}_q$ are such that $X^2 - rX - s$ is irreducible, then the conjugacy class of $\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$ has order q^2-q.

Solution. □

(Hint for Exercises 16.10–16.12: Compute centralizers and use Proposition 3.13.)

Exercise 16.13. (cont.) Verify that what follows is a complete list of the conjugacy classes of G: $q-1$ classes of 1 element each, indexed by elements of $\mathbb{F}_q^\times$; $q-1$ classes of q^2-1 elements each, indexed by elements of $\mathbb{F}_q^\times$; $(q-1)(q-2)/2$ classes of q^2+q elements each, indexed by unordered pairs of distinct elements of $\mathbb{F}_q^\times$; and $q(q-1)/2$ classes of q^2-q elements each, indexed by irreducible monic quadratic polynomials with coefficients in $\mathbb{F}_q$.

Solution. □

Having determined the classes of G, we now turn to finding irreducible characters. Recall that we have an epimorphism $\det : G \to \mathbb{F}_q^\times$. Consequently, we can lift characters of $\mathbb{F}_q^\times$ to characters of G, as in Lemma 15.6. Now $\mathbb{F}_q^\times$ is an abelian group of order $q-1$, and hence it has $q-1$ linear characters, which we denote by $\tau_1, \ldots, \tau_{q-1}$. (In fact, it is true (see [17, p. 132]) that $\mathbb{F}_q^\times \cong \mathbb{Z}_{q-1}$, and so we can, upon fixing a generator of $\mathbb{F}_q^\times$, completely specify the τ_i.) For each $1 \le i \le q-1$, let $\chi_i = \tau_i \circ \det$. We have

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
χ_i	$\tau_i(a)^2$	$\tau_i(a)^2$	$\tau_i(a)\tau_i(b)$	$\tau_i(-s)$

The characters χ_i are linear, and χ_1 is indeed the principal character of G.
As in Chapter 2, let B, U, and T be the subgroups of G consisting, respectively, of all upper triangular, all upper unitriangular, and all diagonal elements of G. We have $T \cong \mathbb{F}_q^\times \times \mathbb{F}_q^\times$, and so we see from Example 15.2 that the irreducible characters of T have the form $\tau_i \times \tau_j$ (after making appropriate identifications). Now $B = U \rtimes T$ by Proposition 5.1, so we can lift each character $\tau_i \times \tau_j$ of $T \cong B/U$ to obtain a character θ_{ij} of B. The epimorphism from B to T sends $\begin{pmatrix} x & y \\ 0 & z \end{pmatrix}$ to $\begin{pmatrix} x & 0 \\ 0 & z \end{pmatrix}$, and consequently we have $\theta_{ij}\begin{pmatrix} x & y \\ 0 & z \end{pmatrix} = (\tau_i \times \tau_j)\begin{pmatrix} x & 0 \\ 0 & z \end{pmatrix} = \tau_i(x)\tau_j(z)$. We now wish to consider the induced characters θ_{ij}^G.

Exercise 16.14. (cont.) Show that

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
θ_{ij}^G	$(q+1)\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(b) + \tau_i(b)\tau_j(a)$	0

Observe that $\theta_{ij}^G = \theta_{ji}^G$.

Solution. □

Exercise 16.15. (cont.) Show that $(\theta_{ij}^G, \theta_{ij}^G) = 1 + \delta_{ij}$ and that $(\theta_{ii}^G, \chi_i) = 1$. Conclude that $\theta_{ii}^G - \chi_i$ is an irreducible character of G for each i and that θ_{ij}^G is an irreducible character of G whenever i and j are distinct.

Solution. □

At this point, we have constructed $q-1$ distinct irreducible characters of degree 1 (the χ_i), $q-1$ distinct irreducible characters of degree q (the $\theta_{ii}^G - \chi_i$), and $(q-1)(q-2)/2$ distinct irreducible characters of degree $q+1$ (the θ_{ij}^G for $i<j$). Our partial character table of G is now

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ s & r \end{pmatrix}$
χ_i	$\tau_i(a)^2$	$\tau_i(a)^2$	$\tau_i(a)\tau_i(b)$	$\tau_i(-s)$
$\theta_{ii}^G - \chi_i$	$q\tau_i(a)^2$	0	$\tau_i(a)\tau_i(b)$	$-\tau_i(-s)$
θ_{ij}^G	$(q+1)\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(a)$	$\tau_i(a)\tau_j(b)+\tau_i(b)\tau_j(a)$	0

To complete the character table, it will be convenient to work with a different set of representatives for the conjugacy classes of those elements whose minimal polynomials are irreducible quadratics. (We shall refer to these classes as being the "fourth-column" classes, owing to their placement in our partial character table above.) By Exercise 4.4, G has a subgroup C which is isomorphic with $\mathbb{F}_{q^2}^\times$, where $\mathbb{F}_{q^2}$ is the field of q^2 elements. As we remarked before, we have $C \cong \mathbb{Z}_{q^2-1}$. It is true, although we are not in a position to prove it, that any subgroup of G that is isomorphic with $\mathbb{F}_{q^2}^\times$ is a conjugate of C. (The proof of this fact requires some familiarity with field theory, but the essential point is that all fields of order q^2 are isomorphic; see [19, Corollary XIV.2.7].) It is also true that C must contain $Z(G) \cong \mathbb{F}_q^\times$, and so we will consider $\mathbb{F}_q^\times$ as being a subgroup of C.

Exercise 16.16. (cont.) Show that the elements of $C - \mathbb{F}_q^\times$ form a double set of representatives of the fourth-column classes, with $x \in C - \mathbb{F}_q^\times$ being conjugate with (but unequal to) x^q. (Hint: Argue that each element of G whose minimal polynomial is an irreducible quadratic determines a subgroup of G that is isomorphic with $\mathbb{F}_{q^2}^\times$.)

Solution. □

We now take $x \in C - \mathbb{F}_q^\times$ as our arbitrary fourth-column class representative. We have $\chi_i(x) = \tau_i(\det x)$, $(\theta_{ii}^G - \chi_i)(x) = -\tau_i(\det x)$, and $\theta_{ij}^G(x) = 0$.
Let $\beta_1, \ldots, \beta_{q^2-1}$ be the irreducible characters of C. (By choosing a generator for C, we can specify the β_i.) We will consider the induced characters β_k^G.

Exercise 16.17. (cont.) Show

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$x \in C - \mathbb{F}_q^\times$
β_k^G	$(q^2-q)\beta_k(a)$	0	0	$\beta_k(x)+\beta_k(x)^q$

Observe that $(\beta_k^q)^G = \beta_k^G$, where by β_k^q we mean the linear character of C defined by $\beta_k^q(x) = \beta_k(x)^q$.

Solution. □

It is not hard to show that there are exactly $q-1$ values of k for which $\beta_k^q = \beta_k$. (This follows, for instance, from the observation made in Example 15.1 that the linear characters of a cyclic group of order n themselves form a cyclic group of order n.)
Choose some k such that $\beta_k^q \neq \beta_k$, and let i_k be such that the restriction of β_k to $\mathbb{F}_q^\times \subseteq C$ is τ_{i_k}. Let $\psi_k = (\theta_{11}^G - \chi_1)\theta_{i_k 1}^G - \theta_{i_k 1}^G - \beta_k^G$; this is a virtual character.

Exercise 16.18. (cont.) Show that

	$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 1 \\ 0 & a \end{pmatrix}$	$\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$	$x \in C - \mathbb{F}_q^\times$
ψ_k	$(q-1)\tau_{i_k}(a)$	$-\tau_{i_k}(a)$	0	$-\beta_k(x)-\beta_k(x)^q$

Solution. □

Exercise 16.19. (cont.) If k is such that $\beta_k^q \neq \beta_k$, show that $(\psi_k, \psi_k) = 1$. Conclude that we have constructed $q(q-1)/2$ distinct irreducible characters of degree $q-1$.

Solution. □

The number of irreducible characters that we have constructed is now equal to the number of conjugacy classes of G, and so by Theorem 14.3 we are done.

Appendix A

Algebraic Integers and Characters

17 Algebraic Integers and Characters

Setting and motivation

- This appendix continues character theory with a different flavour from Chapter 6, leaning on Galois theory and a higher level of algebraic background.
- Logically the appendix follows Section 15.
- Throughout, G is a finite group.
- We say that $x \in \mathbb{C}$ is an *algebraic integer* if there is some monic polynomial $f \in \mathbb{Z}[X]$ such that $f(x) = 0$.

Lemma 17.1. *The set of rational numbers that are also algebraic integers is exactly the set of ordinary integers.*

Proof. □

Algebraic integers form a ring

- The next standard result is used freely; its proof is sketched in the exercises.

Proposition 17.2. *The algebraic integers form a subring of* $\mathbb{C}$.

Proof. □

Characters take algebraic-integer values

- The relevance of algebraic integers to representation theory of finite groups is established by the following basic fact.

Proposition 17.3. *Let* χ *be a character of* G. *Then* $\chi(g)$ *is an algebraic integer for any* $g \in G$.

Proof. □

Class-sum integrality

- The next result is necessary for both intended applications.

Lemma 17.4. *If* χ *is an irreducible character of* G *and* $g \in G$, *then* $|G : C_G(g)|\chi(g)/\chi(1)$ *is an algebraic integer.*

Proof. □

Degrees of irreducible characters divide $|G|$

- This proves the first main result of the appendix, a fact already mentioned in Section 14.

Proposition 17.5. *Let χ be an irreducible character of G. Then $\chi(1)$ divides $|G|$.*

Proof. □

Toward Burnside's $p^a q^b$ theorem

- The goal of the remainder of the appendix is to prove Burnside's Theorem on the solvability of groups of order $p^a q^b$, mentioned in Section 11.

Lemma 17.6. *Let χ be a character of G, let $g \in G$, and define $\gamma = \chi(g)/\chi(1)$. If γ is a non-zero algebraic integer, then $|\gamma| = 1$.*

Proof. □

A classical theorem of Burnside

- We now prove a classical theorem due to Burnside.

Theorem 17.7. *If G has a conjugacy class of non-trivial prime power order, then G is not simple.*

Proof. □

Burnside's famous $p^a q^b$ theorem

- Finally, we have Burnside's famed $p^a q^b$ theorem.

Theorem (Burnside's Theorem)**.** *If $|G| = p^a q^b$, where p and q are primes, then G is solvable.*

Proof. □

Beyond two primes

- It was observed on page 100 that there are non-solvable groups of order $p^a q^b r^c$, where p, q, and r are distinct primes.
- In fact, up to isomorphism there are eight such groups.
- Six of these groups are projective special linear groups. (Recall that A_5 and A_6 are, by Exercises 6.2 and 6.5, isomorphic with projective special linear groups.)
- The remaining two groups are members of a family of groups called the projective special unitary groups (see [24, p. 245]).

Exercises

Exercise 17.1. For $x \in \mathbb{C}$, let $\mathbb{Z}[x] = \{f(x) \mid f(X) \in \mathbb{Z}[X]\}$; this is an abelian subgroup of $\mathbb{C}$. Show that x is an algebraic integer iff $\mathbb{Z}[x]$ is finitely generated.

Solution. □

Exercise 17.2. (cont.) Show that the set of algebraic integers is a subring of $\mathbb{C}$. (Hint: Use Exercise 17.1 and the well-known fact (see [1, p. 145]) that any subgroup of a finitely generated abelian group is finitely generated.)

Solution. □

Exercise 17.3. Let G be a finite group, and define a function $\psi\colon G \to \mathbb{C}$ by setting $\psi(g) = |\{(x, y) \in G \times G \mid [x, y] = g\}|$ for $g \in G$. Show that

$$\psi = \sum_{i=1}^{r} \frac{|G|}{\chi_i(1)} \chi_i,$$

which in light of Proposition 17.5 shows that ψ is a character. (Compare with Exercise 3.6.)

Solution. □